

ELITE IGCSE ACADEMY

Edexcel IGCSE Mathematics A - Unit 1 (4WM1)

# Higher Tier

## Elite IGCSE Classified Unit 1 Expertise Answers (Q20+)

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127 Questions | Question page then worked-solution page

PREPARED & CLASSIFIED BY

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Modular palette | Unit 1 / 4WM1

Student worked-solution book - Q20+ expertise.  
Each question is followed by its worked solution on the next page.

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# Set Notation & Venn Diagrams

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**CLASSIFIED PROBLEM**

Paper: P2H

Session: Nov 2024

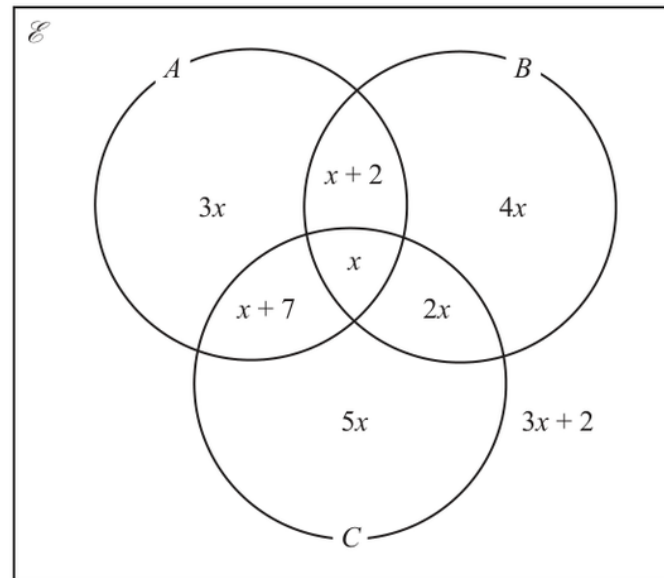
Q#: 21

Marks: 4

TOPIC: **Set Notation & Venn Diagrams**

Source: Nov 2024 P2H

- 21** The Venn diagram shows information about the numbers of items in set  $A$ , set  $B$  and set  $C$ , where  $x$  is an integer.



Given that  $n(A \cup B)' = 26$

find  $n(A' \cap C)$

$n(A' \cap C) = \dots\dots\dots$

**(Total for Question 21 is 4 marks)**

## WORKED SOLUTION

Dr. Eslam Ahmed

## Nov 2024 P2H | Question 21

Set Notation &amp; Venn Diagrams

**1. The complement of contains the**

The complement of  $A \cup B$  contains the regions outside both  $A$  and  $B$ . These are  $5x$  and  $3x + 2$ .

**2. So , hence and**

So  $5x + (3x + 2) = 26$ , hence  $8x + 2 = 26$  and  $x = 3$ .

**3. Now means inside , but**

Now  $A' \cap C$  means inside  $C$ , but not inside  $A$ . These are the regions  $5x$  and  $2x$ .

**4. So**

So  $n(A' \cap C) = 5x + 2x = 7x = 7(3) = 21$ .

**Final answer**

21

EA

# Rounding, Estimation & Bounds

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**CLASSIFIED PROBLEM**

Paper: P1HR

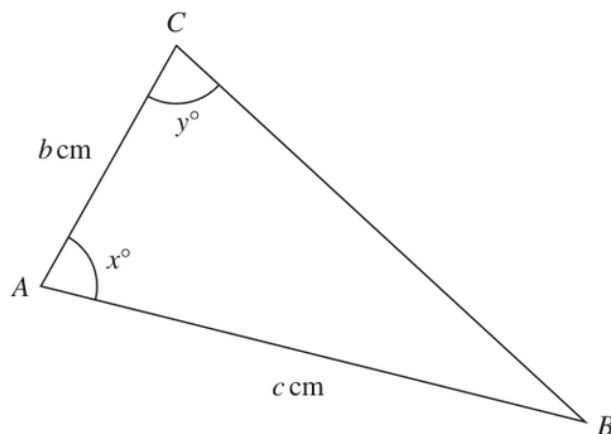
Session: Jan 2022

Q#: 22

Marks: 4

TOPIC: **Rounding, Estimation & Bounds**

Source: Jan 2022 P1HR

**22** The diagram shows triangle  $ABC$ Diagram **NOT**  
accurately drawn

$c = 11.5$  correct to one decimal place  
 $x = 80$  correct to the nearest whole number  
 $y = 75$  correct to the nearest whole number

Calculate the upper bound for the value of  $b$ 

Show your working clearly.

Give your answer correct to 3 significant figures.

---

**(Total for Question 22 is 4 marks)**

## WORKED SOLUTION

Dr. Eslam Ahmed

## Jan 2022 P1HR | Question 22

Rounding, Estimation &amp; Bounds

**1. For an upper bound for**For an upper bound for  $b$ , use the upper bound of  $c$ :

$$c = 11.55$$

**2. Use the lower bounds of**Use the lower bounds of  $x$  and  $y$ , because this makes angle  $B$  as large as possible and makes  $\sin y$  as small as possible.

$$x = 79.5^\circ, \quad y = 74.5^\circ$$

**3. So**

So

$$\angle B = 180^\circ - 79.5^\circ - 74.5^\circ = 26^\circ$$

**4. Using the sine rule,**

Using the sine rule,

$$\frac{b}{\sin 26^\circ} = \frac{11.55}{\sin 74.5^\circ}$$

$$b = \frac{11.55 \sin 26^\circ}{\sin 74.5^\circ} = 5.254\dots$$

**Final answer**

5.25 cm to 3 significant figures.

**CLASSIFIED PROBLEM**

Paper: P2H

Session: Jun 2023

Q#: 25

Marks: 4

TOPIC: **Rounding, Estimation & Bounds**

Source: Jun 2023 P2H

- 25** A solid sphere has a radius of 2.8 centimetres, correct to 1 decimal place.  
The sphere has a mass of  $M\pi$  grams, where  $M = 260$  correct to 2 significant figures.

Work out the upper bound for the density of the sphere.

Give your answer in  $\text{g/cm}^3$  correct to 2 decimal places.

Show your working clearly.

.....  $\text{g/cm}^3$

(Total for Question 25 is 4 marks)

## WORKED SOLUTION

Dr. Eslam Ahmed

## Jun 2023 P2H | Question 25

Rounding, Estimation &amp; Bounds

**1. For the upper bound of**

For the upper bound of density, use the upper bound of the mass and the lower bound of the volume.

**2. The radius is cm correct**

The radius is 2.8 cm correct to 1 decimal place, so the lower bound is

$$r = 2.75$$

**3. correct to 2 significant figures,**

$M = 260$  correct to 2 significant figures, so the upper bound is

$$M = 265$$

**4. The mass is given as**

The mass is given as  $M\pi$  grams, where  $M = 265$  for the upper bound.

**5. The volume of the sphere**

The volume of the sphere is

$$\frac{4}{3}\pi r^3$$

**6. So**

So

$$\text{density} = \frac{M\pi}{\frac{4}{3}\pi r^3}$$

**7. The cancels**

The  $\pi$  cancels:

$$\text{density} = \frac{3M}{4r^3}$$

**8. Upper bound**

Upper bound:

$$\frac{3(265)}{4(2.75)^3} = 9.5567...$$

**Final answer**

9.56 g/cm<sup>3</sup> to 2 decimal places.



**CLASSIFIED PROBLEM**

Paper: P2H

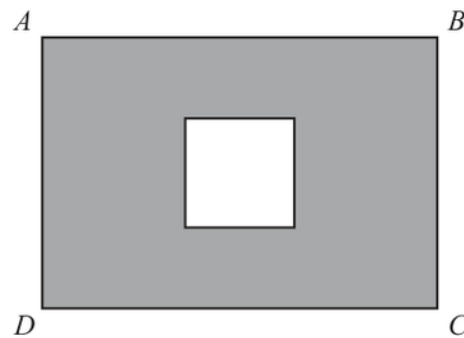
Session: Jun 2025

Q#: 22

Marks: 4

TOPIC: **Rounding, Estimation & Bounds**

Source: Jun 2025 P2H

**22** The diagram shows a square inside rectangle  $ABCD$ Diagram **NOT**  
accurately drawnThe total area of the region shown shaded in the diagram is  $X\text{cm}^2$  $AB = 11.5\text{ cm}$  correct to the nearest  $0.5\text{ cm}$  $BC = 9.2\text{ cm}$  correct to 2 significant figuresside of square =  $4.1\text{ cm}$  correct to 2 significant figuresBy considering bounds, work out the value of  $X$  to a suitable degree of accuracy.  
Show your working clearly. $X = \dots\dots\dots$ **(Total for Question 22 is 4 marks)**

**WORKED SOLUTION***Dr. Eslam Ahmed***Jun 2025 P2H | Question 22**

Rounding, Estimation &amp; Bounds

**1. The shaded area is**

The shaded area is

$$X = \text{area of rectangle} - \text{area of square}$$

**2. Bounds**

Bounds:

$$11.25 \leq AB < 11.75$$

$$9.15 \leq BC < 9.25$$

$$4.05 \leq \text{side of square} < 4.15$$

**3. Lower bound for**Lower bound for  $X$ :

$$11.25(9.15) - 4.15^2 = 85.715$$

**4. Upper bound for**Upper bound for  $X$ :

$$11.75(9.25) - 4.05^2 = 92.285$$

**5. So . Both bounds round**So  $85.715 \leq X < 92.285$ . Both bounds round to 90 to the nearest 10.**Final answer** $X = 90 \text{ cm}^2$  to a suitable degree of accuracy.

**CLASSIFIED PROBLEM**

Paper: P1H

Session: May 2021

Q#: 20

Marks: 3

TOPIC: **Rounding, Estimation & Bounds**

Source: May 2021 P1H

**20**  $P = \frac{t - w}{y}$

 $t = 9.7$  correct to 1 decimal place $w = 5.9$  correct to 1 decimal place $y = 3$  correct to 1 significant figure

Calculate the upper bound for the value of  $P$ .  
Show your working clearly.

---

(Total for Question 20 is 3 marks)

**WORKED SOLUTION***Dr. Eslam Ahmed***May 2021 P1H | Question 20**

Rounding, Estimation &amp; Bounds

**1. Step 1**

$$P = \frac{t - w}{y}$$

**2. For the upper bound of**

For the upper bound of  $P$ , make the numerator as large as possible and the denominator as small as possible.

$$t = 9.75, \quad w = 5.85, \quad y = 2.5$$

$$P = \frac{9.75 - 5.85}{2.5}$$

$$P = \frac{3.90}{2.5} = 1.56$$

**Final answer**

1.56

**CLASSIFIED PROBLEM**

Paper: P1H

Session: Nov 2024

Q#: 21

Marks: 3

TOPIC: **Rounding, Estimation & Bounds**

Source: Nov 2024 P1H

**21**  $T = \frac{x^2 + y^2}{w}$

$x = 28.4$  correct to 1 decimal place.

$y = 17$  correct to 2 significant figures.

$w = 90$  correct to the nearest 5

Calculate the upper bound for the value of  $T$

Give your answer correct to 3 significant figures.

Show your working clearly.

(Total for Question 21 is 3 marks)

## WORKED SOLUTION

Dr. Eslam Ahmed

**Nov 2024 P1H | Question 21**

Rounding, Estimation &amp; Bounds

**1. Step 1**

$$T = \frac{x^2 + y^2}{w}$$

**2. For the upper bound of**

For the upper bound of  $T$ , use the upper bounds of  $x$  and  $y$ , and the lower bound of  $w$ .

$$x = 28.45, \quad y = 17.5, \quad w = 87.5$$

$$T = \frac{28.45^2 + 17.5^2}{87.5}$$

$$T = 12.7503\dots$$

**Final answer**

12.8 to 3 significant figures.

**CLASSIFIED PROBLEM**

Paper: P2H

Session: Nov 2025

Q#: 24

Marks: 5

TOPIC: **Rounding, Estimation & Bounds**

Source: Nov 2025 P2H

**24**  $D = \frac{n}{p - q}$

 $n = 10.3$  correct to 1 decimal place $p = 7.24$  correct to 2 decimal places $q = 4.39$  correct to 2 decimal places

By considering bounds, work out the value of  $D$  to a suitable degree of accuracy.  
Show your working clearly.

(Total for Question 24 is 5 marks)

## WORKED SOLUTION

Dr. Eslam Ahmed

## Nov 2025 P2H | Question 24

Rounding, Estimation &amp; Bounds

## 1. Step 1

$$D = \frac{n}{p - q}$$

## 2. Bounds

Bounds:

$$10.25 \leq n < 10.35$$

$$7.235 \leq p < 7.245$$

$$4.385 \leq q < 4.395$$

## 3. Lower bound

Lower bound:

$$\frac{10.25}{7.245 - 4.385} = 3.5839...$$

## 4. Upper bound

Upper bound:

$$\frac{10.35}{7.235 - 4.395} = 3.6443...$$

## 5. So . Both bounds round

So  $3.5839... \leq D < 3.6443...$  Both bounds round to 3.6 to 1 decimal place.**Final answer** $D = 3.6$  to a suitable degree of accuracy.



# Surds

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**CLASSIFIED PROBLEM**

Paper: P2H

Session: Jan 2020

Q#: 20

Marks: 3

Topic: **Surds**

Source: Jan 2020 P2H

**20** The area of a rectangle is  $18\text{ cm}^2$ The length of the rectangle is  $(\sqrt{7} + 1)\text{ cm}$ .

Without using a calculator and showing each stage of your working,

find the width of the rectangle.

Give your answer in the form  $a\sqrt{b} + c$  where  $a$ ,  $b$  and  $c$  are integers.

..... cm

**(Total for Question 20 is 3 marks)**

## WORKED SOLUTION

Dr. Eslam Ahmed

Jan 2020 P2H | Question 20

Surd

**1. Area of rectangle**

Area of rectangle:

$$18 \text{ cm}^2$$

**2. Length**

Length:

$$\sqrt{7} + 1$$

**3. Width**

Width:

$$\frac{18}{\sqrt{7} + 1}$$

**4. Rationalise the denominator**

Rationalise the denominator:

$$\begin{aligned} & \frac{18}{\sqrt{7} + 1} \times \frac{\sqrt{7} - 1}{\sqrt{7} - 1} \\ &= \frac{18(\sqrt{7} - 1)}{7 - 1} \\ &= \frac{18(\sqrt{7} - 1)}{6} \\ &= 3(\sqrt{7} - 1) \\ &= 3\sqrt{7} - 3 \end{aligned}$$

**Final answer**

$$3\sqrt{7} - 3 \text{ cm}$$

**CLASSIFIED PROBLEM**

Paper: P1HR

Session: Jun 2022

Q#: 21

Marks: 4

TOPIC: **Surds**

Source: Jun 2022 P1HR

**21** Express  $\frac{3 + \sqrt{8}}{(\sqrt{2} - 1)^2}$  in the form  $p + \sqrt{q}$  where  $p$  and  $q$  are integers.

Show each stage of your working clearly.

(Total for Question 21 is 4 marks)

## WORKED SOLUTION

Dr. Eslam Ahmed

## Jun 2022 P1HR | Question 21

Surd

## 1. Step 1

$$\frac{3 + \sqrt{8}}{(\sqrt{2} - 1)^2}$$

## 2. First simplify

First simplify:

$$\sqrt{8} = 2\sqrt{2}$$

## 3. and

and

$$(\sqrt{2} - 1)^2 = 2 - 2\sqrt{2} + 1 = 3 - 2\sqrt{2}$$

## 4. So

So

$$\frac{3 + 2\sqrt{2}}{3 - 2\sqrt{2}}$$

## 5. Rationalise the denominator

Rationalise the denominator:

$$\frac{3 + 2\sqrt{2}}{3 - 2\sqrt{2}} \times \frac{3 + 2\sqrt{2}}{3 + 2\sqrt{2}} = \frac{(3 + 2\sqrt{2})^2}{9 - 8}$$

$$(3 + 2\sqrt{2})^2 = 9 + 12\sqrt{2} + 8 = 17 + 12\sqrt{2}$$

**Final answer**

$$17 + 12\sqrt{2}$$

**CLASSIFIED PROBLEM**

Paper: P2H

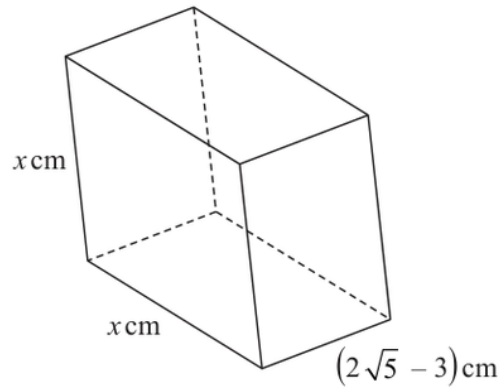
Session: Jun 2023

Q#: 23

Marks: 4

Topic: **Surds**

Source: Jun 2023 P2H

**23** The diagram shows a cuboid with a square cross section.Diagram **NOT**  
accurately drawnThe volume of the cuboid is  $(13 + 6\sqrt{5}) \text{ cm}^3$ Without using a calculator, find the value of  $x$ Give your answer in the form  $a + \sqrt{b}$  where  $a$  and  $b$  are integers.

Show your working clearly.

 $x = \dots\dots\dots$ **(Total for Question 23 is 4 marks)**

## WORKED SOLUTION

Dr. Eslam Ahmed

## Jun 2023 P2H | Question 23

Surd

**1. The cuboid has square cross**

The cuboid has square cross section, so its volume is

$$x^2(2\sqrt{5} - 3)$$

**2. Given**

Given

$$x^2(2\sqrt{5} - 3) = 13 + 6\sqrt{5}$$

**3. So**

So

$$x^2 = \frac{13 + 6\sqrt{5}}{2\sqrt{5} - 3}$$

**4. Rationalise the denominator**

Rationalise the denominator:

$$x^2 = \frac{13 + 6\sqrt{5}}{2\sqrt{5} - 3} \cdot \frac{2\sqrt{5} + 3}{2\sqrt{5} + 3}$$

$$x^2 = \frac{(13 + 6\sqrt{5})(2\sqrt{5} + 3)}{20 - 9}$$

$$x^2 = \frac{99 + 44\sqrt{5}}{11}$$

$$x^2 = 9 + 4\sqrt{5}$$

**5. Now**

Now

$$(2 + \sqrt{5})^2 = 4 + 4\sqrt{5} + 5 = 9 + 4\sqrt{5}$$

**6. Therefore**

Therefore

$$x = 2 + \sqrt{5}$$

**Final answer**

$$\boxed{2 + \sqrt{5}}$$

# Algebraic Roots & Indices

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**CLASSIFIED PROBLEM**

Paper: P2HR

Session: Jan 2020

Q#: 21

Marks: 7

TOPIC: Algebraic Roots &amp; Indices

Source: Jan 2020 P2HR

21 (a) Simplify fully  $\frac{10x^2 + 23x + 12}{4x^2 - 9}$

$$2^{2y} \times 2^{3y+2} = \frac{8^{5y}}{4^n}$$

.....  
(3)

(b) Find an expression for  $n$  in terms of  $y$ .  
Show clear algebraic working and simplify your expression.

.....  
(4)

(Total for Question 21 is 7 marks)

## WORKED SOLUTION

Dr. Eslam Ahmed

## Jan 2020 P2HR | Question 21

Algebraic Roots &amp; Indices

1. (a)

(a)

$$\frac{10x^2 + 23x + 12}{4x^2 - 9} = \frac{(5x + 4)(2x + 3)}{(2x - 3)(2x + 3)}$$

$$= \frac{5x + 4}{2x - 3}$$

2. (b)

(b)

$$2^{2y} \times 2^{3y+2} = 2^{5y+2}$$

3. Also,

Also,

$$\frac{8^{5y}}{4^n} = \frac{(2^3)^{5y}}{(2^2)^n} = 2^{15y-2n}$$

4. Equate powers

Equate powers:

$$5y + 2 = 15y - 2n$$

$$2n = 10y - 2$$

$$n = 5y - 1$$

**Final answer**

$$(a) \frac{5x + 4}{2x - 3}, (b) n = 5y - 1$$

**CLASSIFIED PROBLEM**

Paper: P2H

Session: Jun 2022

Q#: 24

Marks: 3

TOPIC: Algebraic Roots &amp; Indices

Source: Jun 2022 P2H

**24**

$$\frac{18 \times (\sqrt{2})}{6 \times 9^{2n+8}} = 3^x$$

Express  $x$  in terms of  $n$ 

Show your working clearly and simplify your expression.

 $x = \dots\dots\dots$ 

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**(Total for Question 24 is 3 marks)**

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## WORKED SOLUTION

Dr. Eslam Ahmed

**Jun 2022 P2H | Question 24**  
Algebraic Roots & Indices

**1. Step 1**

$$\frac{18(\sqrt{27})^{4n+6}}{6 \times 9^{2n+8}} = 3^x$$

**2. Use powers of 3**

Use powers of 3:

$$18 \div 6 = 3$$

$$\sqrt{27} = (3^3)^{1/2} = 3^{3/2}$$

$$(\sqrt{27})^{4n+6} = 3^{6n+9}$$

$$9^{2n+8} = (3^2)^{2n+8} = 3^{4n+16}$$

**3. So**

So

$$\frac{3 \cdot 3^{6n+9}}{3^{4n+16}} = 3^{1+6n+9-4n-16} = 3^{2n-6}$$

**4. Therefore,**

Therefore,

$$x = 2n - 6$$

**Final answer**

$$x = 2n - 6$$

**CLASSIFIED PROBLEM**

Paper: P2H

Session: Jun 2024

Q#: 23

Marks: 3

TOPIC: Algebraic Roots &amp; Indices

Source: Jun 2024 P2H

23 Simplify  $\frac{30 \times 25^{2x+7}}{\sqrt{180} \times (\sqrt{5})^{4x+9}}$

Give your answer in the form  $5^w$  where  $w$  is an expression in terms of  $x$   
Show each stage of your working clearly.

(Total for Question 23 is 3 marks)

## WORKED SOLUTION

Dr. Eslam Ahmed

Jun 2024 P2H | Question 23  
Algebraic Roots & Indices

## 1. Step 1

$$\frac{30 \times 25^{2x+7}}{\sqrt{180} \times (\sqrt{5})^{4x+9}}$$

$$\frac{30}{\sqrt{180}} = \frac{30}{6\sqrt{5}} = \sqrt{5} = 5^{1/2}$$

$$25^{2x+7} = (5^2)^{2x+7} = 5^{4x+14}$$

$$(\sqrt{5})^{4x+9} = 5^{2x+\frac{9}{2}}$$

## 2. So

So

$$5^{1/2} \times 5^{4x+14} \div 5^{2x+\frac{9}{2}} = 5^{\frac{1}{2}+4x+14-2x-\frac{9}{2}}$$

$$= 5^{2x+10}$$

**Final answer**

$5^{2x+10}$ , so  $w = 2x + 10$

**CLASSIFIED PROBLEM**

Paper: P1HR

Session: May 2023

Q#: 24

Marks: 5

TOPIC: Algebraic Roots &amp; Indices

Source: May 2023 P1HR

**24** Given that

$$2^n = 2^{x^2} \times 16^x \times 8$$

and

$$x > 0$$

find an expression for  $x$  in terms of  $n$ State any restrictions on  $n$ 

---

**(Total for Question 24 is 5 marks)**

## WORKED SOLUTION

Dr. Eslam Ahmed

May 2023 P1HR | Question 24  
Algebraic Roots & Indices

## 1. Step 1

$$2^n = 2^{x^2} \times 16^x \times 8$$

$$16^x = (2^4)^x = 2^{4x}$$

$$8 = 2^3$$

## 2. So

So

$$2^n = 2^{x^2+4x+3}$$

$$n = x^2 + 4x + 3$$

## 3. Complete the square

Complete the square:

$$n = (x + 2)^2 - 1$$

$$(x + 2)^2 = n + 1$$

## 4. Since , take the positive

Since  $x > 0$ , take the positive root:

$$x + 2 = \sqrt{n + 1}$$

$$x = \sqrt{n + 1} - 2$$

## 5. For ,

For  $x > 0$ ,

$$\sqrt{n + 1} > 2$$

$$n > 3$$

**Final answer**

$$x = \sqrt{n + 1} - 2, \text{ with } n > 3$$



**CLASSIFIED PROBLEM**

Paper: P1HR

Session: MayNov 2020

Q#: 21

Marks: 5

TOPIC: Algebraic Roots &amp; Indices

Source: May-Nov 2020 P1HR

**21** Given that  $M = \frac{18^{4n} \times 2^{3(n^2-6n)} \times 3^{2(1-4n)}}{12^2}$

find the values of  $n$  for which  $M = 2$

(Total for Question 21 is 5 marks)

## WORKED SOLUTION

Dr. Eslam Ahmed

May-Nov 2020 P1HR | Question 21  
Algebraic Roots & Indices

## 1. Step 1

$$M = \frac{18^{4n} \times 2^{3(n^2-6n)} \times 3^{2(1-4n)}}{12^2}$$

## 2. Write everything in powers of

Write everything in powers of 2 and 3:

$$18^{4n} = (2 \cdot 3^2)^{4n} = 2^{4n} 3^{8n}$$

$$2^{3(n^2-6n)} = 2^{3n^2-18n}$$

$$3^{2(1-4n)} = 3^{2-8n}$$

$$12^2 = (2^2 \cdot 3)^2 = 2^4 3^2$$

## 3. So

So

$$M = 2^{4n+3n^2-18n-4} \times 3^{8n+2-8n-2}$$

$$M = 2^{3n^2-14n-4}$$

## 4. Given ,

Given  $M = 2$ ,

$$2^{3n^2-14n-4} = 2^1$$

$$3n^2 - 14n - 4 = 1$$

$$3n^2 - 14n - 5 = 0$$

$$(3n + 1)(n - 5) = 0$$

**Final answer**

$$n = 5 \text{ or } n = -\frac{1}{3}$$

# Completing the Square

---

**CLASSIFIED PROBLEM**

Paper: 4WM1H

Session: May2025

Q#: 25

Marks: 5

TOPIC: **Completing the Square**

Source: May 2025 4WM1H

- 25** (a) Express  $8 - 12x - 4x^2$  in the form  $a - b(x + c)^2$   
where  $a$ ,  $b$  and  $c$  are constants.

.....  
(3)

- (b) Hence solve  $8 - 12x - 4x^2 = 0$   
Show your working clearly.  
Give your solutions in surd form.

.....  
(2)

.....  
(Total for Question 25 is 5 marks)

## WORKED SOLUTION

Dr. Eslam Ahmed

## May 2025 4WM1H | Question 25

Completing the Square

## 1. Step 1

$$8x^2 - 12x - 4 = 8\left(x^2 - \frac{3}{2}x\right) - 4$$

## 2. Complete the square

Complete the square.

$$x^2 - \frac{3}{2}x = \left(x - \frac{3}{4}\right)^2 - \frac{9}{16}$$

## 3. So

So

$$8x^2 - 12x - 4 = 8\left(x - \frac{3}{4}\right)^2 - \frac{17}{2}$$

## 4. Now solve

Now solve

$$8x^2 - 12x - 4 = 0$$

$$8\left(x - \frac{3}{4}\right)^2 - \frac{17}{2} = 0$$

$$\left(x - \frac{3}{4}\right)^2 = \frac{17}{16}$$

$$x - \frac{3}{4} = \pm \frac{\sqrt{17}}{4}$$

$$x = \frac{3 \pm \sqrt{17}}{4}$$

**Final answer**

$$8\left(x - \frac{3}{4}\right)^2 - \frac{17}{2}, \text{ and } x = \frac{3 \pm \sqrt{17}}{4}.$$

**CLASSIFIED PROBLEM**

Paper: 4WM1H

Session: Nov2025

Q#: 22

Marks: 2

TOPIC: **Completing the Square**

Source: Nov 2025 4WM1H

**22**  $9x^2 - 12x + q$  can be written in the form  $(px - r)^2$  where  $p$ ,  $q$  and  $r$  are integers.

Find the value of  $q$

$q = \dots\dots\dots$

(Total for Question 22 is 2 marks)

## WORKED SOLUTION

Dr. Eslam Ahmed

Nov 2025 4WM1H | Question 22  
Completing the Square

## 1. Step 1

$$9x^2 - 12x + q = (3x - 2)^2$$

$$(3x - 2)^2 = 9x^2 - 12x + 4$$

**Final answer**

$$q = 4.$$

**CLASSIFIED PROBLEM**

Paper: 4WM1H

Session: Nov2025

Q#: 24

Marks: 3

TOPIC: **Completing the Square**

Source: Nov 2025 4WM1H

**24** Express  $5x^2 - 20x + 23$  in the form  $a(x - b)^2 + c$  where  $a$ ,  $b$  and  $c$  are integers.

---

(Total for Question 24 is 3 marks)



## WORKED SOLUTION

Dr. Eslam Ahmed

Nov 2025 4WM1H | Question 24  
Completing the Square

## 1. Step 1

$$\begin{aligned}5x^2 - 20x + 23 &= 5(x^2 - 4x) + 23 \\&= 5((x - 2)^2 - 4) + 23 \\&= 5(x - 2)^2 + 3\end{aligned}$$

**Final answer**

$$5(x - 2)^2 + 3.$$

EA

# Algebraic Fractions

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**CLASSIFIED PROBLEM**

Paper: 4WM1H

Session: May2025

Q#: 20

Marks: 3

TOPIC: Algebraic Fractions

Source: May 2025 4WM1H

20 Given that  $k = m + n$  and  $m = \frac{3}{4n}$

express  $\frac{7k}{5-m}$  in the form  $\frac{a+bn^2}{cn-d}$  where  $a, b$  and  $c$  are integers and  $d$  is prime.

(Total for Question 20 is 3 marks)

## WORKED SOLUTION

Dr. Eslam Ahmed

May 2025 4WM1H | Question 20  
Algebraic Fractions

## 1. Step 1

$$k = m + n, \quad m = \frac{3}{4n}$$

## 2. So

So

$$k = \frac{3}{4n} + n = \frac{3 + 4n^2}{4n}$$

## 3. Then

Then

$$7k = \frac{21 + 28n^2}{4n}$$

## 4. and

and

$$5 - m = 5 - \frac{3}{4n} = \frac{20n - 3}{4n}$$

## 5. Therefore

Therefore

$$\begin{aligned} \frac{7k}{5 - m} &= \frac{\frac{21 + 28n^2}{4n}}{\frac{20n - 3}{4n}} \\ &= \frac{21 + 28n^2}{20n - 3} \end{aligned}$$

**Final answer**

$$\frac{21 + 28n^2}{20n - 3}$$

**CLASSIFIED PROBLEM**

Paper: 4WM1HR

Session: May2025

Q#: 24

Marks: 4

TOPIC: **Algebraic Fractions**

Source: May 2025 4WM1HR

**24** Write  $(2x - 1) + \frac{2x^2 - 5x - 12}{x^2 - 16} \times \frac{x^2 + 4x}{6x^2 + 11x + 3}$  as a single fraction in its simplest form.

Show clear algebraic working.

(Total for Question 24 is 4 marks)

## WORKED SOLUTION

Dr. Eslam Ahmed

May 2025 4WM1HR | Question 24  
Algebraic Fractions

### 1. Factor the fractional part first

Factor the fractional part first:

$$\begin{aligned} & \frac{2x^2 - 5x - 12}{x^2 - 16} \times \frac{x^2 + 4x}{6x^2 + 11x + 3} \\ &= \frac{(2x + 3)(x - 4)}{(x - 4)(x + 4)} \times \frac{x(x + 4)}{(3x + 1)(2x + 3)} \\ &= \frac{x}{3x + 1} \end{aligned}$$

### 2. So

So

$$\begin{aligned} (2x - 1) + \frac{x}{3x + 1} &= \frac{(2x - 1)(3x + 1) + x}{3x + 1} \\ &= \frac{6x^2 - 1}{3x + 1} \end{aligned}$$

**Final answer**

$$\frac{6x^2 - 1}{3x + 1}.$$

**CLASSIFIED PROBLEM**

Paper: 4WM1H

Session: Nov2025

Q#: 25

Marks: 3

TOPIC: **Algebraic Fractions**

Source: Nov 2025 4WM1H

**25**  $a = \frac{2x+5}{1-x} \quad x = \frac{5-2y}{3y}$

Write  $a$  in the form  $\frac{m+ny}{p(y-1)}$  where  $m, n$  and  $p$  are integers.

Show your working clearly.

$a = \dots\dots\dots$

**(Total for Question 25 is 3 marks)**

## WORKED SOLUTION

Dr. Eslam Ahmed

Nov 2025 4WM1H | Question 25  
Algebraic Fractions

## 1. Step 1

$$a = \frac{2x + 5}{1 - x}, \quad x = \frac{5 - 2y}{3y}$$

## 2. Substitute for

Substitute for  $x$ .

## 3. Numerator

Numerator:

$$\begin{aligned} 2x + 5 &= 2\left(\frac{5 - 2y}{3y}\right) + 5 \\ &= \frac{10 - 4y + 15y}{3y} = \frac{10 + 11y}{3y} \end{aligned}$$

## 4. Denominator

Denominator:

$$\begin{aligned} 1 - x &= 1 - \frac{5 - 2y}{3y} \\ &= \frac{3y - 5 + 2y}{3y} = \frac{5y - 5}{3y} = \frac{5(y - 1)}{3y} \end{aligned}$$

## 5. Therefore

Therefore

$$a = \frac{\frac{10 + 11y}{3y}}{\frac{5(y - 1)}{3y}} = \frac{10 + 11y}{5(y - 1)}$$

**Final answer**  
 $\frac{10 + 11y}{5(y - 1)}$



**CLASSIFIED PROBLEM**

Paper: 4WM1H

Session: Nov2025

Q#: 28

Marks: 4

TOPIC: **Algebraic Fractions**

Source: Nov 2025 4WM1H

**28** Solve  $\frac{6x^3 + 5x^2 + x}{4x^2 - 1} \div \frac{15x^2 - x - 2}{8x - 4} = 2x$

Show clear algebraic working.

 $x = \dots\dots\dots$ 

---

**(Total for Question 28 is 4 marks)**

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## WORKED SOLUTION

Dr. Eslam Ahmed

## Nov 2025 4WM1H | Question 28

Algebraic Fractions

## 1. Factor each expression

Factor each expression:

$$\frac{x(3x+1)(2x+1)}{(2x-1)(2x+1)} \div \frac{(5x-2)(3x+1)}{4(2x-1)} = 2x$$

$$\frac{x(3x+1)}{2x-1} \times \frac{4(2x-1)}{(5x-2)(3x+1)} = 2x$$

$$\frac{4x}{5x-2} = 2x$$

$$4x = 2x(5x-2)$$

$$10x^2 - 8x = 0$$

$$2x(5x-4) = 0$$

## 2. The original fractions do not

The original fractions do not allow  $x = 0$ , so

$$x = \frac{4}{5}$$

## Final answer

$$x = \frac{4}{5}.$$

# Completing the Square

---

**CLASSIFIED PROBLEM**

Paper: P2HR

Session: Jan 2020

Q#: 23

Marks: 3

TOPIC: **Completing the Square**

Source: Jan 2020 P2HR

**23** Express  $7 - 12x - 2x^2$  in the form  $a + b(x + c)^2$  where  $a$ ,  $b$  and  $c$  are integers.

---

(Total for Question 23 is 3 marks)

## WORKED SOLUTION

Dr. Eslam Ahmed

Jan 2020 P2HR | Question 23  
Completing the Square

## 1. Step 1

$$\begin{aligned}7 - 12x - 2x^2 &= -2x^2 - 12x + 7 \\&= -2(x^2 + 6x) + 7 \\&= -2((x + 3)^2 - 9) + 7 \\&= -2(x + 3)^2 + 18 + 7 \\&= 25 - 2(x + 3)^2\end{aligned}$$

**Final answer**  
 $25 - 2(x + 3)^2$

**CLASSIFIED PROBLEM**

Paper: P1H

Session: Jan 2022

Q#: 20

Marks: 4

TOPIC: **Completing the Square**

Source: Jan 2022 P1H

**20** (a) Express  $7 + 12x - 3x^2$  in the form  $a + b(x + c)^2$  where  $a$ ,  $b$  and  $c$  are integers.

.....  
(3)

**C** is the curve with equation  $y = 7 + 12x - 3x^2$

The point **A** is the maximum point on **C**

(b) Use your answer to part (a) to write down the coordinates of **A**

(....., .....)  
(1)

(Total for Question 20 is 4 marks)

## WORKED SOLUTION

Dr. Eslam Ahmed

Jan 2022 P1H | Question 20  
Completing the Square

1. (a)

(a)

$$7 + 12x - 3x^2 = -3x^2 + 12x + 7$$

$$= -3(x^2 - 4x) + 7$$

$$= -3((x - 2)^2 - 4) + 7$$

$$= 19 - 3(x - 2)^2$$

2. This is the same as

This is the same as

$$19 - 3(x + (-2))^2$$

3. (b) The maximum point is

(b) The maximum point is read from the completed square form.

$$y = 19 - 3(x - 2)^2$$

4. The maximum occurs when ,

The maximum occurs when  $x = 2$ , giving  $y = 19$ .

**Final answer**

(a)  $19 - 3(x - 2)^2$ , (b)  $A = (2, 19)$

**CLASSIFIED PROBLEM**

Paper: P1H

Session: Jun 2022

Q#: 24

Marks: 4

TOPIC: **Completing the Square**

Source: Jun 2022 P1H

**24** Express each of  $a$ ,  $b$  and  $c$  in terms of  $q$  so that

$$q + 12x - qx^2$$

can be written as  $a - b(x - c)^2$  $a = \dots\dots\dots$  $b = \dots\dots\dots$  $c = \dots\dots\dots$ **(Total for Question 24 is 4 marks)**



## WORKED SOLUTION

Dr. Eslam Ahmed

## Jun 2022 P1H | Question 24

Completing the Square

## 1. Step 1

$$q + 12x - qx^2 = -q \left( x^2 - \frac{12}{q}x \right) + q$$

## 2. Complete the square

Complete the square:

$$x^2 - \frac{12}{q}x = \left( x - \frac{6}{q} \right)^2 - \frac{36}{q^2}$$

## 3. So

So

$$\begin{aligned} q + 12x - qx^2 &= -q \left[ \left( x - \frac{6}{q} \right)^2 - \frac{36}{q^2} \right] + q \\ &= q + \frac{36}{q} - q \left( x - \frac{6}{q} \right)^2 \end{aligned}$$

## 4. Compare with

Compare with

$$a - b(x - c)^2$$

**Final answer**

$$a = q + \frac{36}{q}, b = q, c = \frac{6}{q}$$

**CLASSIFIED PROBLEM**

Paper: P1HR

Session: Jun 2022

Q#: 23

Marks: 5

TOPIC: **Completing the Square**

Source: Jun 2022 P1HR

**23** (a) Express  $2x^2 - 12x + 3$  in the form  $a(x + b)^2 + c$  where  $a$ ,  $b$  and  $c$  are integers.

.....  
(3)

The curve **C** has equation  $y = 2(x + 4)^2 - 12(x + 4) + 3$

The point  $M$  is the minimum point on **C**

(b) Find the coordinates of  $M$

(..... , .....)  
(2)

(Total for Question 23 is 5 marks)

## WORKED SOLUTION

Dr. Eslam Ahmed

## Jun 2022 P1HR | Question 23

Completing the Square

1. (a)

(a)

$$2x^2 - 12x + 3 = 2(x^2 - 6x) + 3$$

$$= 2((x - 3)^2 - 9) + 3$$

$$= 2(x - 3)^2 - 18 + 3$$

$$= 2(x - 3)^2 - 15$$

2. (b)

(b)

$$y = 2(x + 4)^2 - 12(x + 4) + 3$$

3. Let . From part (a),

Let  $u = x + 4$ . From part (a),

$$2u^2 - 12u + 3 = 2(u - 3)^2 - 15$$

4. So

So

$$y = 2((x + 4) - 3)^2 - 15$$

$$y = 2(x + 1)^2 - 15$$

5. The minimum point is when

The minimum point is when  $x = -1$ , giving  $y = -15$ .**Final answer**(a)  $2(x - 3)^2 - 15$ , (b)  $M = (-1, -15)$

**CLASSIFIED PROBLEM**

Paper: P2H

Session: Jun 2025

Q#: 25

Marks: 6

Topic: **Completing the Square**

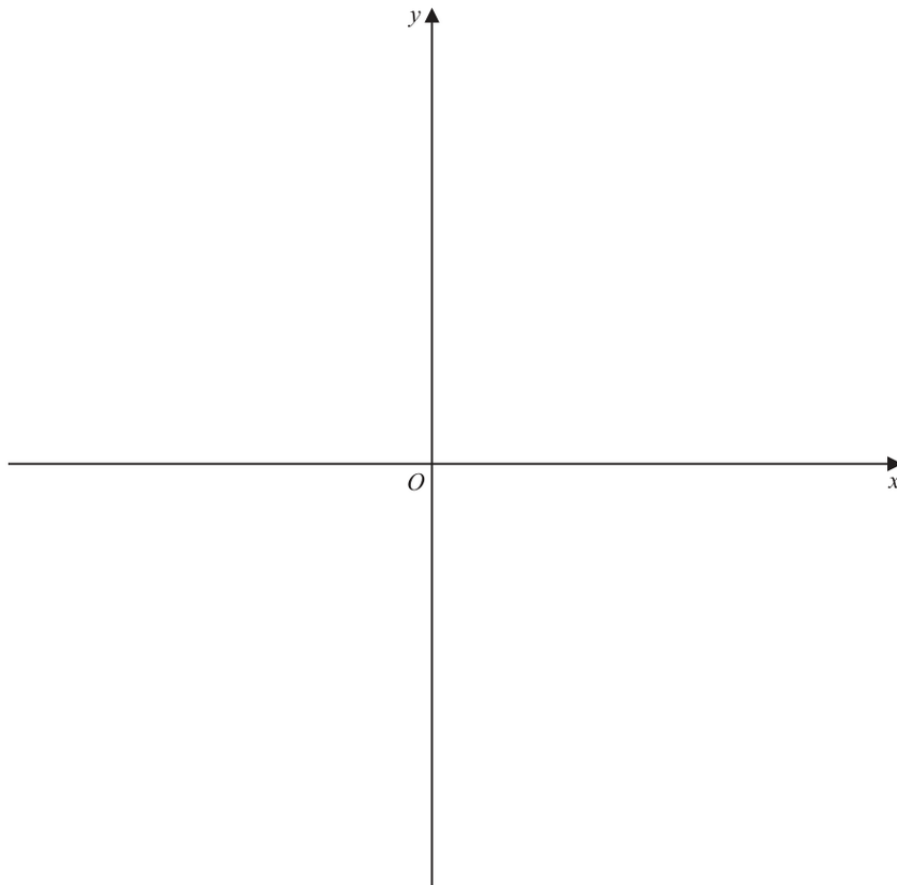
Source: Jun 2025 P2H

25 (a) Write  $28 + 24x - 6x^2$  in the form  $a - b(x - c)^2$  where  $a$ ,  $b$  and  $c$  are integers.

.....  
(3)

(b) On the axes below, sketch the graph of  $y = 28 + 24x - 6x^2$

Show clearly the coordinates of the turning point and the coordinates of the point of intersection of the graph with the  $y$ -axis.



(3)

(Total for Question 25 is 6 marks)

## WORKED SOLUTION

Dr. Eslam Ahmed

## Jun 2025 P2H | Question 25

Completing the Square

1. (a)

(a)

$$28 + 24x - 6x^2 = -6x^2 + 24x + 28$$

$$= -6(x^2 - 4x) + 28$$

$$= -6((x - 2)^2 - 4) + 28$$

$$= 52 - 6(x - 2)^2$$

2. (b)

(b)

$$y = 52 - 6(x - 2)^2$$

3. The graph is a downward

The graph is a downward-opening parabola.

4. The turning point is

The turning point is

$$(2, 52)$$

5. The point of intersection with

The point of intersection with the  $y$ -axis is found by putting  $x = 0$ :

$$y = 28$$

6. So the intercept is

So the  $y$ -intercept is

$$(0, 28)$$

**Final answer**

(a)  $52 - 6(x - 2)^2$ . For the sketch, show a downward parabola with turning point  $(2, 52)$  and  $y$ -intercept  $(0, 28)$ .

**CLASSIFIED PROBLEM**

Paper: P2H

Session: May 2021

Q#: 22

Marks: 4

TOPIC: **Completing the Square**

Source: May 2021 P2H

- 22** The curve **S** has equation  $y = f(x)$  where  $f(x) = x^2$   
The curve **T** has equation  $y = g(x)$  where  $g(x) = 2x^2 - 12x + 13$

By writing  $g(x)$  in the form  $a(x - b)^2 - c$ , where  $a$ ,  $b$  and  $c$  are constants,  
describe fully a series of transformations that map the curve **S** onto the curve **T**.

---

(Total for Question 22 is 4 marks)

## WORKED SOLUTION

Dr. Eslam Ahmed

May 2021 P2H | Question 22  
Completing the Square

## 1. Step 1

$$\begin{aligned}
 g(x) &= 2x^2 - 12x + 13 \\
 &= 2(x^2 - 6x) + 13 \\
 &= 2((x - 3)^2 - 9) + 13 \\
 &= 2(x - 3)^2 - 18 + 13 \\
 &= 2(x - 3)^2 - 5
 \end{aligned}$$

## 2. Starting with

Starting with  $S : y = x^2$ :

## 3. Translate 3 units to the

- Translate 3 units to the right to get  $y = (x - 3)^2$ .
- Stretch parallel to the  $y$ -axis by scale factor 2 to get  $y = 2(x - 3)^2$ .
- Translate 5 units down to get  $y = 2(x - 3)^2 - 5$ .

**Final answer**

$g(x) = 2(x - 3)^2 - 5$ . Transformations: right 3, stretch parallel to the  $y$ -axis by scale factor 2, then down 5.

**CLASSIFIED PROBLEM**

Paper: P1H

Session: May 2024

Q#: 25

Marks: 4

TOPIC: **Completing the Square**

Source: May 2024 P1H

**25**  $f(x) = 17 - 3x^2 + 12x$

Write  $f(x)$  in the form  $a - b(x - c)^2$  where  $a$ ,  $b$  and  $c$  are constants.

$$f(x) = \dots\dots\dots$$

---

**(Total for Question 25 is 4 marks)**

---



## WORKED SOLUTION

Dr. Eslam Ahmed

May 2024 P1H | Question 25  
Completing the Square

## 1. Step 1

$$\begin{aligned}f(x) &= 17 - 3x^2 + 12x \\&= -3x^2 + 12x + 17 \\&= -3(x^2 - 4x) + 17 \\&= -3((x - 2)^2 - 4) + 17 \\&= -3(x - 2)^2 + 12 + 17 \\&= 29 - 3(x - 2)^2\end{aligned}$$

**Final answer**

$$f(x) = 29 - 3(x - 2)^2$$

**CLASSIFIED PROBLEM**

Paper: P1HR

Session: May 2024

Q#: 20

Marks: 5

TOPIC: **Completing the Square**

Source: May 2024 P1HR

- 20 (a) Express  $2x^2 - 11x + 9$  in the form  $a(x - b)^2 - c$  where  $a$ ,  $b$  and  $c$  are numbers to be found.

.....

(3)

The curve **C** has equation  $y = 2(x - 3)^2 - 11(x - 3) + 9$

The point  $P$  is the minimum point on **C**

- (b) Find the coordinates of  $P$

(....., .....)

(2)

(Total for Question 20 is 5 marks)

## WORKED SOLUTION

Dr. Eslam Ahmed

## May 2024 P1HR | Question 20

Completing the Square

**1. For part (a),**

For part (a),

$$2x^2 - 11x + 9$$

**2. Factor out 2 from the**

Factor out 2 from the quadratic terms:

$$2\left(x^2 - \frac{11}{2}x\right) + 9$$

**3. Complete the square**

Complete the square:

$$2\left[\left(x - \frac{11}{4}\right)^2 - \frac{121}{16}\right] + 9$$

$$2\left(x - \frac{11}{4}\right)^2 - \frac{121}{8} + 9$$

$$2\left(x - \frac{11}{4}\right)^2 - \frac{121}{8} + \frac{72}{8}$$

$$2\left(x - \frac{11}{4}\right)^2 - \frac{49}{8}$$

**4. So**

So

$$2x^2 - 11x + 9 = 2\left(x - \frac{11}{4}\right)^2 - \frac{49}{8}$$

**5. For part (b),**

For part (b),

$$y = 2(x - 3)^2 - 11(x - 3) + 9$$

**6. Let**

Let

$$u = x - 3$$

**7. From part (a), the minimum**

From part (a), the minimum occurs when

$$u = \frac{11}{4}$$

8. So

So

$$x - 3 = \frac{11}{4}$$

$$x = \frac{23}{4}$$

9. The minimum value of is

The minimum value of  $y$  is

$$-\frac{49}{8}$$

**Final answer**

$$2\left(x - \frac{11}{4}\right)^2 - \frac{49}{8}, P = \left(\frac{23}{4}, -\frac{49}{8}\right)$$

**CLASSIFIED PROBLEM**

Paper: P1H

Session: Nov 2025

Q#: 21

Marks: 3

TOPIC: **Completing the Square**

Source: Nov 2025 P1H

**21** Express  $5x^2 - 20x + 23$  in the form  $a(x - b)^2 + c$  where  $a$ ,  $b$  and  $c$  are integers.

(Total for Question 21 is 3 marks)

## WORKED SOLUTION

Dr. Eslam Ahmed

Nov 2025 P1H | Question 21  
Completing the Square

## 1. Step 1

$$\begin{aligned}5x^2 - 20x + 23 &= 5(x^2 - 4x) + 23 \\&= 5((x - 2)^2 - 4) + 23 \\&= 5(x - 2)^2 - 20 + 23 \\&= 5(x - 2)^2 + 3\end{aligned}$$

**Final answer**

$$5(x - 2)^2 + 3$$

# Algebraic Fractions

---

**CLASSIFIED PROBLEM**

Paper: P2H

Session: Jan 2020

Q#: 24

Marks: 4

TOPIC: **Algebraic Fractions**

Source: Jan 2020 P2H

**24** Express

$$\left( \frac{4}{2x-5} - \frac{3}{2x-3} \right) \div \frac{9x-4x^3}{6x^2-17x+5}$$

as a single fraction in its simplest form.

---

**(Total for Question 24 is 4 marks)**



## WORKED SOLUTION

Dr. Eslam Ahmed

## Jan 2020 P2H | Question 24

Algebraic Fractions

## 1. Step 1

$$\left( \frac{4}{2x-5} - \frac{3}{2x-3} \right) \div \frac{9x-4x^3}{6x^2-17x+5}$$

## 2. First simplify the bracket

First simplify the bracket:

$$\begin{aligned} \frac{4}{2x-5} - \frac{3}{2x-3} &= \frac{4(2x-3) - 3(2x-5)}{(2x-5)(2x-3)} \\ &= \frac{8x-12-6x+15}{(2x-5)(2x-3)} \\ &= \frac{2x+3}{(2x-5)(2x-3)} \end{aligned}$$

## 3. Now factor the second fraction

Now factor the second fraction:

$$9x-4x^3 = -x(2x-3)(2x+3)$$

$$6x^2-17x+5 = (2x-5)(3x-1)$$

## 4. Dividing by a fraction means

Dividing by a fraction means multiplying by its reciprocal:

$$\frac{2x+3}{(2x-5)(2x-3)} \times \frac{(2x-5)(3x-1)}{-x(2x-3)(2x+3)}$$

## 5. Cancel common factors

Cancel common factors:

$$\begin{aligned} &= -\frac{3x-1}{x(2x-3)^2} \\ &= \frac{1-3x}{x(2x-3)^2} \end{aligned}$$

## Final answer

$$\frac{1-3x}{x(2x-3)^2}$$

**CLASSIFIED PROBLEM**

Paper: P2H

Session: Jan 2021

Q#: 21

Marks: 4

TOPIC: **Algebraic Fractions**

Source: Jan 2021 P2H

**21** Write  $\frac{25x^2 - 64}{5x^2 - 13x - 6} \times \frac{x^2 - 8x + 15}{5x + 8} - (x - 7)$

as a single fraction in its simplest form.  
Show clear algebraic working.

(Total for Question 21 is 4 marks)

## WORKED SOLUTION

Dr. Eslam Ahmed

## Jan 2021 P2H | Question 21

Algebraic Fractions

## 1. Step 1

$$\frac{25x^2 - 64}{5x^2 - 13x - 6} \times \frac{x^2 - 8x + 15}{5x + 8} - (x - 7)$$

## 2. Factor

Factor:

$$25x^2 - 64 = (5x - 8)(5x + 8)$$

$$5x^2 - 13x - 6 = (5x + 2)(x - 3)$$

$$x^2 - 8x + 15 = (x - 3)(x - 5)$$

## 3. So

So

$$\frac{(5x - 8)(5x + 8)}{(5x + 2)(x - 3)} \times \frac{(x - 3)(x - 5)}{5x + 8} - (x - 7)$$

## 4. Cancel common factors

Cancel common factors:

$$\frac{(5x - 8)(x - 5)}{5x + 2} - (x - 7)$$

## 5. Use the common denominator

Use the common denominator  $5x + 2$ :

$$\frac{(5x - 8)(x - 5) - (x - 7)(5x + 2)}{5x + 2}$$

## 6. Expand

Expand:

$$(5x - 8)(x - 5) = 5x^2 - 33x + 40$$

$$(x - 7)(5x + 2) = 5x^2 - 33x - 14$$

## 7. So the numerator is

So the numerator is

$$5x^2 - 33x + 40 - (5x^2 - 33x - 14) = 54$$

**Final answer**

$$\frac{54}{5x + 2}$$

**CLASSIFIED PROBLEM**

Paper: P2HR

Session: Jun 2022

Q#: 23

Marks: 3

TOPIC: **Algebraic Fractions**

Source: Jun 2022 P2HR

**23** Express  $\left(\frac{20}{x^2 - 36} - \frac{2}{x - 6}\right) \times \frac{1}{4 - x}$  as a single fraction in its simplest form.

(Total for Question 23 is 3 marks)

## WORKED SOLUTION

Dr. Eslam Ahmed

Jun 2022 P2HR | Question 23  
Algebraic Fractions

## 1. Step 1

$$\left( \frac{20}{x^2 - 36} - \frac{2}{x - 6} \right) \times \frac{1}{4 - x}$$

## 2. Factor

Factor:

$$x^2 - 36 = (x - 6)(x + 6)$$

## 3. So

So

$$\frac{20}{(x - 6)(x + 6)} - \frac{2}{x - 6}$$

## 4. Use the common denominator

Use the common denominator  $(x - 6)(x + 6)$ :

$$\begin{aligned} & \frac{20 - 2(x + 6)}{(x - 6)(x + 6)} \\ &= \frac{20 - 2x - 12}{(x - 6)(x + 6)} \\ &= \frac{8 - 2x}{(x - 6)(x + 6)} \\ &= \frac{-2(x - 4)}{(x - 6)(x + 6)} \end{aligned}$$

## 5. Now multiply by

Now multiply by

$$\begin{aligned} & \frac{1}{4 - x} = -\frac{1}{x - 4} \\ & \frac{-2(x - 4)}{(x - 6)(x + 6)} \times \left( -\frac{1}{x - 4} \right) \end{aligned}$$

## 6. Cancel

Cancel  $x - 4$ :

$$\begin{aligned} & \frac{2}{(x - 6)(x + 6)} \\ &= \frac{2}{x^2 - 36} \end{aligned}$$

**Final answer**

$$\frac{2}{x^2 - 36}$$

**CLASSIFIED PROBLEM**

Paper: P2H

Session: Jun 2024

Q#: 20

Marks: 3

TOPIC: Algebraic Fractions

Source: Jun 2024 P2H

**20** Given that  $k = x - y$  and  $x = \frac{1}{4y}$

express  $\frac{5k}{x+2}$  in the form  $\frac{a-by^2}{c+dy}$  where  $a, b, c$  and  $d$  are integers.

(Total for Question 20 is 3 marks)



## WORKED SOLUTION

Dr. Eslam Ahmed

Jun 2024 P2H | Question 20  
Algebraic Fractions

## 1. Step 1

$$k = x - y, \quad x = \frac{1}{4y}$$

## 2. So

So

$$k = \frac{1}{4y} - y = \frac{1 - 4y^2}{4y}$$

## 3. Also

Also

$$x + 2 = \frac{1}{4y} + 2 = \frac{1 + 8y}{4y}$$

## 4. Therefore

Therefore

$$\frac{5k}{x + 2} = \frac{5 \left( \frac{1 - 4y^2}{4y} \right)}{\frac{1 + 8y}{4y}}$$

## 5. Cancel

Cancel  $4y$ :

$$\begin{aligned} & \frac{5(1 - 4y^2)}{1 + 8y} \\ &= \frac{5 - 20y^2}{1 + 8y} \end{aligned}$$

**Final answer**  
$$\frac{5 - 20y^2}{1 + 8y}$$

**CLASSIFIED PROBLEM**

Paper: P2HR

Session: Jun 2024

Q#: 26

Marks: 4

TOPIC: **Algebraic Fractions**

Source: Jun 2024 P2HR

**26** Write  $4 - \left[ (3x - 5) \div \frac{3x^2 + x - 10}{4x - 1} \right]$  as a single fraction in its simplest form.

(Total for Question 26 is 4 marks)

## WORKED SOLUTION

Dr. Eslam Ahmed

## Jun 2024 P2HR | Question 26

Algebraic Fractions

## 1. Step 1

$$4 - \left[ (3x - 5) \div \frac{3x^2 + x - 10}{4x - 1} \right]$$

## 2. Factor the quadratic

Factor the quadratic:

$$3x^2 + x - 10 = (3x - 5)(x + 2)$$

## 3. So the expression inside the

So the expression inside the square brackets is

$$(3x - 5) \div \frac{(3x - 5)(x + 2)}{4x - 1}$$

## 4. Dividing by a fraction means

Dividing by a fraction means multiplying by its reciprocal:

$$(3x - 5) \times \frac{4x - 1}{(3x - 5)(x + 2)}$$

## 5. Cancel

Cancel  $3x - 5$ :

$$\frac{4x - 1}{x + 2}$$

## 6. Therefore the full expression is

Therefore the full expression is

$$4 - \frac{4x - 1}{x + 2}$$

## 7. Use the common denominator

Use the common denominator  $x + 2$ :

$$\begin{aligned} & \frac{4(x + 2) - (4x - 1)}{x + 2} \\ &= \frac{4x + 8 - 4x + 1}{x + 2} \\ &= \frac{9}{x + 2} \end{aligned}$$

**Final answer**

$$\frac{9}{x+2}$$

## CLASSIFIED PROBLEM

Paper: P2H

Session: Jun 2025

Q#: 23

Marks: 4

TOPIC: Algebraic Fractions

Source: Jun 2025 P2H

23  $\frac{4x^2 - 4x - 120}{5x^2 - 180} \div \frac{x^2 + 5x}{10x^2 + 60x} = p$  where  $p$  is an integer.

Find the value of  $p$

Show clear algebraic working.

$p =$  .....

(Total for Question 23 is 4 marks)

## WORKED SOLUTION

Dr. Eslam Ahmed

Jun 2025 P2H | Question 23  
Algebraic Fractions

## 1. Step 1

$$\frac{4x^2 - 4x - 120}{5x^2 - 180} \div \frac{x^2 + 5x}{10x^2 + 60x}$$

## 2. Factorise

Factorise:

$$4x^2 - 4x - 120 = 4(x - 6)(x + 5)$$

$$5x^2 - 180 = 5(x - 6)(x + 6)$$

$$x^2 + 5x = x(x + 5)$$

$$10x^2 + 60x = 10x(x + 6)$$

## 3. So

So

$$\begin{aligned} & \frac{4(x - 6)(x + 5)}{5(x - 6)(x + 6)} \div \frac{x(x + 5)}{10x(x + 6)} \\ &= \frac{4(x + 5)}{5(x + 6)} \times \frac{10(x + 6)}{x + 5} = 8 \end{aligned}$$

**Final answer**

$$p = 8$$

**CLASSIFIED PROBLEM**

Paper: P1H

Session: May 2021

Q#: 21

Marks: 4

TOPIC: Algebraic Fractions

Source: May 2021 P1H

**21** Given that  $x = \frac{5}{9y+5}$  and that  $y = \frac{5}{5a-2}$

find an expression for  $x$  in terms of  $a$ .

Give your expression as a single fraction in its simplest form.

(Total for Question 21 is 4 marks)

## WORKED SOLUTION

Dr. Eslam Ahmed

May 2021 P1H | Question 21  
Algebraic Fractions

**1. Given**

Given

$$x = \frac{5}{9y + 5}$$

**2. and**

and

$$y = \frac{5}{5a - 2}$$

**3. Substitute into the expression for**Substitute  $y = \frac{5}{5a-2}$  into the expression for  $x$ :

$$x = \frac{5}{9\left(\frac{5}{5a-2}\right) + 5}$$

$$x = \frac{5}{\frac{45}{5a-2} + 5}$$

**4. Write the denominator as a**

Write the denominator as a single fraction:

$$\begin{aligned} \frac{45}{5a-2} + 5 &= \frac{45 + 5(5a-2)}{5a-2} \\ &= \frac{45 + 25a - 10}{5a-2} \\ &= \frac{25a + 35}{5a-2} \end{aligned}$$

**5. So**

So

$$x = 5 \div \frac{25a + 35}{5a - 2}$$

$$x = 5 \times \frac{5a - 2}{25a + 35}$$

$$x = \frac{5(5a - 2)}{5(5a + 7)}$$

$$x = \frac{5a - 2}{5a + 7}$$



**Final answer**

$$\frac{5a - 2}{5a + 7}$$

**CLASSIFIED PROBLEM**

Paper: P1HR

Session: May 2023

Q#: 23

Marks: 4

TOPIC: Algebraic Fractions

Source: May 2023 P1HR

**23** Simplify  $(x^2 - 4) \div \left( \frac{4x^2 - 7x - 2}{x} \right) - 2x$

Give your answer in the form  $\frac{ax^2}{bx + c}$  where  $a$ ,  $b$  and  $c$  are integers.

(Total for Question 23 is 4 marks)

## WORKED SOLUTION

Dr. Eslam Ahmed

May 2023 P1HR | Question 23  
Algebraic Fractions

## 1. Step 1

$$(x^2 - 4) \div \left( \frac{4x^2 - 7x - 2}{x} \right) - 2x$$

## 2. Change division into multiplication by

Change division into multiplication by the reciprocal:

$$(x^2 - 4) \times \frac{x}{4x^2 - 7x - 2} - 2x$$

## 3. Factorise

Factorise:

$$x^2 - 4 = (x - 2)(x + 2)$$

$$4x^2 - 7x - 2 = (4x + 1)(x - 2)$$

## 4. So

So

$$\begin{aligned} \frac{x(x-2)(x+2)}{(4x+1)(x-2)} - 2x &= \frac{x(x+2)}{4x+1} - 2x \\ \frac{x(x+2) - 2x(4x+1)}{4x+1} &= \frac{x^2 + 2x - 8x^2 - 2x}{4x+1} = \frac{-7x^2}{4x+1} \end{aligned}$$

**Final answer**

$$\frac{-7x^2}{4x+1}$$

**CLASSIFIED PROBLEM**

Paper: P1H

Session: MayNov 2020

Q#: 21

Marks: 5

TOPIC: **Algebraic Fractions**

Source: May-Nov 2020 P1H

**21** Express

$$\frac{1}{3x-2} \times \frac{9x^2-4}{3x^2-13x-10} - \frac{7}{x-1}$$

as a single fraction in its simplest form.

---

**(Total for Question 21 is 5 marks)**

## WORKED SOLUTION

Dr. Eslam Ahmed

## May-Nov 2020 P1H | Question 21

Algebraic Fractions

## 1. Step 1

$$\frac{1}{3x-2} \times \frac{9x^2-4}{3x^2-13x-10} - \frac{7}{x-1}$$

## 2. Factor

Factor:

$$9x^2 - 4 = (3x - 2)(3x + 2)$$

$$3x^2 - 13x - 10 = (3x + 2)(x - 5)$$

## 3. So the first product becomes

So the first product becomes

$$\frac{1}{3x-2} \times \frac{(3x-2)(3x+2)}{(3x+2)(x-5)}$$

## 4. Cancel common factors

Cancel common factors:

$$\frac{1}{x-5}$$

## 5. Now subtract

Now subtract:

$$\frac{1}{x-5} - \frac{7}{x-1}$$

## 6. Use the common denominator

Use the common denominator  $(x-5)(x-1)$ :

$$\begin{aligned} & \frac{x-1-7(x-5)}{(x-5)(x-1)} \\ &= \frac{x-1-7x+35}{(x-5)(x-1)} \\ &= \frac{34-6x}{(x-5)(x-1)} \end{aligned}$$

## Final answer

$$\frac{34-6x}{(x-5)(x-1)}$$

**CLASSIFIED PROBLEM**

Paper: P2H

Session: Nov 2023

Q#: 24

Marks: 5

TOPIC: **Algebraic Fractions**

Source: Nov 2023 P2H

**24** Solve  $\frac{45x^3 - 80x}{3x^2 + x - 4} \times \left( \frac{1}{3x - 4} + \frac{1}{x} \right) = \frac{4(x + 2)}{5x - 8}$

Show clear algebraic working.

$x = \dots\dots\dots$

(Total for Question 24 is 5 marks)

## WORKED SOLUTION

Dr. Eslam Ahmed

## Nov 2023 P2H | Question 24

Algebraic Fractions

## 1. Step 1

$$\frac{45x^3 - 80x}{3x^2 + x - 4} \left( \frac{1}{3x - 4} + \frac{1}{x} \right) = \frac{4(x + 2)}{5x - 8}$$

## 2. Factorise

Factorise:

$$45x^3 - 80x = 5x(3x - 4)(3x + 4)$$

$$3x^2 + x - 4 = (3x + 4)(x - 1)$$

## 3. Also,

Also,

$$\frac{1}{3x - 4} + \frac{1}{x} = \frac{x + 3x - 4}{x(3x - 4)} = \frac{4(x - 1)}{x(3x - 4)}$$

## 4. So the left side becomes

So the left side becomes

$$\frac{5x(3x - 4)(3x + 4)}{(3x + 4)(x - 1)} \times \frac{4(x - 1)}{x(3x - 4)} = 20$$

## 5. Therefore,

Therefore,

$$20 = \frac{4(x + 2)}{5x - 8}$$

$$20(5x - 8) = 4(x + 2)$$

$$100x - 160 = 4x + 8$$

$$96x = 168$$

$$x = \frac{7}{4}$$

**Final answer**

$$x = \frac{7}{4}$$

**CLASSIFIED PROBLEM**

Paper: P1H

Session: Nov 2025

Q#: 23

Marks: 3

TOPIC: **Algebraic Fractions**

Source: Nov 2025 P1H

**23**  $a = \frac{2x+5}{1-x} \quad x = \frac{5-2y}{3y}$

Write  $a$  in the form  $\frac{m+ny}{p(y-1)}$  where  $m, n$  and  $p$  are integers.

Show your working clearly.

$a =$  .....

(Total for Question 23 is 3 marks)



## WORKED SOLUTION

Dr. Eslam Ahmed

## Nov 2025 P1H | Question 23

Algebraic Fractions

## 1. Step 1

$$a = \frac{2x + 5}{1 - x}, \quad x = \frac{5 - 2y}{3y}$$

## 2. Substitute for

Substitute for  $x$ .

## 3. Numerator

Numerator:

$$\begin{aligned} 2x + 5 &= 2 \left( \frac{5 - 2y}{3y} \right) + 5 \\ &= \frac{10 - 4y + 15y}{3y} = \frac{10 + 11y}{3y} \end{aligned}$$

## 4. Denominator

Denominator:

$$\begin{aligned} 1 - x &= 1 - \frac{5 - 2y}{3y} \\ &= \frac{3y - 5 + 2y}{3y} = \frac{5y - 5}{3y} = \frac{5(y - 1)}{3y} \end{aligned}$$

## 5. Therefore

Therefore

$$a = \frac{\frac{10+11y}{3y}}{\frac{5(y-1)}{3y}} = \frac{10 + 11y}{5(y - 1)}$$

**Final answer**

$$\frac{10 + 11y}{5(y - 1)}$$

# Coordinate Geometry

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**CLASSIFIED PROBLEM**

Paper: 4WM1H

Session: May2025

Q#: 24

Marks: 5

TOPIC: **Coordinate Geometry**

Source: May 2025 4WM1H

**24**  $PQRS$  is a square. $PR$  is a diagonal of the square. $P$  is the point with coordinates  $(4, 7)$  $R$  is the point with coordinates  $(8, -5)$ Find an equation of the straight line that passes through the points  $Q$  and  $S$ Give your answer in the form  $ay = bx + c$  where  $a$ ,  $b$  and  $c$  are integers.

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**(Total for Question 24 is 5 marks)**

## WORKED SOLUTION

Dr. Eslam Ahmed

May 2025 4WM1H | Question 24  
Coordinate Geometry

**1. The diagonals of a square**

The diagonals of a square bisect each other and are perpendicular.

**2. Midpoint of and**

Midpoint of  $P(4, 7)$  and  $R(8, -5)$ :

$$\left( \frac{4+8}{2}, \frac{7-5}{2} \right) = (6, 1)$$

**3. Gradient of**

Gradient of  $PR$ :

$$\frac{-5-7}{8-4} = -3$$

**4. So the perpendicular gradient is**

So the perpendicular gradient is  $\frac{1}{3}$ .

**5. The line through and passes**

The line through  $Q$  and  $S$  passes through  $(6, 1)$ , so

$$y - 1 = \frac{1}{3}(x - 6)$$

$$3y - 3 = x - 6$$

$$3y = x - 3$$

**Final answer**

$$3y = x - 3$$

**CLASSIFIED PROBLEM**

Paper: 4WM1HR

Session: May2025

Q#: 23

Marks: 5

TOPIC: **Coordinate Geometry**

Source: May 2025 4WM1HR

**23** A circle  $C$  has centre  $(3q, 4q)$  where  $q$  is an integer.

The straight line with equation  $y = -\frac{5}{4}x + \frac{21}{4}$  is the tangent to  $C$  at the point  $(p, p + 3)$

where  $p$  is an integer.

Work out the radius of  $C$

Give your answer in the form  $\sqrt{a}$  where  $a$  is an integer.

Show your working clearly.

(Total for Question 23 is 5 marks)

## WORKED SOLUTION

Dr. Eslam Ahmed

May 2025 4WM1HR | Question 23  
Coordinate Geometry

**1. The tangent point lies on**

The tangent point lies on the tangent line, so

$$p + 3 = -\frac{5}{4}p + \frac{21}{4}$$

$$p = 1$$

**2. The tangent gradient is ,**

The tangent gradient is  $-\frac{5}{4}$ , so the radius gradient is  $\frac{4}{5}$ .

**3. Using centre and tangent point**

Using centre  $(3q, 4q)$  and tangent point  $(1, 4)$ ,

$$\frac{4 - 4q}{1 - 3q} = \frac{4}{5}$$

$$q = 2$$

**4. So the centre is .**

So the centre is  $(6, 8)$ . Therefore

$$r = \sqrt{(6 - 1)^2 + (8 - 4)^2} = \sqrt{41}$$

**Final answer**  
 $\sqrt{41}$ .

**CLASSIFIED PROBLEM**

Paper: 4WM1H

Session: Nov2025

Q#: 27

Marks: 5

TOPIC: **Coordinate Geometry**

Source: Nov 2025 4WM1H

**27**  $ABCD$  is a rhombus with  $AB = BC = CD = DA$

$A$  has coordinates  $(16, -9)$

$C$  has coordinates  $(24, 15)$

Find an equation of the line  $BD$

Give your answer in the form  $ax + by + c = 0$  where  $a$ ,  $b$  and  $c$  are integers.

Show your working clearly.

**(Total for Question 27 is 5 marks)**

## WORKED SOLUTION

Dr. Eslam Ahmed

## Nov 2025 4WM1H | Question 27

Coordinate Geometry

**1. Diagonals of a rhombus bisect**

Diagonals of a rhombus bisect each other at right angles.

**2. Midpoint of**Midpoint of  $AC$ :

$$\left( \frac{16 + 24}{2}, \frac{-9 + 15}{2} \right) = (20, 3)$$

**3. Gradient of**Gradient of  $AC$ :

$$\frac{15 - (-9)}{24 - 16} = 3$$

**4. So gradient of is**So gradient of  $BD$  is  $-\frac{1}{3}$ .

$$y - 3 = -\frac{1}{3}(x - 20)$$

$$x + 3y - 29 = 0$$

**Final answer**

$$x + 3y - 29 = 0.$$



**CLASSIFIED PROBLEM**

Paper: P1H

Session: Jan 2020

Q#: 22

Marks: 6

TOPIC: **Coordinate Geometry**

Source: Jan 2020 P1H

**22** Triangle  $HJK$  is isosceles with  $HJ = HK$  and  $JK = \sqrt{80}$

$H$  is the point with coordinates  $(-4, 1)$

$J$  is the point with coordinates  $(j, 15)$  where  $j < 0$

$K$  is the point with coordinates  $(6, k)$

$M$  is the midpoint of  $JK$ .

The gradient of  $HM$  is 2

Find the value of  $j$  and the value of  $k$ .

$j = \dots\dots\dots$

$k = \dots\dots\dots$

**(Total for Question 22 is 6 marks)**

## WORKED SOLUTION

Dr. Eslam Ahmed

Jan 2020 P1H | Question 22  
Coordinate Geometry

**1. Since is the midpoint of**

Since  $M$  is the midpoint of  $JK$ ,

$$M = \left( \frac{j+6}{2}, \frac{15+k}{2} \right)$$

**2. The gradient of is 2,**

The gradient of  $HM$  is 2, and  $H = (-4, 1)$ .

$$\frac{\frac{15+k}{2} - 1}{\frac{j+6}{2} + 4} = 2$$

$$\frac{k+13}{j+14} = 2$$

$$k = 2j + 15$$

**3. Also,**

Also,

$$JK = \sqrt{80}$$

**4. So**

So

$$(6-j)^2 + (k-15)^2 = 80$$

**5. Substitute**

Substitute  $k = 2j + 15$ :

$$(6-j)^2 + (2j)^2 = 80$$

$$j^2 - 12j + 36 + 4j^2 = 80$$

$$5j^2 - 12j - 44 = 0$$

$$(5j+10)(j-4.4) = 0$$

**6. Since ,**

Since  $j < 0$ ,

$$j = -2$$

**7. Then**

Then

$$k = 2(-2) + 15 = 11$$

**Final answer**

$$j = -2, k = 11$$

**CLASSIFIED PROBLEM**

Paper: P2H

Session: Jan 2020

Q#: 22

Marks: 5

TOPIC: **Coordinate Geometry**

Source: Jan 2020 P2H

- 22** The line with equation  $y = x + 2$  intersects the curve with equation  $x^2 + y^2 - 2y = 24$  at the points  $A$  and  $B$ .

Find the coordinates of  $A$  and  $B$ .  
Show clear algebraic working.

(..... , .....)

(..... , .....)

**(Total for Question 22 is 5 marks)**

## WORKED SOLUTION

Dr. Eslam Ahmed

Jan 2020 P2H | Question 22  
Coordinate Geometry

**1. The line is**

The line is

$$y = x + 2$$

**2. Substitute this into the circle**

Substitute this into the circle equation:

$$x^2 + y^2 - 2y = 24$$

$$x^2 + (x + 2)^2 - 2(x + 2) = 24$$

$$x^2 + x^2 + 4x + 4 - 2x - 4 = 24$$

$$2x^2 + 2x - 24 = 0$$

$$x^2 + x - 12 = 0$$

$$(x + 4)(x - 3) = 0$$

**3. So**

So

$$x = -4 \quad \text{or} \quad x = 3$$

**4. Using**

Using  $y = x + 2$ :

$$x = -4 \Rightarrow y = -2$$

$$x = 3 \Rightarrow y = 5$$

**Final answer**

$(-4, -2)$  and  $(3, 5)$

**CLASSIFIED PROBLEM**

Paper: P2HR

Session: Jan 2020

Q#: 24

Marks: 5

TOPIC: **Coordinate Geometry**

Source: Jan 2020 P2HR

**24**  $L_1$  and  $L_2$  are two straight lines.

The origin of the coordinate axes is  $O$ .

$L_1$  has equation  $5x + 10y = 8$

$L_2$  is perpendicular to  $L_1$  and passes through the point with coordinates  $(8, 6)$

$L_2$  crosses the  $x$ -axis at the point  $A$ .

$L_2$  intersects the straight line with equation  $x = -3$  at the point  $B$ .

Find the area of triangle  $AOB$ .

Show your working clearly.

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**(Total for Question 24 is 5 marks)**

## WORKED SOLUTION

Dr. Eslam Ahmed

## Jan 2020 P2HR | Question 24

Coordinate Geometry

**1. For ,**For  $L_1$ ,

$$5x + 10y = 8$$

$$10y = -5x + 8$$

$$y = -\frac{1}{2}x + \frac{4}{5}$$

**2. So the gradient of is**So the gradient of  $L_1$  is  $-\frac{1}{2}$ .**3. Since is perpendicular to ,**Since  $L_2$  is perpendicular to  $L_1$ , the gradient of  $L_2$  is 2.**4. Using the point**Using the point  $(8, 6)$ :

$$y - 6 = 2(x - 8)$$

$$y = 2x - 10$$

**5. At , the line crosses**At  $A$ , the line crosses the  $x$ -axis, so  $y = 0$ :

$$0 = 2x - 10$$

$$x = 5$$

**6. So**So  $A = (5, 0)$ .**7. At ,**At  $B$ ,  $x = -3$ :

$$y = 2(-3) - 10 = -16$$

**8. So**So  $B = (-3, -16)$ .**9. Triangle has base and perpendicular**Triangle  $AOB$  has base  $OA = 5$  and perpendicular height 16.

$$\text{Area} = \frac{1}{2} \times 5 \times 16 = 40$$

**Final answer**

40



**CLASSIFIED PROBLEM**

Paper: P1HR

Session: Jan 2021

Q#: 22

Marks: 5

TOPIC: **Coordinate Geometry**

Source: Jan 2021 P1HR

**22**  $ABC$  is an isosceles triangle with  $AB = AC$ .

$B$  is the point with coordinates  $(-1, 5)$

$C$  is the point with coordinates  $(2, 10)$

$M$  is the midpoint of  $BC$ .

Find an equation of the line through the points  $A$  and  $M$ .

Give your answer in the form  $py + qx = r$  where  $p$ ,  $q$  and  $r$  are integers.

(Total for Question 22 is 5 marks)

## WORKED SOLUTION

Dr. Eslam Ahmed

Jan 2021 P1HR | Question 22  
Coordinate Geometry

**1. Since , point lies on**

Since  $AB = AC$ , point  $A$  lies on the perpendicular bisector of  $BC$ .

**2. The midpoint of is**

The midpoint of  $BC$  is

$$M = \left( \frac{-1 + 2}{2}, \frac{5 + 10}{2} \right)$$

$$M = \left( \frac{1}{2}, \frac{15}{2} \right)$$

**3. The gradient of is**

The gradient of  $BC$  is

$$\frac{10 - 5}{2 - (-1)} = \frac{5}{3}$$

**4. So the gradient of the**

So the gradient of the perpendicular bisector is

$$-\frac{3}{5}$$

**5. Use the line through**

Use the line through  $M$ :

$$y - \frac{15}{2} = -\frac{3}{5} \left( x - \frac{1}{2} \right)$$

**6. Multiply by 10**

Multiply by 10:

$$10y - 75 = -6x + 3$$

$$10y + 6x = 78$$

**7. Divide by 2**

Divide by 2:

$$5y + 3x = 39$$

**Final answer**

$$5y + 3x = 39$$

**CLASSIFIED PROBLEM**

Paper: P2HR

Session: Jan 2022

Q#: 22

Marks: 5

TOPIC: **Coordinate Geometry**

Source: Jan 2022 P2HR

**22**  $ABC$  is a triangle in which angle  $ABC = 90^\circ$  $p$  and  $q$  are integers such thatthe coordinates of  $A$  are  $(p, 10)$ the coordinates of  $B$  are  $(-1, -5)$ the coordinates of  $C$  are  $(8, q)$ Given that the gradient of  $AC$  is  $-\frac{6}{7}$ work out the value of  $p$  and the value of  $q$  $p = \dots\dots\dots$  $q = \dots\dots\dots$ **(Total for Question 22 is 5 marks)**

## WORKED SOLUTION

Dr. Eslam Ahmed

Jan 2022 P2HR | Question 22  
Coordinate Geometry

**1. The gradient of is**

The gradient of  $AC$  is  $-\frac{6}{7}$ .

$$\frac{q - 10}{8 - p} = -\frac{6}{7}$$

$$7(q - 10) = -6(8 - p)$$

$$7q - 70 = -48 + 6p$$

$$6p - 7q = -22$$

**2. Since , lines and are**

Since  $\angle ABC = 90^\circ$ , lines  $AB$  and  $BC$  are perpendicular.

$$\text{gradient of } AB = \frac{10 - (-5)}{p - (-1)} = \frac{15}{p + 1}$$

$$\text{gradient of } BC = \frac{q - (-5)}{8 - (-1)} = \frac{q + 5}{9}$$

**3. The product of perpendicular gradients**

The product of perpendicular gradients is  $-1$ :

$$\frac{15}{p + 1} \times \frac{q + 5}{9} = -1$$

$$15(q + 5) = -9(p + 1)$$

$$5(q + 5) = -3(p + 1)$$

$$3p + 5q = -28$$

**4. Now solve**

Now solve

$$6p - 7q = -22$$

$$3p + 5q = -28$$

**5. Double the second equation**

Double the second equation:

$$6p + 10q = -56$$

**6. Subtract**

Subtract:

$$17q = -34$$

$$q = -2$$

**7. Then**

Then

$$3p + 5(-2) = -28$$

$$3p = -18$$

$$p = -6$$

**Final answer**

$$p = -6, q = -2$$

**CLASSIFIED PROBLEM**

Paper: P2H

Session: Jun 2022

Q#: 22

Marks: 7

Topic: **Coordinate Geometry**

Source: Jun 2022 P2H

**22**  $ABCD$  is a kite, with diagonals  $AC$  and  $BD$ , drawn on a centimetre square grid, with a scale of 1 cm for 1 unit on each axis.

$A$  is the point with coordinates  $(-3, 4)$

The diagonals of the kite intersect at the point  $M$  with coordinates  $(0, 2)$

Given that  $AB = AD = 6.5$  cm and the  $x$  coordinate of  $B$  is positive,  
find the coordinates of the points  $B$  and  $D$ .

(..... , .....)

(..... , .....)

**(Total for Question 22 is 7 marks)**

## WORKED SOLUTION

Dr. Eslam Ahmed

## Jun 2022 P2H | Question 22

Coordinate Geometry

**1. The midpoint of is**The midpoint of  $BD$  is  $M = (0, 2)$ .**2. Since , the gradient of**Since  $A = (-3, 4)$ , the gradient of  $AM$  is

$$\frac{2 - 4}{0 - (-3)} = -\frac{2}{3}$$

**3. For the kite, diagonal is**For the kite, diagonal  $AC$  is perpendicular to  $BD$ , so the gradient of  $BD$  is

$$\frac{3}{2}$$

**4. Use direction vector along**Use direction vector  $(2, 3)$  along  $BD$ .**5. Let**

Let

$$B = (2t, 2 + 3t)$$

**6. and**

and

$$D = (-2t, 2 - 3t)$$

**7. Since ,**Since  $AB = 6.5$ ,

$$(2t - (-3))^2 + (2 + 3t - 4)^2 = 6.5^2$$

$$(2t + 3)^2 + (3t - 2)^2 = 42.25$$

$$4t^2 + 12t + 9 + 9t^2 - 12t + 4 = 42.25$$

$$13t^2 + 13 = 42.25$$

$$13t^2 = 29.25$$

$$t^2 = 2.25$$

$$t = 1.5$$

**8. The coordinate of is positive,**

The  $x$ -coordinate of  $B$  is positive, so use  $t = 1.5$ .

$$B = (3, 6.5)$$

$$D = (-3, -2.5)$$

**Final answer**

$$B = (3, 6.5), D = (-3, -2.5)$$



**CLASSIFIED PROBLEM**

Paper: P2HR

Session: Jun 2022

Q#: 20

Marks: 4

TOPIC: **Coordinate Geometry**

Source: Jun 2022 P2HR

**20** The centre  $O$  of a circle has coordinates  $(4, 7)$

The point  $A$ , on the circle, has coordinates  $(6, 11)$  and  $AOP$  is a diameter of the circle.

Find an equation of the tangent to the circle at the point  $P$

(Total for Question 20 is 4 marks)

## WORKED SOLUTION

Dr. Eslam Ahmed

**Jun 2022 P2HR | Question 20**  
Coordinate Geometry

**1. The centre of the circle**

The centre of the circle is

$$O = (4, 7)$$

**2. The point is**

The point  $A$  is

$$A = (6, 11)$$

**3. Since is a diameter, is**

Since  $AOP$  is a diameter,  $O$  is the midpoint of  $AP$ .

**4. Let**

Let  $P = (x, y)$ .

$$\left( \frac{6+x}{2}, \frac{11+y}{2} \right) = (4, 7)$$

$$6+x=8, \quad 11+y=14$$

$$P = (2, 3)$$

**5. The gradient of is**

The gradient of  $OP$  is

$$\frac{3-7}{2-4} = 2$$

**6. The tangent is perpendicular to**

The tangent is perpendicular to the radius, so its gradient is

$$-\frac{1}{2}$$

**7. Using point ,**

Using point  $P = (2, 3)$ ,

$$y-3 = -\frac{1}{2}(x-2)$$

$$y = -\frac{1}{2}x + 4$$

**Final answer**

$$y = -\frac{1}{2}x + 4$$

**CLASSIFIED PROBLEM**

Paper: P2H

Session: Jun 2023

Q#: 24

Marks: 6

TOPIC: **Coordinate Geometry**

Source: Jun 2023 P2H

**24**  $ABCD$  is a kite with  $AB = AD$  and  $CB = CD$  $A$  is the point with coordinates  $(-2, 10)$  $B$  is the point with coordinates  $\left(-\frac{27}{5}, 4\right)$  $C$  is the point with coordinates  $(4, -5)$ Work out the coordinates of  $D$  $(\dots\dots\dots, \dots\dots\dots)$ **(Total for Question 24 is 6 marks)**

## WORKED SOLUTION

Dr. Eslam Ahmed

## Jun 2023 P2H | Question 24

Coordinate Geometry

**1. In the kite, and ,**

In the kite,  $AB = AD$  and  $CB = CD$ , so  $AC$  is the line of symmetry.

**2. Therefore is the reflection of**

Therefore  $D$  is the reflection of  $B$  in the line  $AC$ .

$$A = (-2, 10), \quad B = \left(-\frac{27}{5}, 4\right), \quad C = (4, -5)$$

**3. A direction vector for is**

A direction vector for  $AC$  is

$$\begin{pmatrix} 4 - (-2) \\ -5 - 10 \end{pmatrix} = \begin{pmatrix} 6 \\ -15 \end{pmatrix}$$

**4. Use the simpler direction vector**

Use the simpler direction vector

$$\begin{pmatrix} 2 \\ -5 \end{pmatrix}$$

**5. Let be the foot of**

Let  $H$  be the foot of the perpendicular from  $B$  to  $AC$ .

$$\vec{AB} = \begin{pmatrix} -\frac{17}{5} \\ -6 \end{pmatrix}$$

**6. The fraction along to is**

The fraction along  $AC$  to  $H$  is

$$\begin{aligned} \frac{\vec{AB} \cdot \begin{pmatrix} 2 \\ -5 \end{pmatrix}}{2^2 + (-5)^2} &= \frac{-\frac{34}{5} + 30}{29} \\ &= \frac{\frac{116}{5}}{29} = \frac{4}{5} \end{aligned}$$

**7. So**

So

$$\begin{aligned} H &= A + \frac{4}{5} \begin{pmatrix} 2 \\ -5 \end{pmatrix} \\ H &= \left(-2 + \frac{8}{5}, 10 - 4\right) \end{aligned}$$

$$H = \left(-\frac{2}{5}, 6\right)$$

**8. Since is the midpoint of**

Since  $H$  is the midpoint of  $BD$ ,

$$D = 2H - B$$

$$D = \left(-\frac{4}{5}, 12\right) - \left(-\frac{27}{5}, 4\right)$$

$$D = \left(\frac{23}{5}, 8\right)$$

**Final answer**

$$D = \left(\frac{23}{5}, 8\right)$$

**CLASSIFIED PROBLEM**

Paper: P2HR

Session: Jun 2023

Q#: 21

Marks: 5

TOPIC: **Coordinate Geometry**

Source: Jun 2023 P2HR

**21** Work out the coordinates of the points of intersection of

$$y - 2x = 1 \quad \text{and} \quad y^2 + xy = 7$$

Show clear algebraic working.

(..... , .....)

(..... , .....)

(Total for Question 21 is 5 marks)

## WORKED SOLUTION

Dr. Eslam Ahmed

**Jun 2023 P2HR | Question 21**  
Coordinate Geometry

**1. The line is**

The line is

$$y - 2x = 1$$

**2. So**

So

$$y = 2x + 1$$

**3. Substitute into**

Substitute into

$$y^2 + xy = 7$$

$$(2x + 1)^2 + x(2x + 1) = 7$$

$$4x^2 + 4x + 1 + 2x^2 + x = 7$$

$$6x^2 + 5x - 6 = 0$$

**4. Factorise**

Factorise:

$$(3x - 2)(2x + 3) = 0$$

**5. So**

So

$$x = \frac{2}{3} \quad \text{or} \quad x = -\frac{3}{2}$$

**6. Using**

Using  $y = 2x + 1$ :

$$x = \frac{2}{3} \Rightarrow y = \frac{7}{3}$$

$$x = -\frac{3}{2} \Rightarrow y = -2$$

**Final answer**

$$\left(\frac{2}{3}, \frac{7}{3}\right) \text{ and } \left(-\frac{3}{2}, -2\right)$$



**CLASSIFIED PROBLEM**

Paper: P2HR

Session: Jun 2023

Q#: 25

Marks: 6

TOPIC: **Coordinate Geometry**

Source: Jun 2023 P2HR

- 25** The straight line with equation  $y - 2x = 7$  is the perpendicular bisector of the line  $AB$  where  $A$  is the point with coordinates  $(j, 7)$  and  $B$  is the point with coordinates  $(6, k)$

Find the coordinates of the midpoint of the line  $AB$

Show clear algebraic working.

(....., .....)

(Total for Question 25 is 6 marks)

## WORKED SOLUTION

Dr. Eslam Ahmed

**Jun 2023 P2HR | Question 25**  
Coordinate Geometry

**1. The perpendicular bisector is**

The perpendicular bisector is

$$y - 2x = 7$$

**2. so**

so

$$y = 2x + 7$$

**3. The gradient of the perpendicular**

The gradient of the perpendicular bisector is 2, so the gradient of  $AB$  is

$$-\frac{1}{2}$$

**4. Since and ,**

Since  $A = (j, 7)$  and  $B = (6, k)$ ,

$$\frac{k - 7}{6 - j} = -\frac{1}{2}$$

$$2(k - 7) = -(6 - j)$$

$$j = 2k - 8$$

**5. The midpoint of is**

The midpoint of  $AB$  is

$$\left( \frac{j + 6}{2}, \frac{7 + k}{2} \right)$$

**6. Substitute**

Substitute  $j = 2k - 8$ :

$$\left( k - 1, \frac{k + 7}{2} \right)$$

**7. The midpoint lies on ,**

The midpoint lies on  $y = 2x + 7$ , so

$$\frac{k + 7}{2} = 2(k - 1) + 7$$

$$k + 7 = 4k + 10$$

$$k = -1$$

**8. Then**

Then

$$j = 2(-1) - 8 = -10$$

**9. Therefore the midpoint is**

Therefore the midpoint is

$$\left( \frac{-10 + 6}{2}, \frac{7 + (-1)}{2} \right) = (-2, 3)$$

**Final answer**

$$(-2, 3)$$

**CLASSIFIED PROBLEM**

Paper: P2H

Session: Jun 2024

Q#: 22

Marks: 5

TOPIC: **Coordinate Geometry**

Source: Jun 2024 P2H

**22** The straight line **L** has equation  $x + y = 5$ The curve **C** has equation  $2x^2 + 3y^2 = 210$ Find the coordinates of the points where **L** and **C** intersect.  
Show clear algebraic working.

(....., ..... ) (....., .....)

**(Total for Question 22 is 5 marks)**

## WORKED SOLUTION

Dr. Eslam Ahmed

## Jun 2024 P2H | Question 22

Coordinate Geometry

**1. The line is**

The line is

$$x + y = 5$$

**2. So**

So

$$y = 5 - x$$

**3. Substitute into the curve**

Substitute into the curve:

$$2x^2 + 3y^2 = 210$$

$$2x^2 + 3(5 - x)^2 = 210$$

$$2x^2 + 3(x^2 - 10x + 25) = 210$$

$$5x^2 - 30x + 75 = 210$$

$$5x^2 - 30x - 135 = 0$$

**4. Divide by 5**

Divide by 5:

$$x^2 - 6x - 27 = 0$$

**5. Factorise**

Factorise:

$$(x - 9)(x + 3) = 0$$

**6. So**

So

$$x = 9 \quad \text{or} \quad x = -3$$

**7. Using**Using  $y = 5 - x$ :

$$x = 9 \Rightarrow y = -4$$

$$x = -3 \Rightarrow y = 8$$

**Final answer**

$(9, -4)$  and  $(-3, 8)$

**CLASSIFIED PROBLEM**

Paper: P2HR

Session: Jun 2024

Q#: 21

Marks: 4

TOPIC: **Coordinate Geometry**

Source: Jun 2024 P2HR

**21**  $ABCD$  is a square.The point  $A$  has coordinates  $(-5, 2)$ The point  $B$  has coordinates  $(3, 5)$ Find an equation of the line that passes through  $B$  and  $C$ Give your answer in the form  $ax + by + c = 0$  where  $a$ ,  $b$  and  $c$  are integers.

---

**(Total for Question 21 is 4 marks)**

## WORKED SOLUTION

Dr. Eslam Ahmed

**Jun 2024 P2HR | Question 21**  
Coordinate Geometry

**1. Step 1**

$$A = (-5, 2), \quad B = (3, 5)$$

**2. The gradient of is**

The gradient of  $AB$  is

$$\frac{5 - 2}{3 - (-5)} = \frac{3}{8}$$

**3. Since is a square, is**

Since  $ABCD$  is a square,  $BC$  is perpendicular to  $AB$ .

**4. So the gradient of is**

So the gradient of  $BC$  is

$$-\frac{8}{3}$$

**5. The line through is**

The line through  $B = (3, 5)$  is

$$y - 5 = -\frac{8}{3}(x - 3)$$

**6. Multiply by 3**

Multiply by 3:

$$3y - 15 = -8x + 24$$

$$8x + 3y - 39 = 0$$

**Final answer**

$$8x + 3y - 39 = 0$$



**CLASSIFIED PROBLEM**

Paper: P1H

Session: Jun 2025

Q#: 25

Marks: 5

TOPIC: **Coordinate Geometry**

Source: Jun 2025 P1H

**25**  $PQRS$  is a square. $PR$  is a diagonal of the square. $P$  is the point with coordinates  $(4, 7)$  $R$  is the point with coordinates  $(8, -5)$ Find an equation of the straight line that passes through the points  $Q$  and  $S$ Give your answer in the form  $ay = bx + c$  where  $a$ ,  $b$  and  $c$  are integers.

---

**(Total for Question 25 is 5 marks)**

## WORKED SOLUTION

Dr. Eslam Ahmed

**Jun 2025 P1H | Question 25**  
Coordinate Geometry

**1. Step 1**

$$P = (4, 7), \quad R = (8, -5)$$

**2. The gradient of diagonal is**

The gradient of diagonal  $PR$  is

$$\frac{-5 - 7}{8 - 4} = -3$$

**3. The other diagonal is perpendicular**

The other diagonal  $QS$  is perpendicular to  $PR$ , so its gradient is

$$\frac{1}{3}$$

**4. The diagonals of a square**

The diagonals of a square bisect each other.

**5. The midpoint of is**

The midpoint of  $PR$  is

$$\left( \frac{4 + 8}{2}, \frac{7 + (-5)}{2} \right) = (6, 1)$$

**6. So passes through with gradient**

So  $QS$  passes through  $(6, 1)$  with gradient  $\frac{1}{3}$ .

$$y - 1 = \frac{1}{3}(x - 6)$$

$$y = \frac{1}{3}x - 1$$

**7. Multiply by 3**

Multiply by 3:

$$3y = x - 3$$

**Final answer**

$$3y = x - 3$$

**CLASSIFIED PROBLEM**

Paper: P2HR

Session: Jun 2025

Q#: 24

Marks: 6

TOPIC: **Coordinate Geometry**

Source: Jun 2025 P2HR

**24**  $PQ$  is a straight line drawn on a square grid, with a scale of 1 cm for 1 unit on each axis.

$P$  has coordinates  $(-5, a)$  and  $Q$  has coordinates  $(7, 3a)$  where  $a > 0$

The length of  $PQ$  is  $4\sqrt{10}$  cm

Find an equation of the perpendicular bisector of  $PQ$

Give your answer in the form  $y = mx + c$  where  $m$  and  $c$  are integers.

---

(Total for Question 24 is 6 marks)

## WORKED SOLUTION

Dr. Eslam Ahmed

**Jun 2025 P2HR | Question 24**  
Coordinate Geometry

**1. Step 1**

$$P = (-5, a), \quad Q = (7, 3a)$$

**2. The length of is ,**

The length of  $PQ$  is  $4\sqrt{10}$ , so

$$\sqrt{(7 - (-5))^2 + (3a - a)^2} = 4\sqrt{10}$$

$$\sqrt{12^2 + (2a)^2} = 4\sqrt{10}$$

**3. Square both sides**

Square both sides:

$$144 + 4a^2 = 160$$

$$4a^2 = 16$$

$$a^2 = 4$$

**4. Since ,**

Since  $a > 0$ ,

$$a = 2$$

**5. So**

So

$$P = (-5, 2), \quad Q = (7, 6)$$

**6. The midpoint of is**

The midpoint of  $PQ$  is

$$\left( \frac{-5 + 7}{2}, \frac{2 + 6}{2} \right) = (1, 4)$$

**7. The gradient of is**

The gradient of  $PQ$  is

$$\frac{6 - 2}{7 - (-5)} = \frac{4}{12} = \frac{1}{3}$$

**8. So the gradient of the**

So the gradient of the perpendicular bisector is

$$-3$$

### 9. Using point

Using point  $(1, 4)$ :

$$y - 4 = -3(x - 1)$$

$$y = -3x + 7$$

**Final answer**

$$y = -3x + 7$$

**CLASSIFIED PROBLEM**

Paper: P1H

Session: MayNov 2020

Q#: 22

Marks: 4

TOPIC: **Coordinate Geometry**

Source: May-Nov 2020 P1H

**22**  $ABCD$  is a rhombus.The diagonals,  $AC$  and  $BD$ , intersect at the point  $M$ .The coordinates of  $M$  are  $(6, -11)$ The points  $A$  and  $C$  both lie on the line with equation  $2y + 7x = 20$ Find the exact coordinates of the point where the line through  $B$  and  $D$  intersects the  $y$ -axis.

(....., .....)

**(Total for Question 22 is 4 marks)**

## WORKED SOLUTION

Dr. Eslam Ahmed

## May-Nov 2020 P1H | Question 22

Coordinate Geometry

**1. In a rhombus, the diagonals**

In a rhombus, the diagonals are perpendicular and bisect each other.

**2. The diagonal lies on**The diagonal  $AC$  lies on

$$2y + 7x = 20$$

$$2y = -7x + 20$$

$$y = -\frac{7}{2}x + 10$$

**3. So the gradient of is**So the gradient of  $AC$  is

$$-\frac{7}{2}$$

**4. Therefore the gradient of is**Therefore the gradient of  $BD$  is

$$\frac{2}{7}$$

**5. The line passes through**The line  $BD$  passes through

$$M = (6, -11)$$

**6. So**

So

$$y + 11 = \frac{2}{7}(x - 6)$$

**7. To find where this line**To find where this line intersects the  $y$ -axis, put  $x = 0$ :

$$y + 11 = \frac{2}{7}(0 - 6)$$

$$y + 11 = -\frac{12}{7}$$

$$y = -11 - \frac{12}{7}$$

$$y = -\frac{89}{7}$$

**Final answer**

$$\left(0, -\frac{89}{7}\right)$$



**CLASSIFIED PROBLEM**

Paper: P1H

Session: Nov 2021

Q#: 20

Marks: 7

TOPIC: **Coordinate Geometry**

Source: Nov 2021 P1H

- 20** The straight line **L** passes through point  $A(-6, 2)$  and point  $B(5, 3)$   
The straight line **M** is perpendicular to **L** and passes through the midpoint of  $A$  and  $B$ .  
The line **M** intersects the line  $x = -1$  at point  $C$ .

Calculate the area of triangle  $ABC$ .

(Total for Question 20 is 7 marks)

## WORKED SOLUTION

Dr. Eslam Ahmed

**Nov 2021 P1H** | **Question 20**  
Coordinate Geometry

**1. Step 1**

$$A = (-6, 2), \quad B = (5, 3)$$

**2. The midpoint of is**

The midpoint of  $AB$  is

$$\left( \frac{-6+5}{2}, \frac{2+3}{2} \right) = \left( -\frac{1}{2}, \frac{5}{2} \right)$$

**3. The gradient of is**

The gradient of  $AB$  is

$$\frac{3-2}{5-(-6)} = \frac{1}{11}$$

**4. So the perpendicular gradient is**

So the perpendicular gradient is

$$-11$$

**5. Line passes through , so**

Line  $M$  passes through  $\left(-\frac{1}{2}, \frac{5}{2}\right)$ , so

$$y - \frac{5}{2} = -11 \left( x + \frac{1}{2} \right)$$

**6. At point ,**

At point  $C$ ,  $x = -1$ :

$$y - \frac{5}{2} = -11 \left( -\frac{1}{2} \right)$$

$$y - \frac{5}{2} = \frac{11}{2}$$

$$y = 8$$

**7. So**

So

$$C = (-1, 8)$$

**8. Now use vectors from**

Now use vectors from  $A$ :

$$\vec{AB} = (11, 1), \quad \vec{AC} = (5, 6)$$

**9. Area of triangle is**

Area of triangle  $ABC$  is

$$\begin{aligned} & \frac{1}{2} |11(6) - 1(5)| \\ &= \frac{1}{2} (61) \\ &= 30.5 \end{aligned}$$

**Final answer**

30.5

**CLASSIFIED PROBLEM**

Paper: P1H

Session: Nov 2023

Q#: 22

Marks: 6

TOPIC: **Coordinate Geometry**

Source: Nov 2023 P1H

- 22** [In this question 1 cm = 1 unit on the  $x$ -axis and  
1 cm = 1 unit on the  $y$ -axis]

$P$  is a point on a circle with centre  $(0, 0)$

The coordinates of  $P$  are  $(8, -10)$

The line  $L$  is the tangent to the circle at the point  $P$

$L$  crosses the  $x$ -axis at the point  $Q$  and crosses the  $y$ -axis at the point  $R$

Work out the length of  $RQ$

Give your answer correct to 3 significant figures.

..... cm

**(Total for Question 22 is 6 marks)**

## WORKED SOLUTION

Dr. Eslam Ahmed

## Nov 2023 P1H | Question 22

Coordinate Geometry

**1. The centre of the circle**The centre of the circle is  $(0, 0)$ , and

$$P = (8, -10)$$

**2. The gradient of is**The gradient of  $OP$  is

$$\frac{-10}{8} = -\frac{5}{4}$$

**3. So the gradient of the**

So the gradient of the tangent is

$$\frac{4}{5}$$

**4. The tangent through is**The tangent through  $P$  is

$$y + 10 = \frac{4}{5}(x - 8)$$

$$y = \frac{4}{5}x - \frac{82}{5}$$

**5. At the axis,**At the  $x$ -axis,  $y = 0$ :

$$0 = \frac{4}{5}x - \frac{82}{5}$$

$$x = \frac{41}{2}$$

**6. So**

So

$$Q = \left(\frac{41}{2}, 0\right)$$

**7. At the axis,**At the  $y$ -axis,  $x = 0$ :

$$R = \left(0, -\frac{82}{5}\right)$$

**8. Therefore**

Therefore

$$RQ = \sqrt{\left(\frac{41}{2}\right)^2 + \left(\frac{82}{5}\right)^2}$$

$$RQ = 26.252\dots$$

**Final answer**

26.3 cm

**CLASSIFIED PROBLEM**

Paper: P2H

Session: Nov 2023

Q#: 21

Marks: 5

TOPIC: **Coordinate Geometry**

Source: Nov 2023 P2H

- 21** The line with equation  $x + 2y = 5$  intersects the curve with equation  $x^2 + 3y^2 = 13$  at the points  $A$  and  $B$

Find the coordinates of  $A$  and the coordinates of  $B$   
Show clear algebraic working.

(..... , .....)

(..... , .....)

(Total for Question 21 is 5 marks)

## WORKED SOLUTION

Dr. Eslam Ahmed

**Nov 2023 P2H** | **Question 21**  
Coordinate Geometry

**1. The line is**

The line is

$$x + 2y = 5$$

**2. So**

So

$$x = 5 - 2y$$

**3. Substitute into**

Substitute into

$$x^2 + 3y^2 = 13$$

$$(5 - 2y)^2 + 3y^2 = 13$$

$$25 - 20y + 4y^2 + 3y^2 = 13$$

$$7y^2 - 20y + 12 = 0$$

**4. Factorise**

Factorise:

$$(7y - 6)(y - 2) = 0$$

**5. So**

So

$$y = \frac{6}{7} \quad \text{or} \quad y = 2$$

**6. Using**Using  $x = 5 - 2y$ :

$$y = \frac{6}{7} \Rightarrow x = \frac{23}{7}$$

$$y = 2 \Rightarrow x = 1$$

**Final answer**  
 $\left(\frac{23}{7}, \frac{6}{7}\right)$  and  $(1, 2)$



**CLASSIFIED PROBLEM**

Paper: P1H

Session: Nov 2025

Q#: 20

Marks: 3

TOPIC: **Coordinate Geometry**

Source: Nov 2025 P1H

**20** The straight line **L** has equation  $y = 4x + 7$

The straight line **M** is perpendicular to **L** and passes through the point with coordinates (8, 1)

Find an equation for **M**

Give your answer in the form  $y = mx + c$

---

(Total for Question 20 is 3 marks)

## WORKED SOLUTION

Dr. Eslam Ahmed

**Nov 2025 P1H** | **Question 20**  
Coordinate Geometry

**1. Line has equation**Line  $L$  has equation

$$y = 4x + 7$$

**2. So its gradient is**

So its gradient is

$$4$$

**3. Line is perpendicular to ,**Line  $M$  is perpendicular to  $L$ , so its gradient is

$$-\frac{1}{4}$$

**4. Line passes through**Line  $M$  passes through  $(8, 1)$ .

$$y - 1 = -\frac{1}{4}(x - 8)$$

$$y - 1 = -\frac{1}{4}x + 2$$

$$y = -\frac{1}{4}x + 3$$

**Final answer**

$$y = -\frac{1}{4}x + 3$$

**CLASSIFIED PROBLEM**

Paper: P1H

Session: Nov 2025

Q#: 28

Marks: 6

TOPIC: **Coordinate Geometry**

Source: Nov 2025 P1H

- 28** The curve **C** has equation  $x^2 + y^2 = d - 11x$  where  $d$  is an integer.  
The line **L** has equation  $y = 3x + e$  where  $e$  is an integer.

**C** and **L** intersect at the point  $A$  and at the point  $B$

The coordinates of  $A$  are  $(0.2, 2.6)$

Work out the coordinates of  $B$   
Show your working clearly.

---



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( ..... , ..... )

**(Total for Question 28 is 6 marks)**

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## WORKED SOLUTION

Dr. Eslam Ahmed

**Nov 2025 P1H** | **Question 28**  
Coordinate Geometry

**1. The line is**

The line is

$$y = 3x + e$$

**2. Point lies on the line**

Point  $A = (0.2, 2.6)$  lies on the line.

$$2.6 = 3(0.2) + e$$

$$2.6 = 0.6 + e$$

$$e = 2$$

**3. So**

So

$$y = 3x + 2$$

**4. The curve is**

The curve is

$$x^2 + y^2 = d - 11x$$

**5. Use point**

Use point  $A = (0.2, 2.6)$ :

$$0.2^2 + 2.6^2 = d - 11(0.2)$$

$$0.04 + 6.76 = d - 2.2$$

$$d = 9$$

**6. Now substitute into the curve**

Now substitute  $y = 3x + 2$  into the curve:

$$x^2 + (3x + 2)^2 = 9 - 11x$$

$$x^2 + 9x^2 + 12x + 4 = 9 - 11x$$

$$10x^2 + 23x - 5 = 0$$

**7. One root is**

One root is  $0.2 = \frac{1}{5}$ .

**8. The product of the roots**

The product of the roots is

$$\frac{-5}{10} = -\frac{1}{2}$$

**9. So the other root is**

So the other root is

$$\frac{-\frac{1}{2}}{\frac{1}{5}} = -\frac{5}{2}$$

**10. Then**

Then

$$y = 3\left(-\frac{5}{2}\right) + 2$$

$$y = -\frac{11}{2}$$

**Final answer**

$$B = \left(-\frac{5}{2}, -\frac{11}{2}\right)$$

# Graphs of Functions

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**CLASSIFIED PROBLEM**

Paper: P1H

Session: Jan 2020

Q#: 20

Marks: 4

TOPIC: **Graphs of Functions**

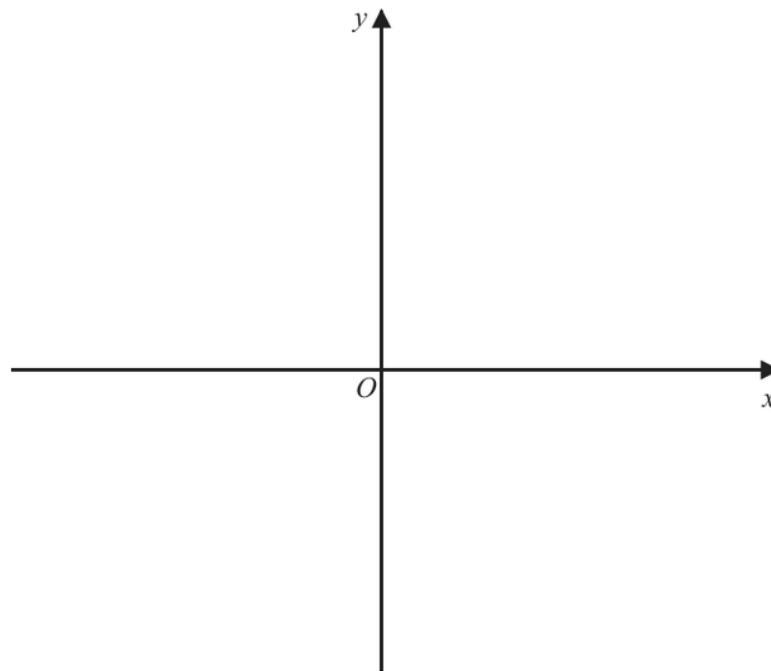
Source: Jan 2020 P1H

**20** The curve **C** has equation  $y = 4(x - 1)^2 - a$  where  $a > 4$

Using the axes below, sketch the curve **C**.

On your sketch show clearly, in terms of  $a$ ,

- (i) the coordinates of any points of intersection of **C** with the coordinate axes,
- (ii) the coordinates of the turning point.



(Total for Question 20 is 4 marks)

## WORKED SOLUTION

Dr. Eslam Ahmed

Jan 2020 P1H | Question 20  
Graphs of Functions

## 1. Step 1

$$y = 4(x - 1)^2 - a$$

## 2. The turning point is read

The turning point is read directly from the completed square form:

$$(1, -a)$$

## 3. For the intercept, put

For the  $y$ -intercept, put  $x = 0$ :

$$y = 4(0 - 1)^2 - a$$

$$y = 4 - a$$

## 4. So the intercept is

So the  $y$ -intercept is

$$(0, 4 - a)$$

## 5. For the intercepts, put

For the  $x$ -intercepts, put  $y = 0$ :

$$4(x - 1)^2 - a = 0$$

$$4(x - 1)^2 = a$$

$$(x - 1)^2 = \frac{a}{4}$$

$$x - 1 = \pm \frac{\sqrt{a}}{2}$$

$$x = 1 \pm \frac{\sqrt{a}}{2}$$

## 6. So the intercepts are

So the  $x$ -intercepts are

$$\left(1 - \frac{\sqrt{a}}{2}, 0\right) \quad \text{and} \quad \left(1 + \frac{\sqrt{a}}{2}, 0\right)$$

## 7. Since , the turning point



Since  $a > 4$ , the turning point is below the  $x$ -axis and the curve crosses the  $x$ -axis twice.

**Final answer**

turning point  $(1, -a)$ ,  $y$ -intercept  $(0, 4 - a)$ ,  $x$ -intercepts  $\left(1 - \frac{\sqrt{a}}{2}, 0\right)$  and  $\left(1 + \frac{\sqrt{a}}{2}, 0\right)$

**CLASSIFIED PROBLEM**

Paper: P2H

Session: Jan 2022

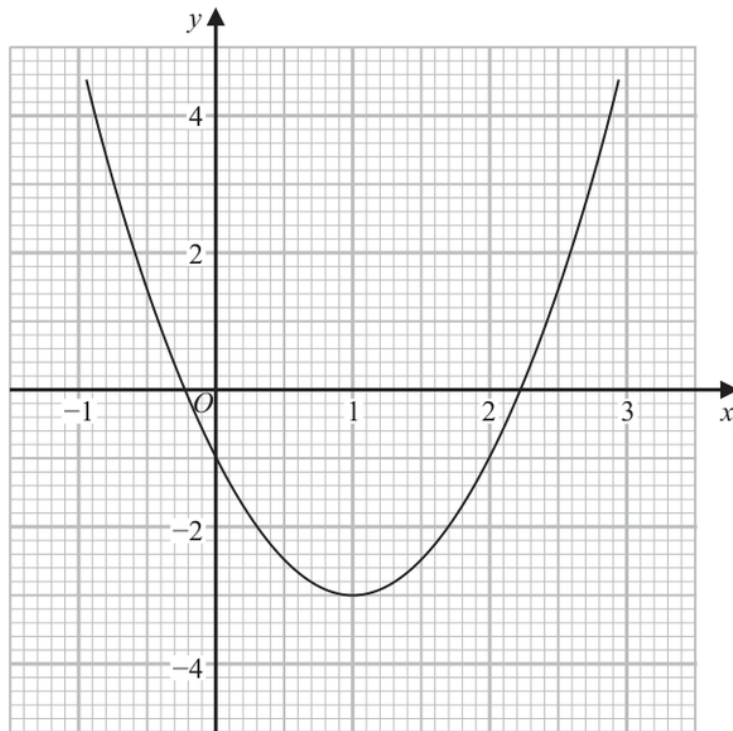
Q#: 21

Marks: 5

TOPIC: **Graphs of Functions**

Source: Jan 2022 P2H

**21** Part of the graph of  $y = 2x^2 - 4x - 1$  is shown on the grid.



- (a) Use the graph to find estimates for the solutions of the equation  $2x^2 - 4x - 1 = 0$   
Give your solutions correct to one decimal place.

.....  
(2)

- (b) By drawing a suitable straight line on the grid, find estimates for the solutions of the equation  $x^2 - x - 1 = 0$   
Show your working clearly.  
Give your solutions correct to one decimal place.

.....  
(3)

(Total for Question 21 is 5 marks)

## WORKED SOLUTION

Dr. Eslam Ahmed

## Jan 2022 P2H | Question 21

Graphs of Functions

**1. For part (a), solve**

For part (a), solve

$$2x^2 - 4x - 1 = 0$$

**2. Using the graph gives approximately**

Using the graph gives approximately

$$x = -0.2, \quad x = 2.2$$

**3. For part (b), we need**

For part (b), we need to solve

$$x^2 - x - 1 = 0$$

**4. Multiply by 2**

Multiply by 2:

$$2x^2 - 2x - 2 = 0$$

**5. The given graph is**

The given graph is

$$y = 2x^2 - 4x - 1$$

**6. Now**

Now

$$2x^2 - 2x - 2 = (2x^2 - 4x - 1) + 2x - 1$$

**7. So draw the straight line**

So draw the straight line

$$y = 1 - 2x$$

**8. The intersections of this line**

The intersections of this line with the curve give approximately

$$x = -0.6, \quad x = 1.6$$

**Final answer**(a)  $-0.2, 2.2$ ; (b) draw  $y = 1 - 2x$ , giving  $-0.6, 1.6$

**CLASSIFIED PROBLEM**

Paper: P2H

Session: Jun 2022

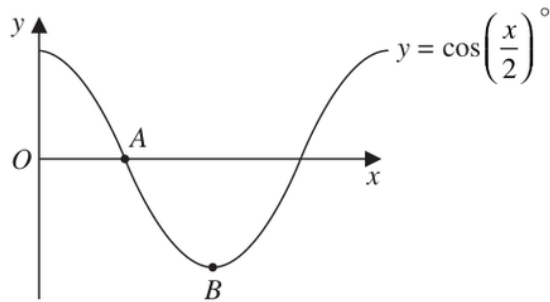
Q#: 23

Marks: 2

Topic: **Graphs of Functions**

Source: Jun 2022 P2H

23 The diagram shows a sketch of the graph of  $y = \cos\left(\frac{x}{2}\right)^\circ$



(i) Find the coordinates of the point A

(....., .....)  
(1)

(ii) Find the coordinates of the point B

(....., .....)  
(1)

(Total for Question 23 is 2 marks)

## WORKED SOLUTION

Dr. Eslam Ahmed

**Jun 2022 P2H | Question 23**  
Graphs of Functions

**1. The graph is**

The graph is

$$y = \cos\left(\frac{x}{2}\right)^\circ$$

**2. Point is the first positive**

Point  $A$  is the first positive  $x$ -intercept, so  $y = 0$ .

$$\cos\left(\frac{x}{2}\right) = 0$$

$$\frac{x}{2} = 90^\circ$$

$$x = 180$$

**3. So**

So

$$A = (180, 0)$$

**4. Point is the minimum point**

Point  $B$  is the minimum point.

**5. The minimum value of cosine**

The minimum value of cosine is  $-1$ , which occurs when

$$\frac{x}{2} = 180^\circ$$

$$x = 360$$

**6. So**

So

$$B = (360, -1)$$

**Final answer**

$$A = (180, 0), B = (360, -1)$$

**CLASSIFIED PROBLEM**

Paper: P2HR

Session: Jun 2023

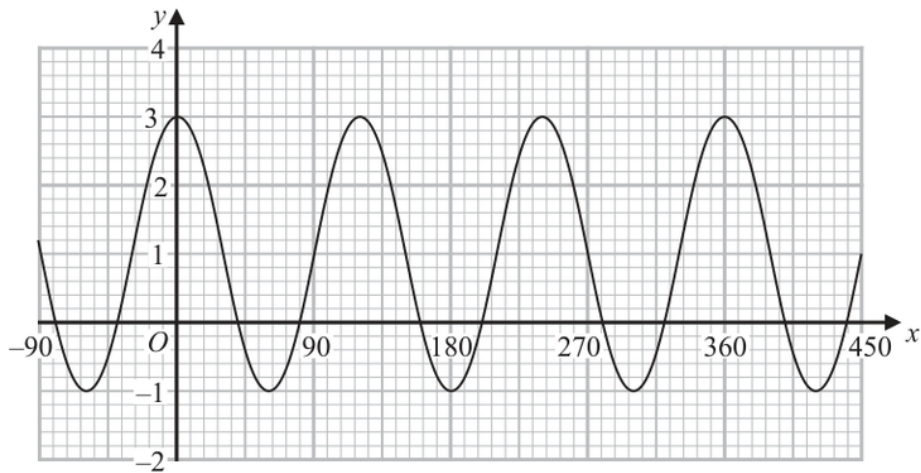
Q#: 26

Marks: 3

TOPIC: **Graphs of Functions**

Source: Jun 2023 P2HR

**26** Here is a sketch of the curve with equation  $y = a \cos bx^\circ + c$  where  $-90 \leq x \leq 450$



Find the value of  $a$ , the value of  $b$  and the value of  $c$

$a =$  .....

$b =$  .....

$c =$  .....

(Total for Question 26 is 3 marks)

## WORKED SOLUTION

Dr. Eslam Ahmed

## Jun 2023 P2HR | Question 26

Graphs of Functions

**1. The curve is**

The curve is

$$y = a \cos bx^\circ + c$$

**2. From the graph, the maximum**

From the graph, the maximum value is

$$3$$

**3. and the minimum value is**

and the minimum value is

$$-1$$

**4. The amplitude is half the**

The amplitude is half the difference:

$$a = \frac{3 - (-1)}{2} = 2$$

**5. The midline is the average**

The midline is the average:

$$c = \frac{3 + (-1)}{2} = 1$$

**6. The distance between consecutive maxima**The distance between consecutive maxima is  $120^\circ$ , so the period is

$$120^\circ$$

**7. For , the period is**For  $y = a \cos bx^\circ + c$ , the period is

$$\frac{360^\circ}{b}$$

**8. So**

So

$$\frac{360}{b} = 120$$

$$b = 3$$

**Final answer**

$$a = 2, b = 3, c = 1$$



**CLASSIFIED PROBLEM**

Paper: P2H

Session: Jun 2024

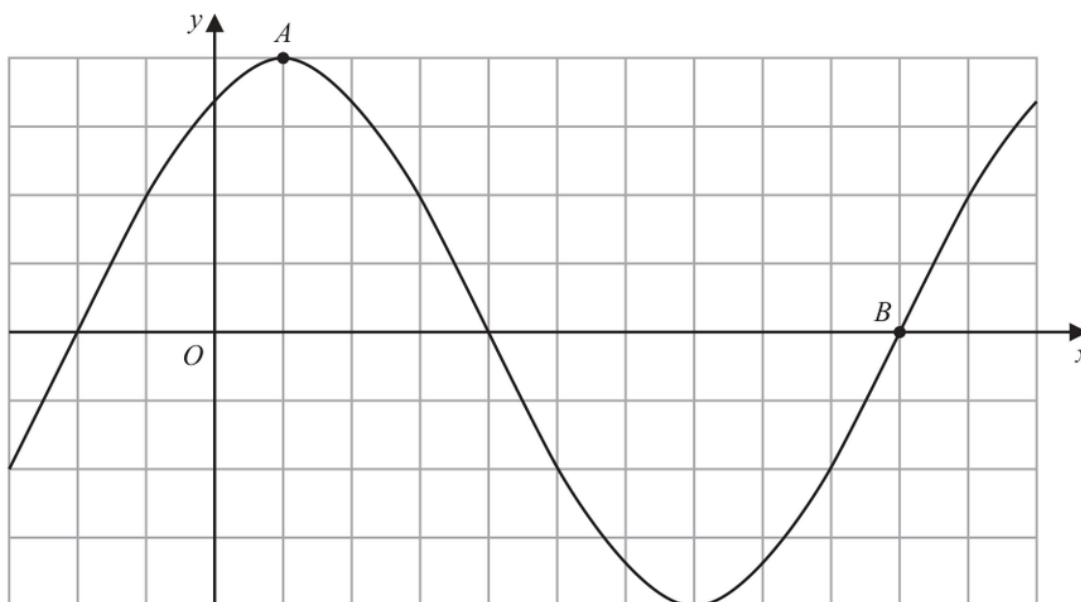
Q#: 25

Marks: 2

TOPIC: **Graphs of Functions**

Source: Jun 2024 P2H

25 The diagram shows a sketch of the graph of  $y = 2\sin(x + 60)^\circ$



(i) Find the coordinates of the point  $A$

(....., .....)  
(1)

(ii) Find the coordinates of the point  $B$

(....., .....)  
(1)

(Total for Question 25 is 2 marks)

## WORKED SOLUTION

Dr. Eslam Ahmed

**Jun 2024 P2H | Question 25**  
 Graphs of Functions

**1. Step 1**

$$y = 2 \sin(x + 60)^\circ$$

**2. Point is the maximum point**

Point  $A$  is the maximum point.

**3. The maximum value is**

The maximum value is

$$y = 2$$

**4. This happens when**

This happens when

$$x + 60 = 90$$

$$x = 30$$

**5. So**

So

$$A = (30, 2)$$

**6. Point is the positive intercept**

Point  $B$  is the positive  $x$ -intercept after the minimum.

$$2 \sin(x + 60)^\circ = 0$$

$$\sin(x + 60)^\circ = 0$$

**7. For this point,**

For this point,

$$x + 60 = 360$$

$$x = 300$$

**8. So**

So

$$B = (300, 0)$$

**Final answer**

$$A = (30, 2), B = (300, 0)$$

**CLASSIFIED PROBLEM**

Paper: P2HR

Session: Jun 2025

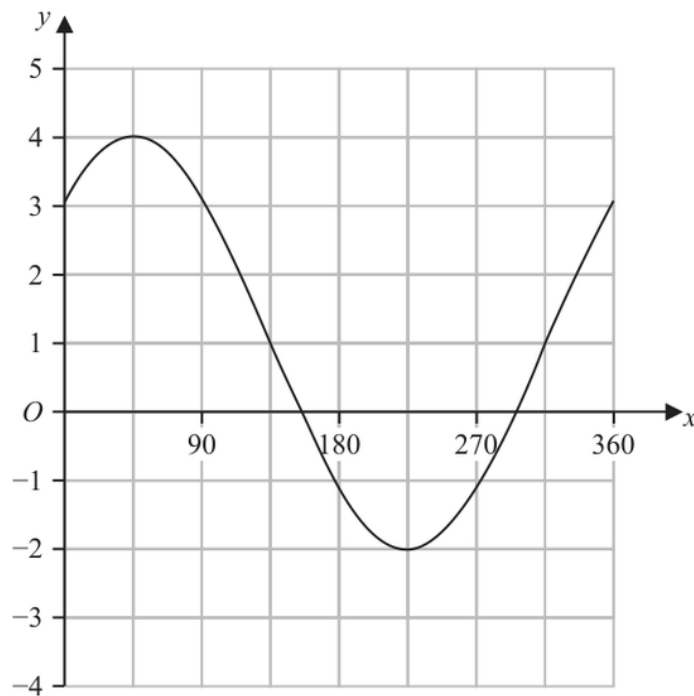
Q#: 25

Marks: 3

Topic: **Graphs of Functions**

Source: Jun 2025 P2HR

**25** The graph of  $y = a\sin(x + b)^\circ + c$  for  $0 \leq x \leq 360$  is drawn on the grid below.



Find a suitable value for  $a$ , for  $b$  and for  $c$

$a = \dots\dots\dots$

$b = \dots\dots\dots$

$c = \dots\dots\dots$

**(Total for Question 25 is 3 marks)**

## WORKED SOLUTION

Dr. Eslam Ahmed

**Jun 2025 P2HR | Question 25**  
 Graphs of Functions

**1. From the graph,**

From the graph,

$$\text{maximum} = 4$$

$$\text{minimum} = -2$$

**2. The amplitude is**

The amplitude is

$$a = \frac{4 - (-2)}{2} = 3$$

**3. The centre line is**

The centre line is

$$c = \frac{4 + (-2)}{2} = 1$$

**4. The maximum is about .**

The maximum is about  $45^\circ$ . For a sine graph, a maximum happens when the angle inside the sine is  $90^\circ$ , so

$$45 + b = 90$$

$$b = 45$$

**Final answer**

$$a = 3, b = 45, c = 1$$

**CLASSIFIED PROBLEM**

Paper: P1H

Session: Nov 2024

Q#: 23

Marks: 5

Topic: **Graphs of Functions**

Source: Nov 2024 P1H

- 23** The curve **C** has equation  $y = x^2 - 8x - 9$   
 The straight line **L** has equation  $y = k$  where  $k$  is an integer.

**C** and **L** intersect at the points  $A$  and  $B$

The coordinates of point  $A$  are  $(p, k)$

The coordinates of point  $B$  are  $(q, k)$

Given that  $p - q = 14$

find the value of  $k$

Show clear algebraic working.

$k = \dots\dots\dots$

**(Total for Question 23 is 5 marks)**

## WORKED SOLUTION

Dr. Eslam Ahmed

## Nov 2024 P1H | Question 23

Graphs of Functions

**1. The curve is**

The curve is

$$y = x^2 - 8x - 9$$

**2. The line is**

The line is

$$y = k$$

**3. At the intersection points,**

At the intersection points,

$$x^2 - 8x - 9 = k$$

$$x^2 - 8x - (9 + k) = 0$$

**4. The two roots are and**The two roots are  $p$  and  $q$ .**5. For this quadratic,**

For this quadratic,

$$p + q = 8$$

**6. We are also given**

We are also given

$$p - q = 14$$

**7. Solve the two equations**

Solve the two equations:

$$2p = 22$$

$$p = 11$$

$$q = -3$$

**8. Use one root in**Use one root in  $y = x^2 - 8x - 9$ :

$$k = 11^2 - 8(11) - 9$$

$$k = 121 - 88 - 9$$

$$k = 24$$

**Final answer**

$$k = 24$$

EA



# Standard & Compound Units

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**CLASSIFIED PROBLEM**

Paper: P2HR

Session: Jan 2022

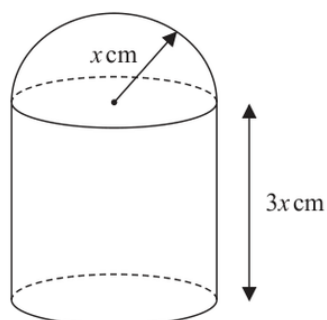
Q#: 21

Marks: 6

TOPIC: **Standard & Compound Units**

Source: Jan 2022 P2HR

- 21 The diagram shows a solid made from a cylinder and a hemisphere.  
The cylinder and the hemisphere are both made from the same metal.

Diagram **NOT**  
accurately drawn

The plane face of the hemisphere coincides with the upper plane face of the cylinder.

The radius of the cylinder and the radius of the hemisphere are both  $x$  cm.

The height of the cylinder is  $3x$  cm.

The total surface area of the solid is  $81\pi\text{ cm}^2$

The mass of the solid is 840 grams.

The following table gives the density of each of four metals.

| Metal     | Density ( $\text{g/cm}^3$ ) |
|-----------|-----------------------------|
| Aluminium | 2.7                         |
| Nickel    | 8.9                         |
| Gold      | 19.3                        |
| Silver    | 10.5                        |

The metal used to make the solid is one of the metals in the table.

Determine the metal used to make the solid.

Show your working clearly.

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(Total for Question 21 is 6 marks)

## WORKED SOLUTION

Dr. Eslam Ahmed

## Jan 2022 P2HR | Question 21

Standard &amp; Compound Units

**1. The outside surface area is**

The outside surface area is:

**2. curved surface area of cylinder**

curved surface area of cylinder:

$$2\pi x(3x) = 6\pi x^2$$

**3. bottom of cylinder**

bottom of cylinder:

$$\pi x^2$$

**4. curved surface area of hemisphere**

curved surface area of hemisphere:

$$2\pi x^2$$

**5. So**

So

$$6\pi x^2 + \pi x^2 + 2\pi x^2 = 81\pi$$

$$9\pi x^2 = 81\pi$$

$$x^2 = 9$$

$$x = 3$$

**6. Volume of the cylinder**

Volume of the cylinder:

$$\pi x^2(3x) = 3\pi x^3$$

**7. Volume of the hemisphere**

Volume of the hemisphere:

$$\frac{2}{3}\pi x^3$$

**8. Total volume**

Total volume:

$$3\pi(3)^3 + \frac{2}{3}\pi(3)^3$$

$$= 81\pi + 18\pi = 99\pi$$

### 9. Density

Density:

$$\frac{840}{99\pi} = 2.70\dots$$

### 10. This matches aluminium, whose density

This matches aluminium, whose density is  $2.7 \text{ g/cm}^3$ .

**Final answer**

Aluminium

**CLASSIFIED PROBLEM**

Paper: P2H

Session: Nov 2025

Q#: 26

Marks: 6

TOPIC: **Standard & Compound Units**

Source: Nov 2025 P2H

**26** A solid cone is joined to a solid hemisphere to make the solid shown below.

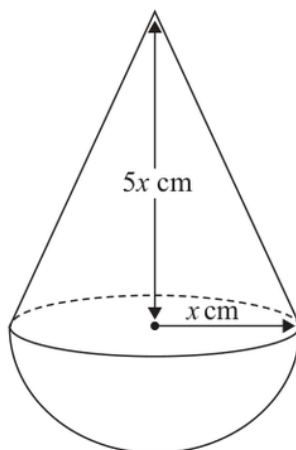


Diagram **NOT**  
accurately drawn

The cone is made from copper.  
The density of copper is  $9 \text{ g/cm}^3$

The hemisphere is made from a different metal.

The total mass of the solid is  $4752\pi$  grams  
The total volume of the solid is  $504\pi \text{ cm}^3$

Work out the density of the hemisphere.  
Show your working clearly.

.....  $\text{g/cm}^3$

(Total for Question 26 is 6 marks)

## WORKED SOLUTION

Dr. Eslam Ahmed

## Nov 2025 P2H | Question 26

Standard &amp; Compound Units

**1. Volume of cone**

Volume of cone:

$$\frac{1}{3}\pi x^2(5x) = \frac{5}{3}\pi x^3$$

**2. Volume of hemisphere**

Volume of hemisphere:

$$\frac{2}{3}\pi x^3$$

**3. Total volume**

Total volume:

$$\frac{5}{3}\pi x^3 + \frac{2}{3}\pi x^3 = 504\pi$$

$$\frac{7}{3}\pi x^3 = 504\pi$$

$$x^3 = 216$$

$$x = 6$$

**4. Cone volume**

Cone volume:

$$\frac{5}{3}\pi(6)^3 = 360\pi$$

**5. Mass of copper cone**

Mass of copper cone:

$$360\pi \times 9 = 3240\pi$$

**6. Mass of hemisphere**

Mass of hemisphere:

$$4752\pi - 3240\pi = 1512\pi$$

**7. Hemisphere volume**

Hemisphere volume:

$$\frac{2}{3}\pi(6)^3 = 144\pi$$

**8. Density of the hemisphere**

Density of the hemisphere:

$$\frac{1512\pi}{144\pi} = 10.5$$

**Final answer**10.5 g/cm<sup>3</sup>

EA

# Area & Perimeter

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**CLASSIFIED PROBLEM**

Paper: 4WM1H

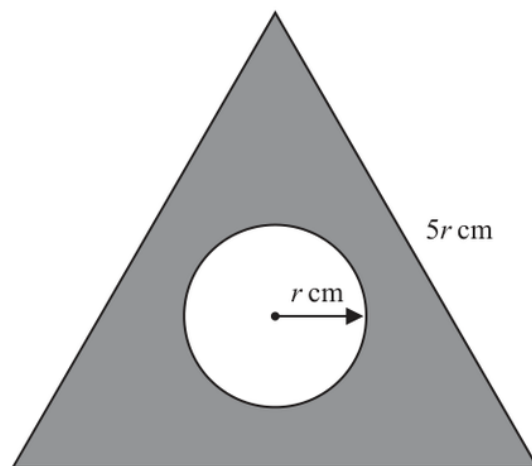
Session: Nov2025

Q#: 21

Marks: 4

TOPIC: **Area & Perimeter**

Source: Nov 2025 4WM1H

**21** The diagram shows an equilateral triangle and a circle.Diagram **NOT**  
accurately drawnThe equilateral triangle has sides of length  $5r \text{ cm}$ The circle has radius  $r \text{ cm}$ The area of the shaded region is  $610\pi \text{ cm}^2$ Work out the value of  $r$ 

Give your answer correct to 3 significant figures.

 $r = \dots\dots\dots$ **(Total for Question 21 is 4 marks)**

## WORKED SOLUTION

Dr. Eslam Ahmed

## Nov 2025 4WM1H | Question 21

Area &amp; Perimeter

## 1. Triangle area

Triangle area:

$$\frac{\sqrt{3}}{4}(5r)^2 = \frac{25\sqrt{3}}{4}r^2$$

## 2. Circle area

Circle area:

$$\pi r^2$$

## 3. So

So

$$\left(\frac{25\sqrt{3}}{4} - \pi\right)r^2 = 610\pi$$

$$r \approx 15.7926$$

**Final answer** $r = 15.8$  to 3 significant figures.

# Circles, Arcs & Sectors

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**CLASSIFIED PROBLEM**

Paper: 4WM1HR

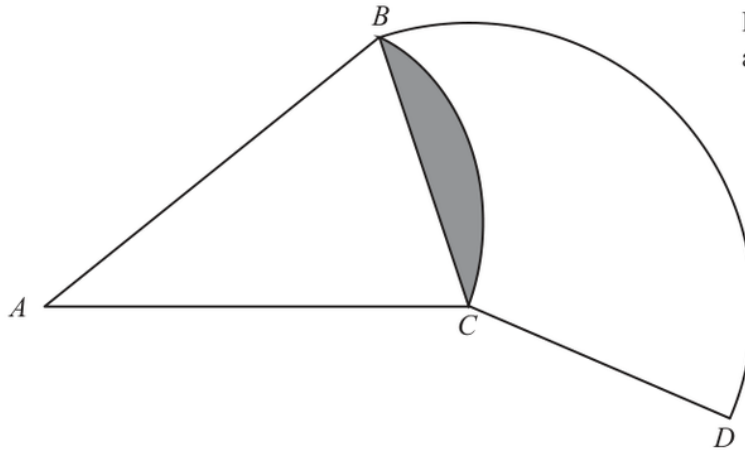
Session: May2025

Q#: 22

Marks: 6

TOPIC: **Circles, Arcs & Sectors**

Source: May 2025 4WM1HR

**22**Diagram **NOT**  
accurately drawn $BAC$  is a sector of a circle, centre  $A$  $BCD$  is a sector of a circle, centre  $C$ Angle  $BAC = 40^\circ$ Angle  $BCD = 130^\circ$ Area of shaded segment =  $28 \text{ cm}^2$ Find the length of the arc  $BD$ 

Give your answer correct to 3 significant figures.

..... cm

(Total for Question 22 is 6 marks)

## WORKED SOLUTION

Dr. Eslam Ahmed

## May 2025 4WM1HR | Question 22

Circles, Arcs &amp; Sectors

**1. Let the radius of sector**Let the radius of sector  $BAC$  be  $R$ .**2. The shaded segment is the**The shaded segment is the segment cut off by chord  $BC$  in the sector with centre  $A$ .**3. Its area is**

Its area is

sector area – triangle area

$$28 = \frac{40}{360}\pi R^2 - \frac{1}{2}R^2 \sin 40^\circ$$

$$28 = R^2 \left( \frac{\pi}{9} - \frac{1}{2} \sin 40^\circ \right)$$

$$R = 31.8096 \dots$$

**4. Now find the chord**Now find the chord  $BC$ :

$$BC = 2R \sin 20^\circ$$

$$BC = 2(31.8096 \dots) \sin 20^\circ = 21.7591 \dots$$

**5. In sector , the centre**In sector  $BCD$ , the centre is  $C$ , so the radius is

$$CB = CD = 21.7591 \dots$$

**6. The arc has angle ,**The arc  $BD$  has angle  $130^\circ$ , so

$$\text{arc } BD = \frac{130}{360} \times 2\pi(21.7591 \dots)$$

$$= 49.3697 \dots$$

**Final answer**

49.4 cm

# Area & Perimeter

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**CLASSIFIED PROBLEM**

Paper: P2H

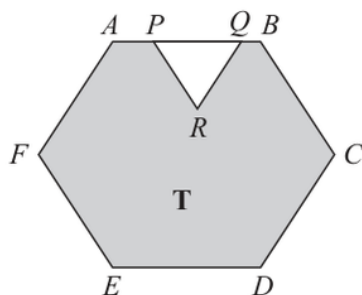
Session: Jan 2021

Q#: 20

Marks: 4

TOPIC: **Area & Perimeter**

Source: Jan 2021 P2H

**20**Diagram **NOT**  
accurately drawn

The diagram shows a shaded region **T** formed by removing an equilateral triangle  $PQR$  from a regular hexagon  $ABCDEF$ .

The points  $P$  and  $Q$  lie on  $AB$  such that  $AB = 1.5 \times PQ$

Given that the area of region **T** is  $72\sqrt{3} \text{ cm}^2$

work out the length of  $PQ$ .

..... cm

**(Total for Question 20 is 4 marks)**

## WORKED SOLUTION

Dr. Eslam Ahmed

Jan 2021 P2H | Question 20  
Area & Perimeter

**1. Let**

Let

$$PQ = p$$

**2. Given**

Given

$$AB = 1.5PQ$$

**3. so the side length of**

so the side length of the regular hexagon is

$$1.5p = \frac{3}{2}p$$

**4. Area of a regular hexagon**Area of a regular hexagon with side length  $s$  is

$$\frac{3\sqrt{3}}{2}s^2$$

**5. So the area of the**

So the area of the hexagon is

$$\begin{aligned} & \frac{3\sqrt{3}}{2} \left( \frac{3}{2}p \right)^2 \\ &= \frac{27\sqrt{3}}{8}p^2 \end{aligned}$$

**6. The removed triangle is equilateral,**The removed triangle  $PQR$  is equilateral, so its area is

$$\frac{\sqrt{3}}{4}p^2$$

**7. The shaded area is**

The shaded area is

$$\begin{aligned} & \frac{27\sqrt{3}}{8}p^2 - \frac{\sqrt{3}}{4}p^2 \\ &= \frac{25\sqrt{3}}{8}p^2 \end{aligned}$$

**8. This is , so**



This is  $72\sqrt{3}$ , so

$$\frac{25\sqrt{3}}{8}p^2 = 72\sqrt{3}$$

$$\frac{25}{8}p^2 = 72$$

$$p^2 = \frac{576}{25}$$

$$p = 4.8$$

**Final answer**

$$PQ = 4.8 \text{ cm}$$

**CLASSIFIED PROBLEM**

Paper: P1HR

Session: MayNov 2020

Q#: 22

Marks: 6

Topic: **Area & Perimeter**

Source: May-Nov 2020 P1HR

**22** The diagram shows a regular octagon  $ABCDEFGH$ .

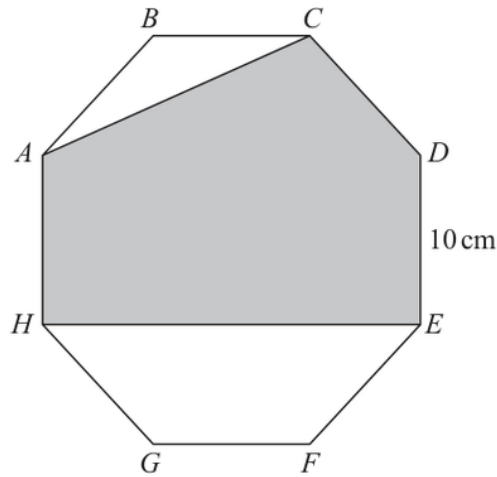


Diagram **NOT**  
accurately drawn

Each side of the octagon has length 10 cm.

Find the area of the shaded region  $ACDEH$ .  
Give your answer correct to the nearest  $\text{cm}^2$

.....  $\text{cm}^2$

(Total for Question 22 is 6 marks)

## WORKED SOLUTION

Dr. Eslam Ahmed

## May-Nov 2020 P1HR | Question 22

Area &amp; Perimeter

**1. Each side of the regular**

Each side of the regular octagon is 10 cm.

**2. For a regular octagon with**For a regular octagon with side length  $s$ ,

$$\text{area} = 2(1 + \sqrt{2})s^2$$

**3. So the area of the**

So the area of the whole octagon is

$$2(1 + \sqrt{2})(10)^2 = 200 + 200\sqrt{2}$$

**4. The unshaded triangle has**The unshaded triangle  $ABC$  has

$$AB = 10, \quad BC = 10, \quad \angle ABC = 135^\circ$$

**5. So**

So

$$\text{area of triangle } ABC = \frac{1}{2}(10)(10) \sin 135^\circ = 25\sqrt{2}$$

**6. At the bottom, the unshaded**At the bottom, the unshaded trapezium  $HEFG$  has parallel sides

$$HE = 10 + 10\sqrt{2}, \quad GF = 10$$

**7. and height**

and height

$$5\sqrt{2}$$

**8. So**

So

$$\begin{aligned} \text{area of trapezium } HEFG &= \frac{1}{2}(10 + 10 + 10\sqrt{2})(5\sqrt{2}) \\ &= 50 + 50\sqrt{2} \end{aligned}$$

**9. The shaded area is**

The shaded area is

$$(200 + 200\sqrt{2}) - 25\sqrt{2} - (50 + 50\sqrt{2})$$

$$= 150 + 125\sqrt{2}$$

$$= 326.7766 \dots$$

**Final answer**

$$327 \text{ cm}^2$$

EA

# Circles, Arcs & Sectors

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**CLASSIFIED PROBLEM**

Paper: P2H

Session: Jan 2021

Q#: 22

Marks: 6

TOPIC: **Circles, Arcs & Sectors**

Source: Jan 2021 P2H

**22** The diagram shows a sector  $OBC$  of a circle with centre  $O$  and radius  $(6 + x)$  cm.

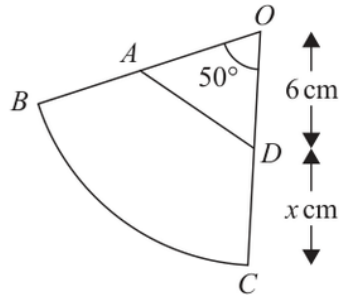


Diagram **NOT**  
accurately drawn

$A$  is the point on  $OB$  and  $D$  is the point on  $OC$  such that  $OA = OD = 6$  cm

Angle  $BOC = 50^\circ$

Given that

the perimeter of sector  $OBC = 2 \times$  the perimeter of triangle  $OAD$

find the value of  $x$ .

Give your answer correct to 3 significant figures.

$x = \dots\dots\dots$

(Total for Question 22 is 6 marks)

## WORKED SOLUTION

Dr. Eslam Ahmed

## Jan 2021 P2H | Question 22

Circles, Arcs &amp; Sectors

**1. The radius of sector is**The radius of sector  $OBC$  is

$$6 + x$$

**2. The angle is , so**The angle is  $50^\circ$ , so the arc length of sector  $OBC$  is

$$\frac{50}{360} \times 2\pi(6 + x)$$

**3. Therefore the perimeter of sector**Therefore the perimeter of sector  $OBC$  is

$$\begin{aligned} 2(6 + x) + \frac{50}{360} \times 2\pi(6 + x) \\ = (6 + x) \left( 2 + \frac{5\pi}{18} \right) \end{aligned}$$

**4. In triangle ,**In triangle  $OAD$ ,

$$OA = OD = 6, \quad \angle AOD = 50^\circ$$

**5. So**

So

$$AD = 2(6) \sin 25^\circ = 12 \sin 25^\circ$$

**6. The perimeter of triangle is**The perimeter of triangle  $OAD$  is

$$12 + 12 \sin 25^\circ$$

**7. Given that**

Given that

$$\text{perimeter of sector } OBC = 2 \times \text{perimeter of triangle } OAD$$

$$(6 + x) \left( 2 + \frac{5\pi}{18} \right) = 2(12 + 12 \sin 25^\circ)$$

$$x = \frac{2(12 + 12 \sin 25^\circ)}{2 + \frac{5\pi}{18}} - 6$$

$$x = 5.8854 \dots$$

**Final answer**

|      |
|------|
| 5.89 |
|------|



**CLASSIFIED PROBLEM**

Paper: P2HR

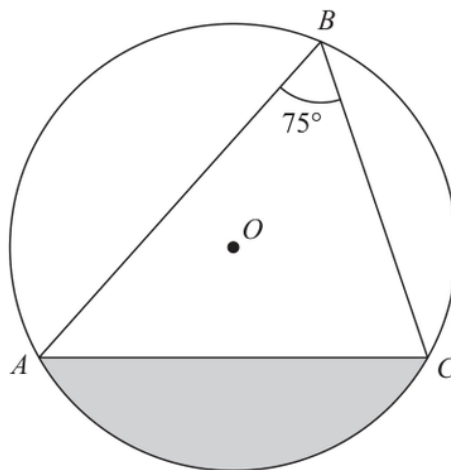
Session: Jan 2021

Q#: 20

Marks: 5

TOPIC: **Circles, Arcs & Sectors**

Source: Jan 2021 P2HR

**20**  $A, B$  and  $C$  are points on a circle with centre  $O$ .Diagram **NOT**  
accurately drawnAngle  $ABC = 75^\circ$ The area of the shaded segment is  $200 \text{ cm}^2$ 

Calculate the radius of the circle.

Give your answer correct to 3 significant figures.

..... cm

(Total for Question 20 is 5 marks)

## WORKED SOLUTION

Dr. Eslam Ahmed

## Jan 2021 P2HR | Question 20

Circles, Arcs &amp; Sectors

## 1. Step 1

$$\angle ABC = 75^\circ$$

## 2. The angle at the centre

The angle at the centre is twice the angle at the circumference, so

$$\angle AOC = 2(75^\circ) = 150^\circ$$

## 3. The shaded segment is

The shaded segment is:

$$\text{sector } AOC - \text{triangle } AOC$$

## 4. Let the radius be

Let the radius be  $r$ .

## 5. Sector area

Sector area:

$$\frac{150}{360}\pi r^2$$

## 6. Triangle area

Triangle area:

$$\frac{1}{2}r^2 \sin 150^\circ$$

## 7. So

So

$$\frac{150}{360}\pi r^2 - \frac{1}{2}r^2 \sin 150^\circ = 200$$

$$r^2 \left( \frac{5\pi}{12} - \frac{1}{4} \right) = 200$$

$$r^2 = \frac{200}{\frac{5\pi}{12} - \frac{1}{4}}$$

$$r = 13.7426 \dots$$

**Final answer**

13.7 cm

**CLASSIFIED PROBLEM**

Paper: P2H

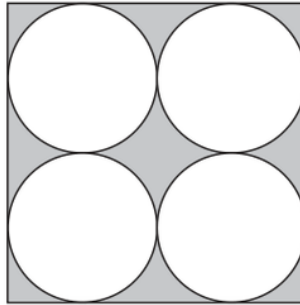
Session: Jun 2022

Q#: 20

Marks: 4

TOPIC: **Circles, Arcs & Sectors**

Source: Jun 2022 P2H

**20** The diagram shows four identical circles drawn inside a square.Diagram **NOT**  
accurately drawn

Each circle touches two other circles and two sides of the square.

The region inside the square that is outside the circles, shown shaded in the diagram, has a total area of  $40\text{ cm}^2$ 

Work out the perimeter of the square.

Give your answer correct to 3 significant figures.

..... cm

**(Total for Question 20 is 4 marks)**

## WORKED SOLUTION

Dr. Eslam Ahmed

## Jun 2022 P2H | Question 20

Circles, Arcs &amp; Sectors

**1. Let the radius of each**Let the radius of each circle be  $r$ .**2. There are two circles across**

There are two circles across the width of the square, so the side length of the square is

$$4r$$

**3. Area of the square**

Area of the square:

$$(4r)^2 = 16r^2$$

**4. Area of the four circles**

Area of the four circles:

$$4\pi r^2$$

**5. The shaded area is ,**The shaded area is  $40 \text{ cm}^2$ , so

$$16r^2 - 4\pi r^2 = 40$$

$$r^2(16 - 4\pi) = 40$$

$$r = \sqrt{\frac{40}{16 - 4\pi}} = 3.4131 \dots$$

**6. The perimeter of the square**

The perimeter of the square is

$$4(4r) = 16r$$

$$16r = 54.6101 \dots$$

**Final answer**

54.6 cm

**CLASSIFIED PROBLEM**

Paper: P2H

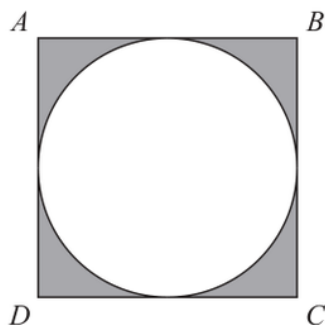
Session: Jun 2024

Q#: 21

Marks: 5

TOPIC: **Circles, Arcs & Sectors**

Source: Jun 2024 P2H

**21** The diagram shows a square  $ABCD$  and a circle.Diagram **NOT**  
accurately drawn

The sides of the square are tangents to the circle.

The total area of the shaded regions is  $80\text{ cm}^2$ Work out the length of  $AC$ 

Give your answer correct to 3 significant figures.

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..... cm

**(Total for Question 21 is 5 marks)**

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## WORKED SOLUTION

Dr. Eslam Ahmed

## Jun 2024 P2H | Question 21

Circles, Arcs &amp; Sectors

**1. Let the side length of**Let the side length of the square be  $s$ .**2. The circle is inscribed in**

The circle is inscribed in the square, so its radius is

$$\frac{s}{2}$$

**3. The shaded area is the**

The shaded area is the area of the square minus the area of the circle:

$$s^2 - \pi \left(\frac{s}{2}\right)^2 = 80$$

$$s^2 - \frac{\pi s^2}{4} = 80$$

$$s^2 \left(1 - \frac{\pi}{4}\right) = 80$$

$$s = \sqrt{\frac{80}{1 - \frac{\pi}{4}}} = 19.3076 \dots$$

**4. The diagonal of the square**

The diagonal of the square is

$$AC = s\sqrt{2}$$

$$AC = 19.3076 \dots \sqrt{2} = 27.3051 \dots$$

**Final answer****27.3 cm**

**CLASSIFIED PROBLEM**

Paper: P1HR

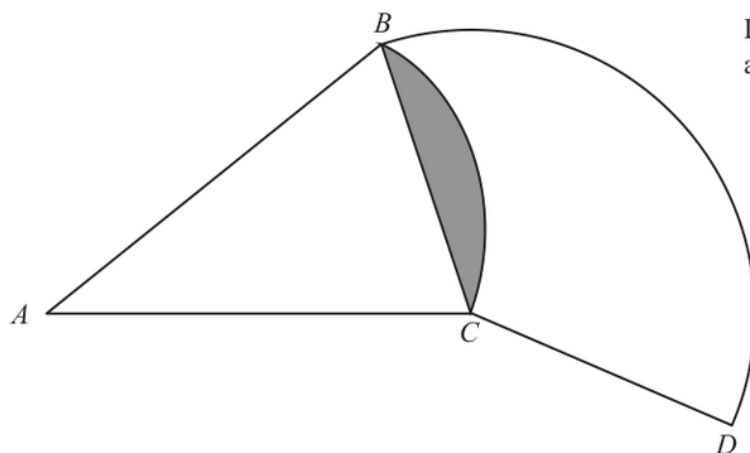
Session: Jun 2025

Q#: 25

Marks: 6

TOPIC: **Circles, Arcs & Sectors**

Source: Jun 2025 P1HR

**25**Diagram **NOT**  
accurately drawn $BAC$  is a sector of a circle, centre  $A$  $BCD$  is a sector of a circle, centre  $C$ Angle  $BAC = 40^\circ$ Angle  $BCD = 130^\circ$ Area of shaded segment =  $28 \text{ cm}^2$ Find the length of the arc  $BD$ 

Give your answer correct to 3 significant figures.

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..... cm

**(Total for Question 25 is 6 marks)**


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## WORKED SOLUTION

Dr. Eslam Ahmed

## Jun 2025 P1HR | Question 25

Circles, Arcs &amp; Sectors

**1. Let the radius of sector**Let the radius of sector  $BAC$  be  $R$ .**2. The shaded segment is the**The shaded segment is the segment cut off by chord  $BC$  in the sector with centre  $A$ .**3. Its area is**

Its area is

sector area – triangle area

$$28 = \frac{40}{360}\pi R^2 - \frac{1}{2}R^2 \sin 40^\circ$$

$$28 = R^2 \left( \frac{\pi}{9} - \frac{1}{2} \sin 40^\circ \right)$$

$$R = 31.8096 \dots$$

**4. Now find the chord**Now find the chord  $BC$ :

$$BC = 2R \sin 20^\circ$$

$$BC = 2(31.8096 \dots) \sin 20^\circ = 21.7591 \dots$$

**5. In sector , the centre**In sector  $BCD$ , the centre is  $C$ , so the radius is

$$CB = CD = 21.7591 \dots$$

**6. The arc has angle ,**The arc  $BD$  has angle  $130^\circ$ , so

$$\text{arc } BD = \frac{130}{360} \times 2\pi(21.7591 \dots)$$

$$= 49.3697 \dots$$

**Final answer**

49.4 cm



**CLASSIFIED PROBLEM**

Paper: P1H

Session: May 2021

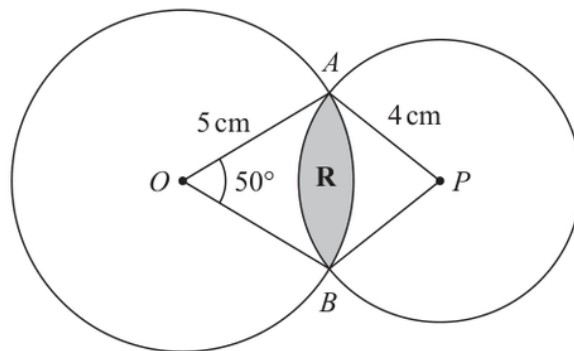
Q#: 25

Marks: 6

TOPIC: **Circles, Arcs & Sectors**

Source: May 2021 P1H

- 25** The diagram shows two circles such that the region **R**, shown shaded in the diagram, is the region common to both circles.

Diagram **NOT**  
accurately drawn

One of the circles has centre  $O$  and radius  $5\text{ cm}$ .

The other circle has centre  $P$  and radius  $4\text{ cm}$ .

Angle  $AOB = 50^\circ$

Calculate the area of region **R**.

Give your answer correct to 3 significant figures.

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.....  $\text{cm}^2$

(Total for Question 25 is 6 marks)

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## WORKED SOLUTION

Dr. Eslam Ahmed

## May 2021 P1H | Question 25

Circles, Arcs &amp; Sectors

**1. For the circle with centre**For the circle with centre  $O$ :

$$\angle AOB = 50^\circ, \quad OA = OB = 5$$

**2. The half chord length is**

The half chord length is

$$\frac{AB}{2} = 5 \sin 25^\circ$$

**3. For the circle with centre**For the circle with centre  $P$ , let

$$\angle APB = \theta$$

**4. The radius is , so**

The radius is 4, so

$$4 \sin \left( \frac{\theta}{2} \right) = 5 \sin 25^\circ$$

$$\sin \left( \frac{\theta}{2} \right) = \frac{5 \sin 25^\circ}{4}$$

$$\theta = 63.7777 \dots^\circ$$

**5. The common region is made**

The common region is made from two minor segments.

**6. For the circle with centre**For the circle with centre  $O$ :

$$\text{segment area} = \frac{50}{360} \pi (5)^2 - \frac{1}{2} (5)^2 \sin 50^\circ = 1.3328 \dots$$

**7. For the circle with centre**For the circle with centre  $P$ :

$$\text{segment area} = \frac{\theta}{360} \pi (4)^2 - \frac{1}{2} (4)^2 \sin \theta$$

$$= 1.7284 \dots$$

**8. Total area of region**Total area of region  $R$ :

$$1.3328 \dots + 1.7284 \dots = 3.0611 \dots$$

**Final answer**

$$3.06 \text{ cm}^2$$

**CLASSIFIED PROBLEM**

Paper: P1H

Session: May 2023

Q#: 22

Marks: 6

TOPIC: **Circles, Arcs & Sectors**

Source: May 2023 P1H

**22** The diagram shows two circles with centre  $O$  and a regular pentagon  $ABCDE$

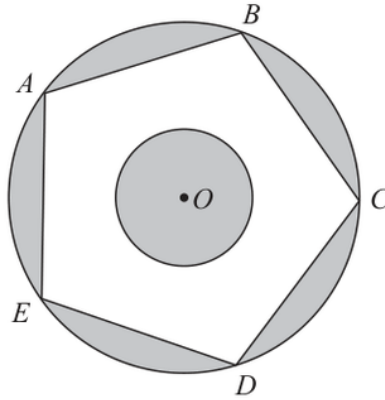


Diagram **NOT**  
accurately drawn

$A$ ,  $B$ ,  $C$ ,  $D$  and  $E$  are points on the larger circle.  
The pentagon has sides of length 8 cm.

The diagram is shaded such that

shaded area = unshaded area

Work out the radius of the smaller circle.  
Give your answer correct to 3 significant figures.

..... cm

(Total for Question 22 is 6 marks)

## WORKED SOLUTION

Dr. Eslam Ahmed

## May 2023 P1H | Question 22

Circles, Arcs &amp; Sectors

**1. The regular pentagon has side**

The regular pentagon has side length 8 cm.

**2. For a regular pentagon,**

For a regular pentagon,

$$R = \frac{s}{2 \sin 36^\circ}$$

**3. where is the radius of**where  $R$  is the radius of the larger circle.

$$R = \frac{8}{2 \sin 36^\circ} = 6.8052 \dots$$

**4. Area of the larger circle**

Area of the larger circle:

$$\pi R^2 = 145.4898 \dots$$

**5. Area of the regular pentagon**

Area of the regular pentagon:

$$\frac{5s^2}{4 \tan 36^\circ} = \frac{5(8)^2}{4 \tan 36^\circ} = 110.1106 \dots$$

**6. Let the radius of the**Let the radius of the smaller circle be  $r$ .**7. The shaded area is**

The shaded area is:

$$\text{larger circle area} - \text{pentagon area} + \pi r^2$$

**8. The unshaded area is**

The unshaded area is:

$$\text{pentagon area} - \pi r^2$$

**9. These are equal, so**

These are equal, so

$$145.4898 - 110.1106 + \pi r^2 = 110.1106 - \pi r^2$$

$$2\pi r^2 = 2(110.1106) - 145.4898$$

$$\pi r^2 = 37.3657 \dots$$

$$r = \sqrt{\frac{37.3657 \dots}{\pi}} = 3.4487 \dots$$

**Final answer**

3.45 cm

**CLASSIFIED PROBLEM**

Paper: P1H

Session: May 2024

Q#: 20

Marks: 4

TOPIC: **Circles, Arcs & Sectors**

Source: May 2024 P1H

**20** The diagram shows a sector  $OABC$  of a circle centre  $O$

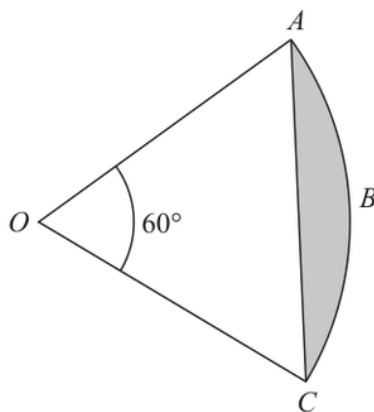


Diagram **NOT**  
accurately drawn

Angle  $AOC = 60^\circ$

The area of the shaded segment  $ABC$  is  $38 \text{ cm}^2$

Work out the perimeter of the shaded segment  $ABC$

Give your answer correct to one decimal place.

..... cm

(Total for Question 20 is 4 marks)

## WORKED SOLUTION

Dr. Eslam Ahmed

## May 2024 P1H | Question 20

Circles, Arcs &amp; Sectors

**1. Let the radius of the**Let the radius of the sector be  $r$ .**2. The shaded segment is**

The shaded segment is

$$\text{sector area} - \text{triangle area}$$

**3. The angle is , so**The angle is  $60^\circ$ , so

$$38 = \frac{60}{360} \pi r^2 - \frac{1}{2} r^2 \sin 60^\circ$$

$$38 = r^2 \left( \frac{\pi}{6} - \frac{\sqrt{3}}{4} \right)$$

$$r = 20.4815 \dots$$

**4. The perimeter of the shaded**The perimeter of the shaded segment is the arc  $AC$  plus the chord  $AC$ .**5. Arc length**

Arc length:

$$\frac{60}{360} \times 2\pi r = \frac{\pi r}{3}$$

**6. Chord length**

Chord length:

$$2r \sin 30^\circ = r$$

**7. So the perimeter is**

So the perimeter is

$$\frac{\pi r}{3} + r$$

$$= \frac{\pi(20.4815 \dots)}{3} + 20.4815 \dots$$

$$= 41.9296 \dots$$

**Final answer**

41.9 cm



**CLASSIFIED PROBLEM**

Paper: P2H

Session: Nov 2023

Q#: 20

Marks: 4

TOPIC: **Circles, Arcs & Sectors**

Source: Nov 2023 P2H

- 20** The diagram shows equilateral triangle  $ABC$  with sides of length 10 cm. A circle is drawn inside the triangle.

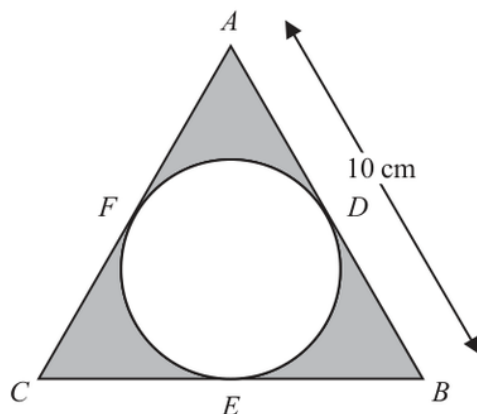


Diagram **NOT**  
accurately drawn

$D$ ,  $E$  and  $F$  are points on the circle.

$ADB$ ,  $BEC$  and  $CFA$  are tangents to the circle.

Calculate the total area of the regions shown shaded in the diagram.  
Give your answer correct to 3 significant figures.

.....  $\text{cm}^2$

(Total for Question 20 is 4 marks)

## WORKED SOLUTION

Dr. Eslam Ahmed

## Nov 2023 P2H | Question 20

Circles, Arcs &amp; Sectors

**1. The triangle is equilateral with**

The triangle is equilateral with side length 10 cm.

**2. Area of the triangle**

Area of the triangle:

$$\frac{\sqrt{3}}{4}(10)^2 = 25\sqrt{3}$$

**3. For an equilateral triangle, the**

For an equilateral triangle, the inradius is

$$r = \frac{s\sqrt{3}}{6}$$

**4. So**

So

$$r = \frac{10\sqrt{3}}{6} = \frac{5\sqrt{3}}{3}$$

**5. Area of the circle**

Area of the circle:

$$\pi r^2 = \pi \left( \frac{5\sqrt{3}}{3} \right)^2 = \frac{25\pi}{3}$$

**6. The shaded area is**

The shaded area is

$$25\sqrt{3} - \frac{25\pi}{3} = 17.1213\dots$$

**Final answer**17.1 cm<sup>2</sup>

**CLASSIFIED PROBLEM**

Paper: P2H

Session: Nov 2024

Q#: 25

Marks: 5

TOPIC: **Circles, Arcs & Sectors**

Source: Nov 2024 P2H

**25** The diagram shows an equilateral triangle  $ABC$  and a circle with centre  $O$

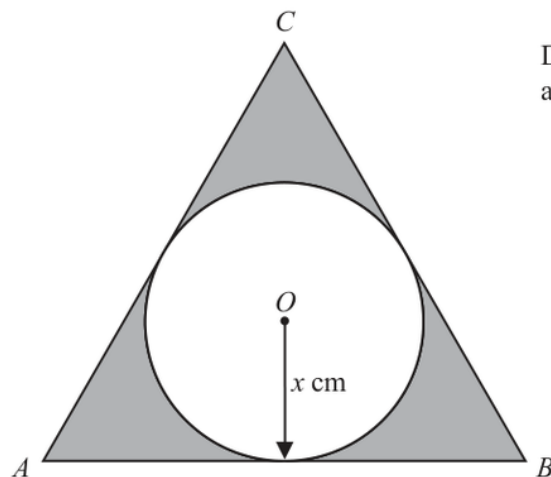


Diagram **NOT**  
accurately drawn

$AB$ ,  $BC$  and  $CA$  are tangents to the circle.

The radius of the circle is  $x$  cm

The total area, in  $\text{cm}^2$ , of the regions shown shaded in the diagram is  $nx^2$

Find the value of  $n$

Give your answer correct to 3 significant figures.

$n = \dots\dots\dots$

(Total for Question 25 is 5 marks)

## WORKED SOLUTION

Dr. Eslam Ahmed

## Nov 2024 P2H | Question 25

Circles, Arcs &amp; Sectors

**1. The circle has radius ,**

The circle has radius  $x$ , so the inradius of the equilateral triangle is  $x$ .

**2. For an equilateral triangle with**

For an equilateral triangle with side length  $a$ ,

$$\text{inradius} = \frac{a\sqrt{3}}{6}$$

**3. So**

So

$$x = \frac{a\sqrt{3}}{6}$$

$$a = \frac{6x}{\sqrt{3}} = 2\sqrt{3}x$$

**4. Area of the triangle**

Area of the triangle:

$$\begin{aligned} \frac{\sqrt{3}}{4}a^2 &= \frac{\sqrt{3}}{4}(2\sqrt{3}x)^2 \\ &= \frac{\sqrt{3}}{4}(12x^2) = 3\sqrt{3}x^2 \end{aligned}$$

**5. Area of the circle**

Area of the circle:

$$\pi x^2$$

**6. The shaded area is**

The shaded area is

$$\begin{aligned} 3\sqrt{3}x^2 - \pi x^2 \\ = (3\sqrt{3} - \pi)x^2 \end{aligned}$$

**7. Therefore**

Therefore

$$n = 3\sqrt{3} - \pi$$

**Final answer**

$$3\sqrt{3} - \pi$$

EA

# Sine, Cosine Rule & Area of Triangles

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**CLASSIFIED PROBLEM**

Paper: 4WM1H

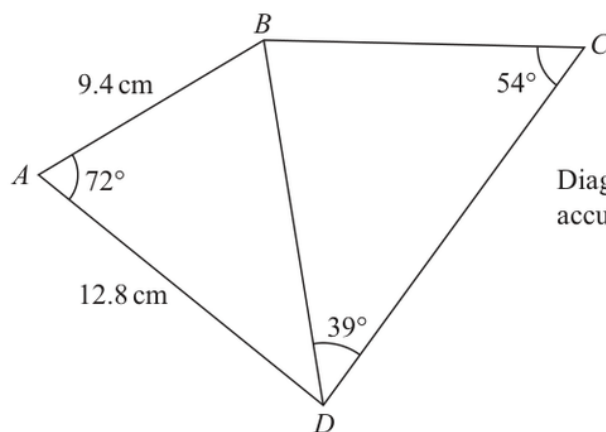
Session: May2025

Q#: 21

Marks: 5

TOPIC: Sine, Cosine Rule &amp; Area of Triangles

Source: May 2025 4WM1H

**21**Diagram **NOT**  
accurately drawn

Work out the length of  $BC$   
Give your answer correct to 3 significant figures.

..... cm

(Total for Question 21 is 5 marks)

## WORKED SOLUTION

Dr. Eslam Ahmed

## May 2025 4WM1H | Question 21

Sine, Cosine Rule &amp; Area of Triangles

## 1. First find in triangle using

First find  $BD$  in triangle  $ABD$  using the cosine rule.

$$BD^2 = 9.4^2 + 12.8^2 - 2(9.4)(12.8) \cos 72^\circ$$

$$BD = 13.3356 \dots$$

## 2. In triangle , use the

In triangle  $BCD$ , use the sine rule.

$$\frac{BC}{\sin 39^\circ} = \frac{BD}{\sin 54^\circ}$$

$$BC = 13.3356 \dots \times \frac{\sin 39^\circ}{\sin 54^\circ}$$

$$BC = 10.3735 \dots$$

## 3. To 3 significant figures,

To 3 significant figures,

$$BC = 10.4$$

**Final answer**

10.4 cm



# 3D Pythagoras & Trigonometry

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**CLASSIFIED PROBLEM**

Paper: 4WM1H

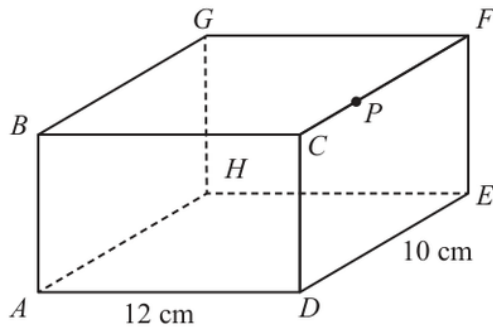
Session: May2025

Q#: 23

Marks: 6

TOPIC: 3D Pythagoras &amp; Trigonometry

Source: May 2025 4WM1H

**23** The diagram shows cuboid  $ABCDEFGH$ Diagram **NOT**  
accurately drawnThe point  $P$  lies on  $CF$ 

$$AD = 12\text{ cm} \quad DE = 10\text{ cm} \quad CP = 3\text{ cm}$$

The angle of elevation of  $P$  from  $A$  is  $24^\circ$ Calculate the size of angle  $APG$ 

Give your answer correct to the nearest degree.

Show your working clearly.

(Total for Question 23 is 6 marks)

## WORKED SOLUTION

Dr. Eslam Ahmed

## May 2025 4WM1H | Question 23

3D Pythagoras &amp; Trigonometry

**1. Point is cm from along**Point  $P$  is 3 cm from  $C$  along  $CF$ . The horizontal distance from  $A$  to the point directly below  $P$  is

$$\sqrt{12^2 + 3^2} = \sqrt{153}$$

**2. Using the angle of elevation,**

Using the angle of elevation,

$$h = \sqrt{153} \tan 24^\circ = 5.507 \dots$$

**3. So**

So

$$AP = \frac{\sqrt{153}}{\cos 24^\circ} = 13.5399 \dots$$

**4. On the top face,**

On the top face,

$$PG = \sqrt{12^2 + 7^2} = \sqrt{193} = 13.8924 \dots$$

**5. Also,**

Also,

$$AG = \sqrt{12^2 + 10^2 + h^2} = 16.561 \dots$$

**6. Use the cosine rule in**Use the cosine rule in triangle  $APG$ .

$$\cos \angle APG = \frac{AP^2 + PG^2 - AG^2}{2(AP)(PG)}$$

$$\angle APG = 49.163 \dots^\circ$$

**7. To the nearest degree,**

To the nearest degree,

$$49^\circ$$

**Final answer** $49^\circ$

**CLASSIFIED PROBLEM**

Paper: 4WM1HR

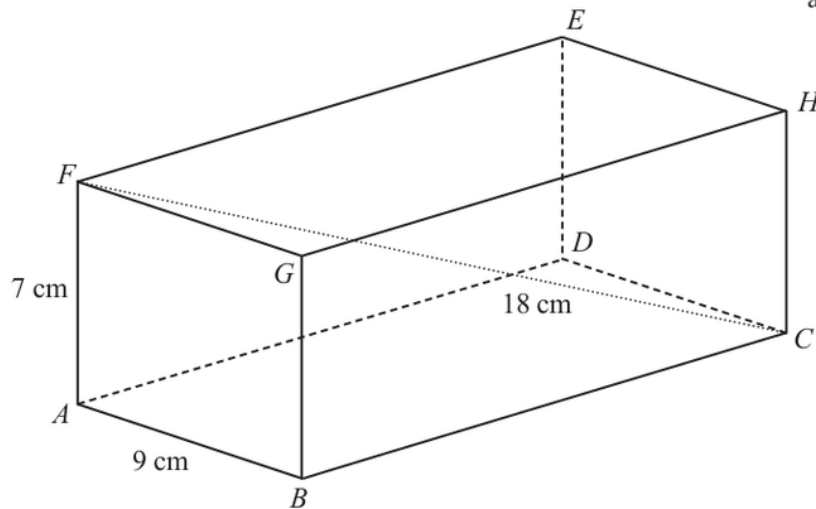
Session: May2025

Q#: 20

Marks: 3

TOPIC: 3D Pythagoras &amp; Trigonometry

Source: May 2025 4WM1HR

**20** The diagram shows cuboid  $ABCDEFGH$ Diagram **NOT**  
accurately drawn

$$AB = 9\text{ cm} \quad AF = 7\text{ cm} \quad FC = 18\text{ cm}$$

Calculate the length of  $BC$ 

Give your answer correct to 3 significant figures.

..... cm

**(Total for Question 20 is 3 marks)**

## WORKED SOLUTION

Dr. Eslam Ahmed

## May 2025 4WM1HR | Question 20

3D Pythagoras &amp; Trigonometry

**1. In the rectangle , use**In the rectangle  $ABCF$ , use Pythagoras with the diagonal  $FC$ :

$$AC^2 = FC^2 - AF^2$$

$$AC^2 = 18^2 - 7^2 = 324 - 49 = 275$$

**2. Now use Pythagoras in triangle**Now use Pythagoras in triangle  $ABC$ :

$$BC^2 = AC^2 - AB^2$$

$$BC^2 = 275 - 9^2 = 275 - 81 = 194$$

$$BC = \sqrt{194} = 13.928\dots$$

**Final answer**

13.9 cm to 3 significant figures

**CLASSIFIED PROBLEM**

Paper: 4WM1H

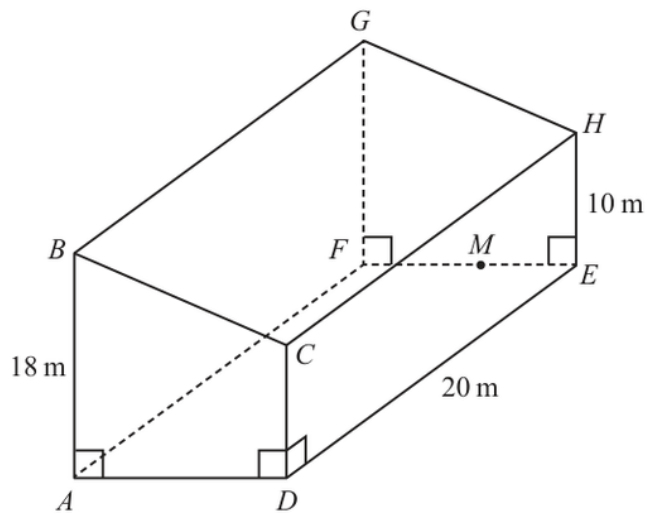
Session: Nov2025

Q#: 26

Marks: 5

TOPIC: 3D Pythagoras &amp; Trigonometry

Source: Nov 2025 4WM1H

**26** The diagram shows prism  $ABCDEFGH$ Diagram **NOT**  
accurately drawn

$$AB = FG = 18 \text{ m} \quad DC = EH = 10 \text{ m} \quad AD = FE$$

$$DE = CH = AF = BG = 20 \text{ m}$$

$$\text{angle } BAD = \text{angle } ADC = \text{angle } CDE = 90^\circ$$

$M$  is the point on  $FE$  such that  $FM = \frac{3}{5}FE$

Given that angle  $GMF = 60^\circ$

calculate the angle of elevation of  $G$  from  $D$

Give your answer correct to one decimal place.

Show your working clearly.

(Total for Question 26 is 5 marks)

## WORKED SOLUTION

Dr. Eslam Ahmed

## Nov 2025 4WM1H | Question 26

3D Pythagoras &amp; Trigonometry

## 1. From triangle ,

From triangle  $GMF$ ,

$$\tan 60^\circ = \frac{FG}{FM} = \frac{18}{FM}$$

$$FM = \frac{18}{\tan 60^\circ} = 6\sqrt{3}$$

## 2. Since ,

Since  $FM = \frac{3}{5}FE$ ,

$$FE = 10\sqrt{3}$$

## 3. The horizontal distance from to

The horizontal distance from  $D$  to the point below  $G$  is

$$\sqrt{20^2 + (10\sqrt{3})^2} = 10\sqrt{7}$$

## 4. If is the angle of

If  $\theta$  is the angle of elevation,

$$\tan \theta = \frac{18}{10\sqrt{7}}$$

$$\theta \approx 34.2289^\circ$$

**Final answer**

34.2°.

# Sine, Cosine Rule & Area of Triangles

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**CLASSIFIED PROBLEM**

Paper: P1HR

Session: Jan 2020

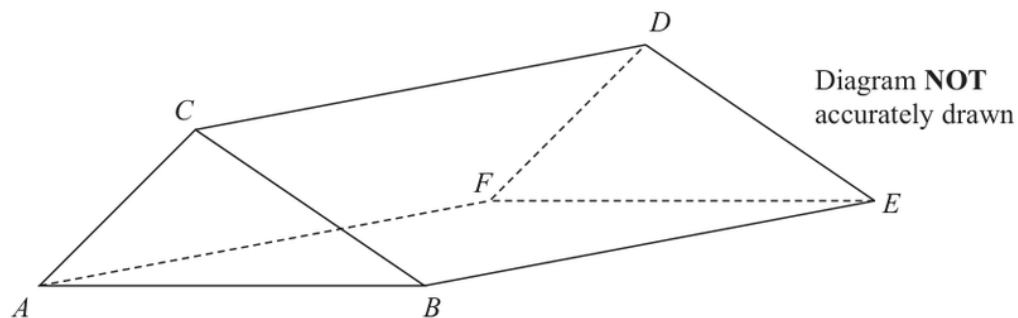
Q#: 21

Marks: 6

TOPIC: Sine, Cosine Rule &amp; Area of Triangles

Source: Jan 2020 P1HR

21 The diagram shows the prism  $ABCDEF$  with cross section triangle  $ABC$ .



Angle  $BEC = 40^\circ$  and angle  $ACB$  is obtuse.  
 $AC = 6\text{ cm}$  and  $CE = 13\text{ cm}$

The area of triangle  $ABC$  is  $22\text{ cm}^2$

Calculate the length of  $AB$ .

Give your answer correct to one decimal place.

..... cm

(Total for Question 21 is 6 marks)

## WORKED SOLUTION

Dr. Eslam Ahmed

**Jan 2020 P1HR | Question 21**

Sine, Cosine Rule &amp; Area of Triangles

**1. In triangle , and**In triangle  $BEC$ ,  $CE = 13$  and  $\angle BEC = 40^\circ$ .**2. Since this is a prism,**Since this is a prism,  $BE$  is perpendicular to the cross section, so triangle  $BEC$  is right-angled at  $B$ .

$$BC = 13 \sin 40^\circ$$

$$BC = 8.356238 \dots$$

**3. The area of triangle is**The area of triangle  $ABC$  is  $22 \text{ cm}^2$ .**4. Using**

Using

$$\text{Area} = \frac{1}{2}ab \sin C$$

$$22 = \frac{1}{2}(AC)(BC) \sin(\angle ACB)$$

$$22 = \frac{1}{2}(6)(8.356238 \dots) \sin(\angle ACB)$$

$$\sin(\angle ACB) = 0.877587 \dots$$

**5. The angle is obtuse, so**The angle  $ACB$  is obtuse, so

$$\angle ACB = 180^\circ - \sin^{-1}(0.877587 \dots)$$

$$\angle ACB = 118.647 \dots^\circ$$

**6. Use the cosine rule**

Use the cosine rule:

$$AB^2 = AC^2 + BC^2 - 2(AC)(BC) \cos(\angle ACB)$$

$$AB^2 = 6^2 + (8.356238 \dots)^2 - 2(6)(8.356238 \dots) \cos(118.647 \dots^\circ)$$

$$AB = 12.4056 \dots$$

**7. Correct to one decimal place**

Correct to one decimal place:

12.4

**Final answer**

12.4 cm

**CLASSIFIED PROBLEM**

Paper: P2H

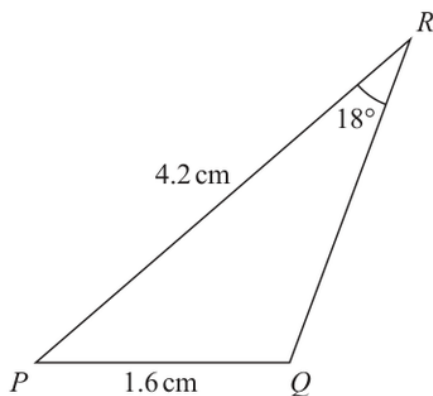
Session: Jan 2022

Q#: 23

Marks: 6

TOPIC: Sine, Cosine Rule &amp; Area of Triangles

Source: Jan 2022 P2H

**23** The diagram shows triangle  $PQR$ Diagram **NOT**  
accurately drawn

$PQ = 1.6 \text{ cm}$

$PR = 4.2 \text{ cm}$

$\text{Angle } PRQ = 18^\circ$

Given that angle  $PQR$  is obtuse,work out the area of triangle  $PQR$ 

Give your answer correct to 3 significant figures.

.....  $\text{cm}^2$ **(Total for Question 23 is 6 marks)**

## WORKED SOLUTION

Dr. Eslam Ahmed

## Jan 2022 P2H | Question 23

Sine, Cosine Rule &amp; Area of Triangles

**1. In triangle ,**In triangle  $PQR$ ,

$$PQ = 1.6, \quad PR = 4.2, \quad \angle PRQ = 18^\circ$$

**2. Use the sine rule**

Use the sine rule:

$$\frac{\sin Q}{PR} = \frac{\sin R}{PQ}$$

$$\frac{\sin Q}{4.2} = \frac{\sin 18^\circ}{1.6}$$

$$\sin Q = \frac{4.2 \sin 18^\circ}{1.6}$$

$$\sin Q = 0.811169 \dots$$

**3. Since angle is obtuse,**Since angle  $PQR$  is obtuse,

$$Q = 180^\circ - \sin^{-1}(0.811169 \dots)$$

$$Q = 125.7896 \dots^\circ$$

**4. Now find angle**Now find angle  $P$ :

$$P = 180^\circ - 18^\circ - 125.7896 \dots^\circ$$

$$P = 36.2103 \dots^\circ$$

**5. Area of triangle**Area of triangle  $PQR$ :

$$\frac{1}{2}(PQ)(PR) \sin P$$

$$= \frac{1}{2}(1.6)(4.2) \sin(36.2103 \dots^\circ)$$

$$= 1.984925 \dots$$

**6. Correct to 3 significant figures**

Correct to 3 significant figures:

1.98

**Final answer**

1.98 cm<sup>2</sup>

**CLASSIFIED PROBLEM**

Paper: P2H

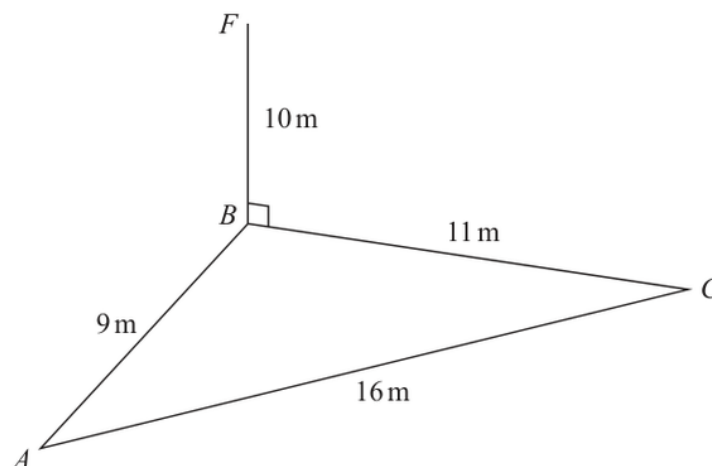
Session: Jun 2023

Q#: 22

Marks: 5

TOPIC: Sine, Cosine Rule &amp; Area of Triangles

Source: Jun 2023 P2H

**22** The diagram shows a triangle  $ABC$  and a flagpole  $BF$ Diagram **NOT**  
accurately drawn $A$ ,  $B$  and  $C$  are points on horizontal ground. $BF$  is vertical.

$$AB = 9\text{ m} \quad BC = 11\text{ m} \quad AC = 16\text{ m} \quad BF = 10\text{ m}$$

 $D$  is the point on  $AC$  such that angle  $BDC = 90^\circ$ 

Work out the size of the angle of elevation of the point  $F$  from the point  $D$   
 Give your answer correct to one decimal place.

(Total for Question 22 is 5 marks)

## WORKED SOLUTION

Dr. Eslam Ahmed

**Jun 2023 P2H | Question 22**

Sine, Cosine Rule &amp; Area of Triangles

**1. In triangle ,**In triangle  $ABC$ ,

$$AB = 9, \quad BC = 11, \quad AC = 16$$

**2. Use Heron's formula**

Use Heron's formula.

$$s = \frac{9 + 11 + 16}{2} = 18$$

$$\begin{aligned} \text{area} &= \sqrt{18(18 - 9)(18 - 11)(18 - 16)} \\ &= \sqrt{18 \cdot 9 \cdot 7 \cdot 2} = 47.6235 \dots \end{aligned}$$

**3. is the foot of the** $D$  is the foot of the perpendicular from  $B$  to  $AC$ , so  $BD$  is the height of triangle  $ABC$ .

$$\text{area} = \frac{1}{2} \times AC \times BD$$

$$47.6235 \dots = \frac{1}{2} \times 16 \times BD$$

$$BD = 5.9529 \dots$$

**4. In right angled triangle ,**In right-angled triangle  $BDF$ ,

$$\tan \theta = \frac{BF}{BD}$$

$$\tan \theta = \frac{10}{5.9529 \dots}$$

$$\theta = \tan^{-1} \left( \frac{10}{5.9529 \dots} \right) = 59.2349 \dots^\circ$$

**Final answer****59.2°**



**CLASSIFIED PROBLEM**

Paper: P1H

Session: Jun 2025

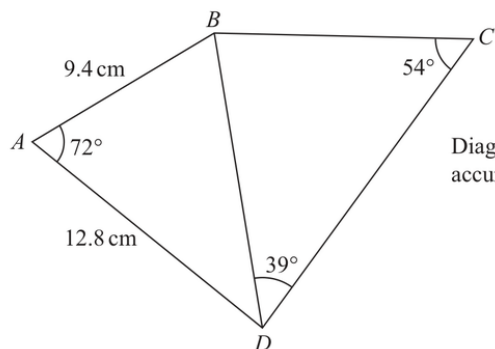
Q#: 20

Marks: 5

TOPIC: Sine, Cosine Rule &amp; Area of Triangles

Source: Jun 2025 P1H

20

Diagram **NOT**  
accurately drawn

Work out the length of  $BC$   
Give your answer correct to 3 significant figures.

..... cm

(Total for Question 20 is 5 marks)

## WORKED SOLUTION

Dr. Eslam Ahmed

**Jun 2025 P1H | Question 20**

Sine, Cosine Rule &amp; Area of Triangles

**1. First use triangle**First use triangle  $ABD$ .

$$AB = 9.4, \quad AD = 12.8, \quad \angle BAD = 72^\circ$$

**2. Use the cosine rule to**Use the cosine rule to find  $BD$ :

$$BD^2 = 9.4^2 + 12.8^2 - 2(9.4)(12.8) \cos 72^\circ$$

$$BD = 13.3356 \dots$$

**3. In triangle ,**In triangle  $BCD$ ,

$$\angle BDC = 39^\circ, \quad \angle BCD = 54^\circ$$

**4. Use the sine rule**

Use the sine rule:

$$\frac{BC}{\sin 39^\circ} = \frac{BD}{\sin 54^\circ}$$

$$BC = \frac{13.3356 \dots \sin 39^\circ}{\sin 54^\circ}$$

$$BC = 10.3735 \dots$$

**5. Correct to 3 significant figures,**

Correct to 3 significant figures,

$$BC = 10.4$$

**Final answer**

10.4 cm

**CLASSIFIED PROBLEM**

Paper: P1HR

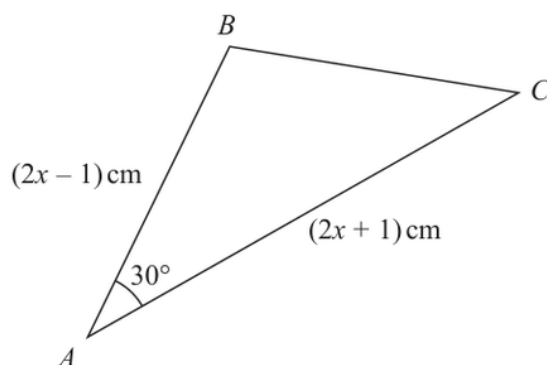
Session: May 2023

Q#: 25

Marks: 6

TOPIC: Sine, Cosine Rule &amp; Area of Triangles

Source: May 2023 P1HR

**25** Here is a triangle  $ABC$ Diagram **NOT**  
accurately drawnThe area of the triangle is  $(x^2 + x - 3.75)\text{ cm}^2$ Find the size of the largest angle in triangle  $ABC$   
Give your answer correct to the nearest degree.

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o

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**(Total for Question 25 is 6 marks)**

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## WORKED SOLUTION

Dr. Eslam Ahmed

## May 2023 P1HR | Question 25

Sine, Cosine Rule &amp; Area of Triangles

**1. The two sides around the**The two sides around the  $30^\circ$  angle are

$$2x - 1 \quad \text{and} \quad 2x + 1$$

**2. Using the area formula,**

Using the area formula,

$$\text{area} = \frac{1}{2}ab \sin C$$

$$x^2 + x - 3.75 = \frac{1}{2}(2x - 1)(2x + 1) \sin 30^\circ$$

**3. Since ,**Since  $\sin 30^\circ = \frac{1}{2}$ ,

$$x^2 + x - 3.75 = \frac{1}{4}(2x - 1)(2x + 1)$$

$$x^2 + x - 3.75 = \frac{1}{4}(4x^2 - 1)$$

$$x^2 + x - 3.75 = x^2 - 0.25$$

$$x = 3.5$$

**4. So**

So

$$AB = 2x - 1 = 6, \quad AC = 2x + 1 = 8$$

**5. Use the cosine rule to**Use the cosine rule to find  $BC$ :

$$BC^2 = 6^2 + 8^2 - 2(6)(8) \cos 30^\circ$$

$$BC = 4.1063 \dots$$

**6. The largest side is ,**The largest side is  $AC = 8$ , so the largest angle is  $\angle B$ .**7. Using the cosine rule again**

Using the cosine rule again:

$$\cos B = \frac{6^2 + BC^2 - 8^2}{2(6)(BC)}$$

$$B = 103.0643 \dots^\circ$$

**Final answer****103°**

EA

## 3D Pythagoras & Trigonometry

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**CLASSIFIED PROBLEM**

Paper: P2HR

Session: Jan 2021

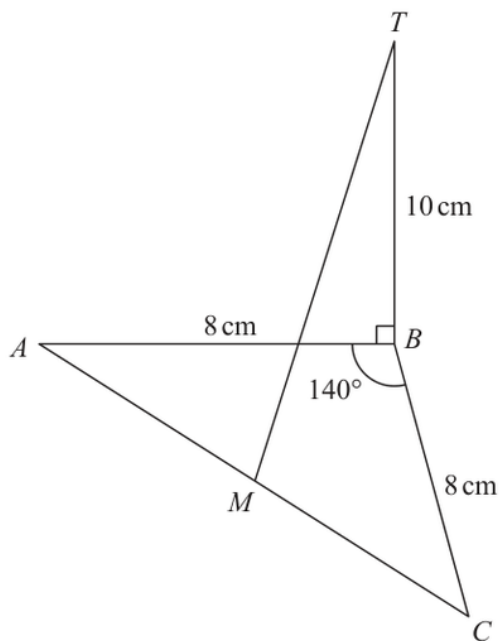
Q#: 22

Marks: 4

TOPIC: 3D Pythagoras &amp; Trigonometry

Source: Jan 2021 P2HR

- 22  $ABC$  is an isosceles triangle in a horizontal plane.  
The point  $T$  is vertically above  $B$ .

Diagram **NOT**  
accurately drawnAngle  $ABC = 140^\circ$  $AB = BC = 8 \text{ cm}$  $TB = 10 \text{ cm}$  $M$  is the midpoint of  $AC$ .

Calculate the size of the angle between  $MT$  and the horizontal plane  $ABC$ .  
Give your answer correct to one decimal place.

(Total for Question 22 is 4 marks)

## WORKED SOLUTION

Dr. Eslam Ahmed

Jan 2021 P2HR | Question 22  
3D Pythagoras & Trigonometry

1. Since , triangle is isosceles

Since  $AB = BC = 8$ , triangle  $ABC$  is isosceles.

2. The line from to the

The line from  $B$  to the midpoint  $M$  bisects angle  $ABC$ .

$$\angle ABM = \frac{140^\circ}{2} = 70^\circ$$

3. In right triangle ,

In right triangle  $ABM$ ,

$$\cos 70^\circ = \frac{BM}{8}$$

$$BM = 8 \cos 70^\circ$$

4. Since is vertically above ,

Since  $T$  is vertically above  $B$ , the projection of  $MT$  onto the horizontal plane is  $MB$ .

5. Let the required angle be

Let the required angle be  $\theta$ .

$$\tan \theta = \frac{TB}{BM}$$

$$\tan \theta = \frac{10}{8 \cos 70^\circ}$$

$$\theta = 74.7^\circ$$

**Final answer**

$74.7^\circ$



## CLASSIFIED PROBLEM

Paper: P1H

Session: Jan 2022

Q#: 21

Marks: 5

TOPIC: 3D Pythagoras &amp; Trigonometry

Source: Jan 2022 P1H

21 The diagram shows the prism  $ABCDEFGHJK$  with horizontal base  $AEFG$

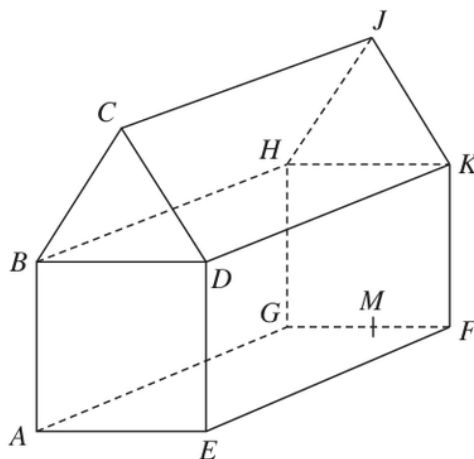


Diagram **NOT**  
accurately drawn

$ABCDE$  is a cross section of the prism where  
 $ABDE$  is a square  
 $BCD$  is an equilateral triangle

$$EF = 2 \times AE$$

$M$  is the midpoint of  $GF$  so that  $JM$  is vertical.

Angle  $MAJ = y^\circ$

Given that  $\tan y^\circ = T$

find the value of  $T$ , giving your answer in the form  $\frac{\sqrt{p} + \sqrt{q}}{17}$  where  $p$  and  $q$  are integers.

$T = \dots\dots\dots$

(Total for Question 21 is 5 marks)

## WORKED SOLUTION

Dr. Eslam Ahmed

**Jan 2022 P1H | Question 21**  
3D Pythagoras & Trigonometry

**1. Let**

Let

$$AE = x$$

**2. Since is a square,**

Since  $ABDE$  is a square,

$$AB = BD = DE = AE = x$$

**3. Since is an equilateral triangle,**

Since  $BCD$  is an equilateral triangle, the height from  $C$  to  $BD$  is

$$\frac{\sqrt{3}}{2}x$$

**4. Therefore the vertical height from**

Therefore the vertical height from the base to  $J$  is

$$JM = x + \frac{\sqrt{3}}{2}x$$

$$JM = \frac{x(2 + \sqrt{3})}{2}$$

**5. Also,**

Also,

$$EF = 2AE = 2x$$

**6. Since is the midpoint of**

Since  $M$  is the midpoint of  $GF$ , the horizontal distance from  $A$  to  $M$  is

$$AM = \sqrt{\left(\frac{x}{2}\right)^2 + (2x)^2}$$

$$AM = \sqrt{\frac{x^2}{4} + 4x^2}$$

$$AM = \frac{x\sqrt{17}}{2}$$

**7. In right triangle ,**

In right triangle  $AMJ$ ,

$$\tan y^\circ = \frac{JM}{AM}$$

$$T = \frac{\frac{x(2+\sqrt{3})}{2}}{\frac{x\sqrt{17}}{2}}$$

$$T = \frac{2 + \sqrt{3}}{\sqrt{17}}$$

### 8. Rationalise the denominator

Rationalise the denominator:

$$T = \frac{(2 + \sqrt{3})\sqrt{17}}{17}$$

$$T = \frac{2\sqrt{17} + \sqrt{51}}{17}$$

$$T = \frac{\sqrt{68} + \sqrt{51}}{17}$$

**Final answer**

$$T = \frac{\sqrt{68} + \sqrt{51}}{17}$$

**CLASSIFIED PROBLEM**

Paper: P2HR

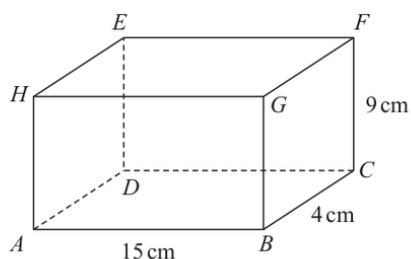
Session: Jun 2023

Q#: 22

Marks: 5

TOPIC: 3D Pythagoras &amp; Trigonometry

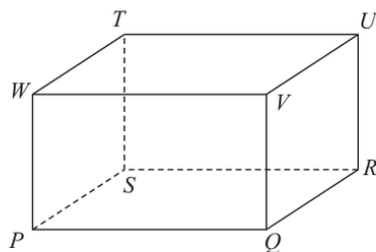
Source: Jun 2023 P2HR

22 Here is a cuboid  $ABCDEFGH$ Diagram **NOT**  
accurately drawn

$$AB = 15 \text{ cm} \quad BC = 4 \text{ cm} \quad CF = 9 \text{ cm}$$

- (a) Work out the length of  $BE$   
Give your answer correct to 3 significant figures.

..... cm  
(2)

Here is a cuboid  $PQRSTUVW$ Diagram **NOT**  
accurately drawn

$$PR = 42 \text{ cm}$$

The size of the angle between  $PU$  and the plane  $PQRS$  is  $30^\circ$  $M$  is the midpoint of  $PR$ 

- (b) Work out the size of angle  $UMR$   
Give your answer correct to 3 significant figures.

.....  
(3)

(Total for Question 22 is 5 marks)

## WORKED SOLUTION

Dr. Eslam Ahmed

## Jun 2023 P2HR | Question 22

3D Pythagoras &amp; Trigonometry

**1. For part (a), is the**For part (a),  $BE$  is the space diagonal using the three perpendicular lengths:

$$AB = 15, \quad BC = 4, \quad CF = 9$$

$$BE^2 = 15^2 + 4^2 + 9^2$$

$$BE^2 = 225 + 16 + 81 = 322$$

$$BE = \sqrt{322} = 17.9 \text{ cm}$$

**2. For part (b), the projection**For part (b), the projection of  $PU$  onto the plane  $PQRS$  is  $PR$ .

$$PR = 42$$

**3. The angle between and the**The angle between  $PU$  and the plane is  $30^\circ$ , so

$$\tan 30^\circ = \frac{RU}{PR}$$

$$RU = 42 \tan 30^\circ$$

**4. Since is the midpoint of**Since  $M$  is the midpoint of  $PR$ ,

$$MR = 21$$

**5. In right triangle ,**In right triangle  $MRU$ ,

$$\tan \angle UMR = \frac{RU}{MR}$$

$$\tan \angle UMR = \frac{42 \tan 30^\circ}{21}$$

$$\tan \angle UMR = 2 \tan 30^\circ$$

$$\angle UMR = 49.1^\circ$$

**Final answer**

$$BE = 17.9 \text{ cm}, \angle UMR = 49.1^\circ$$

**CLASSIFIED PROBLEM**

Paper: P1HR

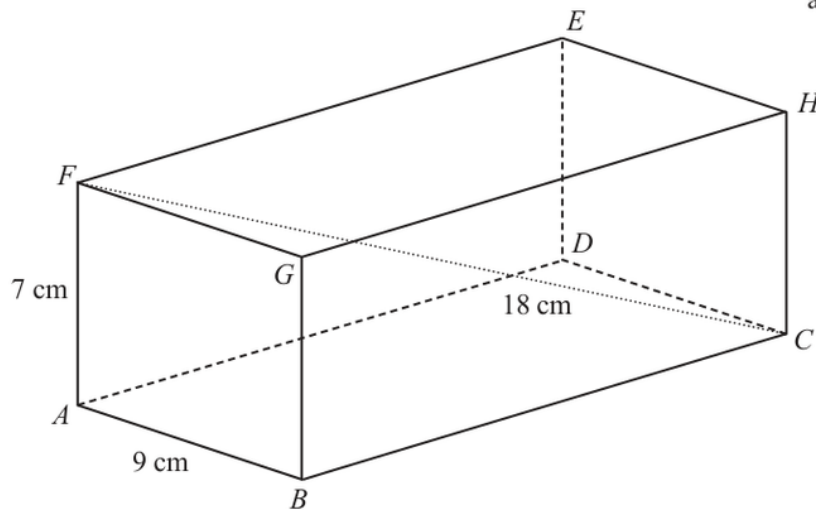
Session: Jun 2025

Q#: 20

Marks: 3

TOPIC: 3D Pythagoras &amp; Trigonometry

Source: Jun 2025 P1HR

**20** The diagram shows cuboid  $ABCDEFGH$ Diagram **NOT**  
accurately drawn

$$AB = 9 \text{ cm} \quad AF = 7 \text{ cm} \quad FC = 18 \text{ cm}$$

Calculate the length of  $BC$ 

Give your answer correct to 3 significant figures.

..... cm

(Total for Question 20 is 3 marks)

## WORKED SOLUTION

Dr. Eslam Ahmed

## Jun 2025 P1HR | Question 20

3D Pythagoras &amp; Trigonometry

## 1. In the rectangle , use

In the rectangle  $ABCF$ , use Pythagoras with the diagonal  $FC$ :

$$AC^2 = FC^2 - AF^2$$

$$AC^2 = 18^2 - 7^2 = 324 - 49 = 275$$

## 2. Now use Pythagoras in triangle

Now use Pythagoras in triangle  $ABC$ :

$$BC^2 = AC^2 - AB^2$$

$$BC^2 = 275 - 9^2 = 275 - 81 = 194$$

$$BC = \sqrt{194} = 13.928\dots$$

**Final answer**

13.9 cm to 3 significant figures

**CLASSIFIED PROBLEM**

Paper: P2H

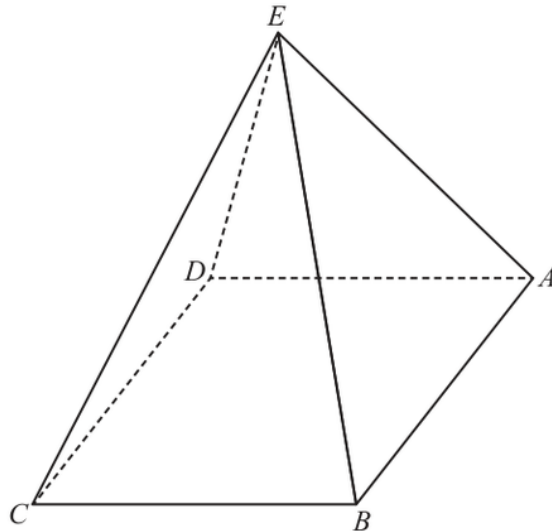
Session: Jun 2025

Q#: 24

Marks: 4

TOPIC: 3D Pythagoras &amp; Trigonometry

Source: Jun 2025 P2H

**24** The diagram shows a square-based pyramid  $ABCDE$ Diagram **NOT**  
accurately drawn

$$EA = EB = EC = ED$$

 $M$  is the centre of the horizontal square base  $ABCD$  $Q$  is the midpoint of  $AB$ 

$$\text{Angle } EQM = 80^\circ$$

$$EA : AB = n : 1$$

Find the value of  $n$ 

Give your answer correct to 3 significant figures.

$$n = \dots\dots\dots$$

(Total for Question 24 is 4 marks)



## WORKED SOLUTION

Dr. Eslam Ahmed

## Jun 2025 P2H | Question 24

3D Pythagoras &amp; Trigonometry

## 1. Let

Let

$$AB = s$$

2. Since  $M$  is the centre ofSince  $M$  is the centre of the square base and  $Q$  is the midpoint of  $AB$ ,

$$QM = \frac{s}{2}$$

3. Since  $E$  is vertically above  $M$ ,Since  $EA = EB = EC = ED$ , the point  $E$  is vertically above  $M$ .4. In right triangle  $EQM$ ,In right triangle  $EQM$ ,

$$\tan 80^\circ = \frac{EM}{QM}$$

$$EM = \frac{s}{2} \tan 80^\circ$$

5. The distance from the centre  $M$  to a vertex  $A$  is

The distance from the centre of a square to a vertex is half the diagonal:

$$AM = \frac{s\sqrt{2}}{2}$$

6. In right triangle  $EMA$ ,In right triangle  $EMA$ ,

$$EA^2 = EM^2 + AM^2$$

$$EA^2 = \left(\frac{s}{2} \tan 80^\circ\right)^2 + \left(\frac{s\sqrt{2}}{2}\right)^2$$

$$EA = s \sqrt{\frac{\tan^2 80^\circ}{4} + \frac{1}{2}}$$

## 7. So

So

$$\frac{EA}{AB} = \sqrt{\frac{\tan^2 80^\circ}{4} + \frac{1}{2}}$$

$$n = 2.92$$

**Final answer**

$$n = 2.92$$

**CLASSIFIED PROBLEM**

Paper: P2HR

Session: Jun 2025

Q#: 23

Marks: 3

TOPIC: 3D Pythagoras &amp; Trigonometry

Source: Jun 2025 P2HR

- 23** The diagram shows a triangular prism,  $ABCDEF$ , with a horizontal rectangular base  $ABCD$

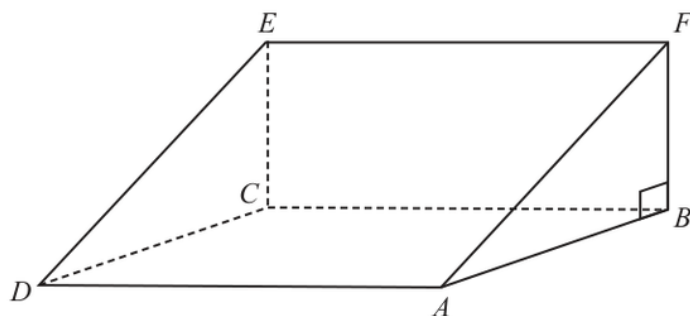


Diagram **NOT**  
accurately drawn

$M$  is the midpoint of the line  $AC$

$AC = 40 \text{ cm}$     angle  $CAE = 35^\circ$     angle  $ABF = 90^\circ$

Work out the size of angle  $CME$

Give your answer correct to 3 significant figures.

°

(Total for Question 23 is 3 marks)

## WORKED SOLUTION

Dr. Eslam Ahmed

**Jun 2025 P2HR | Question 23**  
3D Pythagoras & Trigonometry

**1. Step 1**

$$AC = 40$$

**2. The point is vertically above**

The point  $E$  is vertically above  $C$ , so triangle  $ACE$  is right-angled at  $C$ .

$$\angle CAE = 35^\circ$$

$$\tan 35^\circ = \frac{CE}{AC}$$

$$CE = 40 \tan 35^\circ$$

**3. Since is the midpoint of**

Since  $M$  is the midpoint of  $AC$ ,

$$CM = 20$$

**4. In right triangle ,**

In right triangle  $CME$ ,

$$\tan \angle CME = \frac{CE}{CM}$$

$$\tan \angle CME = \frac{40 \tan 35^\circ}{20}$$

$$\tan \angle CME = 2 \tan 35^\circ$$

$$\angle CME = 54.5^\circ$$

**Final answer**

54.5°

**CLASSIFIED PROBLEM**

Paper: P1H

Session: May 2021

Q#: 22

Marks: 4

TOPIC: 3D Pythagoras &amp; Trigonometry

Source: May 2021 P1H

**22** The diagram shows a triangular prism  $ABCDEF$  with a horizontal base  $ABEF$ .

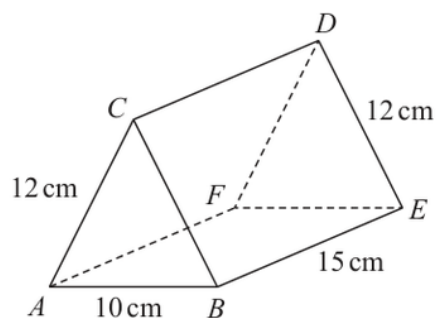


Diagram **NOT**  
accurately drawn

$$AC = BC = FD = ED = 12 \text{ cm} \quad AB = 10 \text{ cm} \quad BE = 15 \text{ cm}$$

Calculate the size of the angle between  $AD$  and the base  $ABEF$ .

Give your answer correct to 3 significant figures.

(Total for Question 22 is 4 marks)

°

## WORKED SOLUTION

Dr. Eslam Ahmed

**May 2021 P1H | Question 22**  
3D Pythagoras & Trigonometry

**1. The triangular end is isosceles,**

The triangular end is isosceles, with

$$AC = BC = 12, \quad AB = 10$$

**2. The perpendicular height of the**

The perpendicular height of the triangle is found by halving  $AB$ :

$$AM = 5$$

$$CM^2 = 12^2 - 5^2 = 144 - 25 = 119$$

$$CM = \sqrt{119}$$

**3. The projection of on the**

The projection of  $AD$  on the horizontal base goes from  $A$  to the midpoint of  $FE$ . Its length is

$$\sqrt{15^2 + 5^2} = \sqrt{250}$$

**4. So, for the angle between**

So, for the angle  $\theta$  between  $AD$  and the base,

$$\tan \theta = \frac{\sqrt{119}}{\sqrt{250}}$$

$$\theta = \tan^{-1} \left( \frac{\sqrt{119}}{\sqrt{250}} \right) = 34.602 \dots$$

**Final answer**

$34.6^\circ$

**CLASSIFIED PROBLEM**

Paper: P1HR

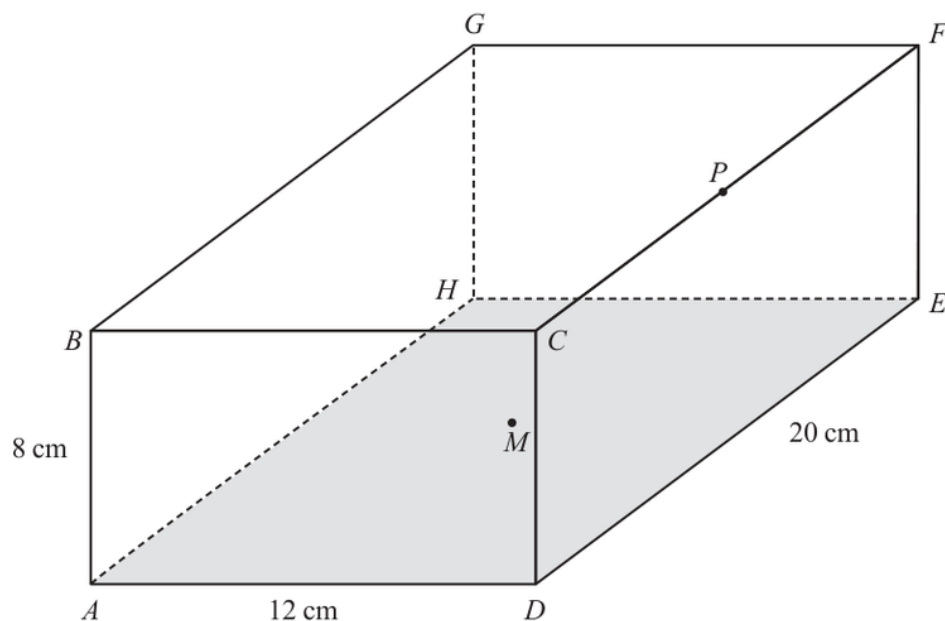
Session: May 2024

Q#: 22

Marks: 6

TOPIC: 3D Pythagoras &amp; Trigonometry

Source: May 2024 P1HR

**22** The diagram shows a cuboid  $ABCDEFGH$  with horizontal base  $ADEH$ Diagram **NOT**  
accurately drawn

$AB = 8 \text{ cm}$

$AD = 12 \text{ cm}$

$DE = 20 \text{ cm}$

 $M$  is the midpoint of the base  $ADEH$  and  $P$  is the midpoint of the edge  $CF$ Work out the size of angle  $BMP$ 

Give your answer correct to one decimal place.

(Total for Question 22 is 6 marks)

## WORKED SOLUTION

Dr. Eslam Ahmed

May 2024 P1HR | Question 22  
3D Pythagoras & Trigonometry

**1. Use coordinates with**

Use coordinates with

$$A = (0, 0, 0)$$

**2. Since**

Since

$$AD = 12, \quad DE = 20, \quad AB = 8$$

**3. take**

take

$$B = (0, 0, 8)$$

**4. The centre of the rectangular**

The centre of the rectangular base  $ADEH$  is

$$M = (6, 10, 0)$$

**5. Point is the midpoint of**

Point  $P$  is the midpoint of  $CF$ .

$$C = (12, 0, 8), \quad F = (12, 20, 8)$$

$$P = (12, 10, 8)$$

**6. Now**

Now

$$\vec{MB} = B - M = (-6, -10, 8)$$

$$\vec{MP} = P - M = (6, 0, 8)$$

**7. Use the scalar product formula**

Use the scalar product formula:

$$\vec{MB} \cdot \vec{MP} = (-6)(6) + (-10)(0) + (8)(8)$$

$$\vec{MB} \cdot \vec{MP} = 28$$

$$|\vec{MB}| = \sqrt{(-6)^2 + (-10)^2 + 8^2} = \sqrt{200}$$



$$|\vec{MP}| = \sqrt{6^2 + 0^2 + 8^2} = 10$$

$$\cos \angle BMP = \frac{28}{10\sqrt{200}}$$

$$\angle BMP = 78.6^\circ$$

**Final answer**

$78.6^\circ$

**CLASSIFIED PROBLEM**

Paper: P1H

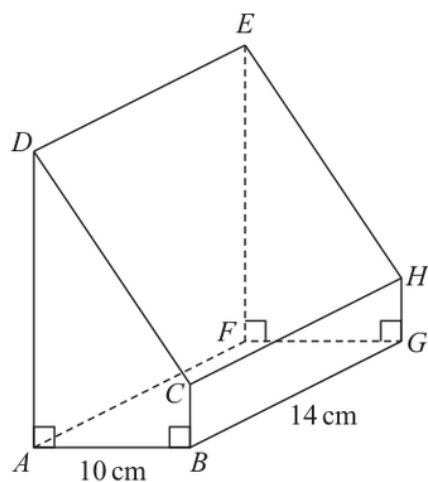
Session: Nov 2025

Q#: 26

Marks: 5

TOPIC: 3D Pythagoras &amp; Trigonometry

Source: Nov 2025 P1H

**26** Here is a prism  $ABCDEFGH$  with a horizontal, rectangular base  $ABGF$ Diagram **NOT**  
accurately drawn

$$AB = 10 \text{ cm} \quad BG = 14 \text{ cm}$$

$$\text{angle } DAB = \text{angle } ABC = \text{angle } EFG = \text{angle } FGH = 90^\circ$$

$$BC = \frac{1}{5} AD$$

The angle of elevation of  $E$  from  $B$  is  $50^\circ$ 

Calculate the volume of the prism.

Give your answer correct to 3 significant figures.

Show your working clearly.

.....cm<sup>3</sup>

(Total for Question 26 is 5 marks)

## WORKED SOLUTION

Dr. Eslam Ahmed

## Nov 2025 P1H | Question 26

3D Pythagoras &amp; Trigonometry

**1. The horizontal base is a**The horizontal base  $ABGF$  is a rectangle.

$$AB = 10, \quad BG = 14$$

**2. So the horizontal distance from**So the horizontal distance from  $B$  to  $F$  is

$$BF = \sqrt{10^2 + 14^2} = \sqrt{296}$$

**3. The angle of elevation of**The angle of elevation of  $E$  from  $B$  is  $50^\circ$ .**4. Since is vertically above ,**Since  $E$  is vertically above  $F$ ,

$$\tan 50^\circ = \frac{FE}{BF}$$

$$FE = \sqrt{296} \tan 50^\circ$$

**5. The prism has the same**

The prism has the same cross section throughout, so

$$AD = FE = \sqrt{296} \tan 50^\circ$$

**6. Also**

Also

$$BC = \frac{1}{5}AD$$

**7. Area of the trapezium cross**Area of the trapezium cross section  $ABCD$ :

$$\begin{aligned} & \frac{1}{2}(AD + BC) \times AB \\ &= \frac{1}{2} \left( AD + \frac{1}{5}AD \right) \times 10 \\ &= 6AD \end{aligned}$$

**8. Volume of the prism**

Volume of the prism:

$$6AD \times 14 = 84AD$$

$$= 84\sqrt{296} \tan 50^\circ$$

$$= 1722.311 \dots$$

### 9. Correct to 3 significant figures

Correct to 3 significant figures:

$$1720$$

**Final answer**

$$1720 \text{ cm}^3$$

EA

# Histograms

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**CLASSIFIED PROBLEM**

Paper: 4WM1HR

Session: May2025

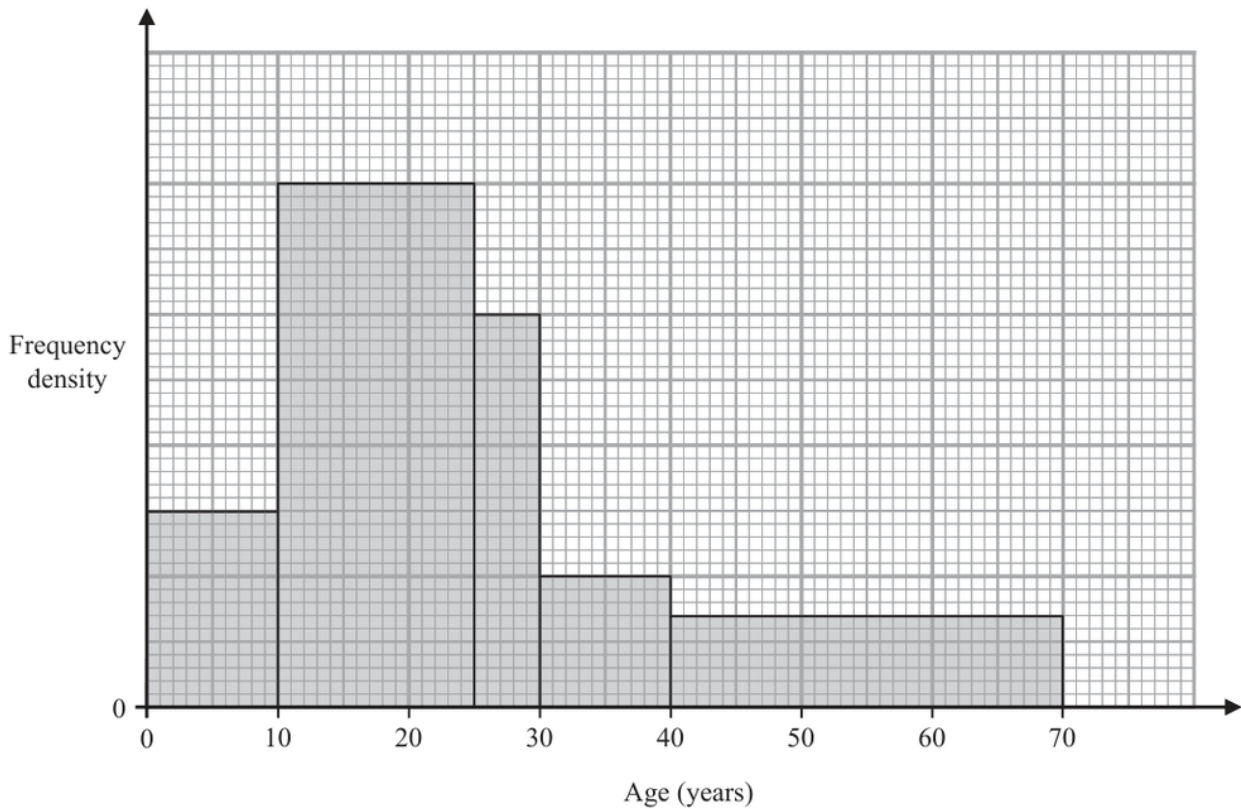
Q#: 21

Marks: 3

Topic: Histograms

Source: May 2025 4WM1HR

**21** The histogram shows information about the ages of the members of a chess club.



There are 60 members aged between 10 and 25

There are no members older than 70

Work out an estimate for the number of members who are over 35 years of age.

**(Total for Question 21 is 3 marks)**

## WORKED SOLUTION

Dr. Eslam Ahmed

May 2025 4WM1HR | Question 21  
Histograms

**1. The interval to has width**

The interval 10 to 25 has width 15 and frequency 60, so its frequency density is

$$\frac{60}{15} = 4$$

**2. Using that scale from the**

Using that scale from the histogram:

$$35 \text{ to } 40 : 5 \times 1 = 5$$

$$40 \text{ to } 70 : 30 \times 0.75 = 22.5$$

$$5 + 22.5 = 27.5$$

**Final answer**  
about 28 members.

**CLASSIFIED PROBLEM**

Paper: 4WM1H

Session: Nov2025

Q#: 20

Marks: 3

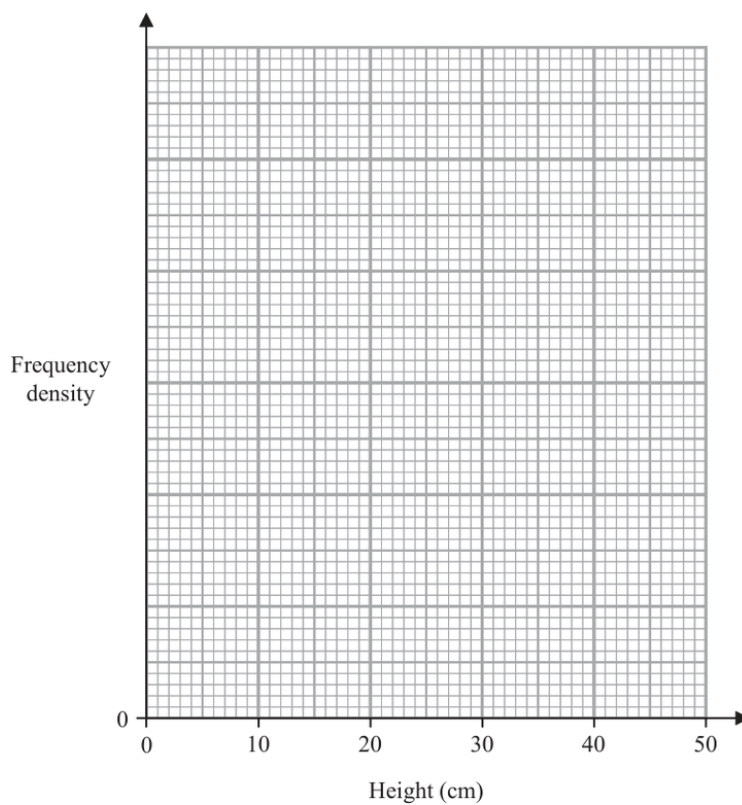
Topic: Histograms

Source: Nov 2025 4WM1H

**20** The table gives information about the heights, in centimetres, of some plants.

| Height ( $h$ cm) | Frequency |
|------------------|-----------|
| $0 < h \leq 5$   | 12        |
| $5 < h \leq 15$  | 34        |
| $15 < h \leq 30$ | 45        |
| $30 < h \leq 50$ | 28        |

On the grid, draw a histogram for this information.  
You must include a scale on the vertical axis.



(Total for Question 20 is 3 marks)



## WORKED SOLUTION

Dr. Eslam Ahmed

Nov 2025 4WM1H | Question 20  
Histograms

**1. Frequency density**

$$\text{Frequency density} = \frac{\text{frequency}}{\text{class width}}.$$

$$0 < h \leq 5 : \frac{12}{5} = 2.4$$

$$5 < h \leq 15 : \frac{34}{10} = 3.4$$

$$15 < h \leq 30 : \frac{45}{15} = 3$$

$$30 < h \leq 50 : \frac{28}{20} = 1.4$$

**Final answer**

draw bars with densities 2.4, 3.4, 3, 1.4.

# Probability Toolkit

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**CLASSIFIED PROBLEM**

Paper: 4WM1H

Session: May2025

Q#: 22

Marks: 3

TOPIC: **Probability Toolkit**

Source: May 2025 4WM1H

**22** A box contains 20 counters.

9 of the counters are red

7 of the counters are yellow

4 of the counters are green

Alex takes at random three counters from the box.

Work out the probability that exactly two of the three counters are the same colour.

---

**(Total for Question 22 is 3 marks)**

## WORKED SOLUTION

Dr. Eslam Ahmed

May 2025 4WM1H | Question 22  
Probability Toolkit

**1. Total ways to choose counters**

Total ways to choose 3 counters:

$$\binom{20}{3} = 1140$$

**2. Exactly two red**

Exactly two red:

$$\binom{9}{2} \binom{11}{1} = 396$$

**3. Exactly two yellow**

Exactly two yellow:

$$\binom{7}{2} \binom{13}{1} = 273$$

**4. Exactly two green**

Exactly two green:

$$\binom{4}{2} \binom{16}{1} = 96$$

**5. Favourable outcomes**

Favourable outcomes:

$$396 + 273 + 96 = 765$$

$$P = \frac{765}{1140} = \frac{51}{76}$$

**Final answer**

$$\frac{51}{76}$$

# Histograms

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**CLASSIFIED PROBLEM**

Paper: P1HR

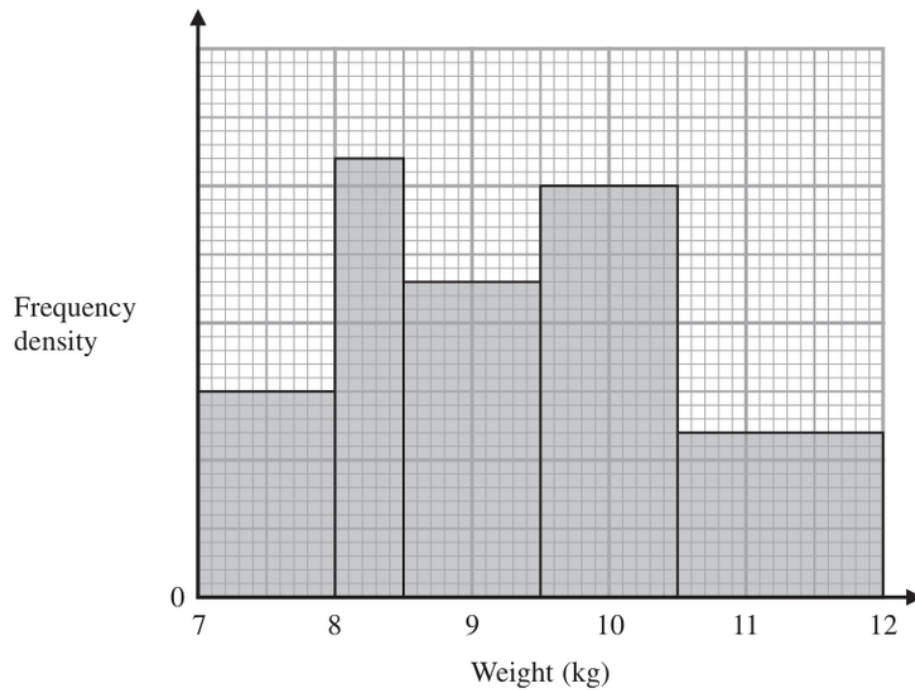
Session: Jan 2022

Q#: 21

Marks: 3

TOPIC: Histograms

Source: Jan 2022 P1HR

**21**

The histogram gives information about the weights, in kg, of all the watermelons in a field.

There are 16 watermelons with a weight between 8 kg and 8.5 kg

Work out the total number of watermelons in the field.

(Total for Question 21 is 3 marks)

## WORKED SOLUTION

Dr. Eslam Ahmed

Jan 2022 P1HR | Question 21  
Histograms

**1. For a histogram,**

For a histogram,

$$\text{frequency} = \text{class width} \times \text{frequency density}$$

**2. For , the frequency is**

For  $8 \leq w < 8.5$ , the frequency is 16 and the class width is 0.5, so

$$\text{frequency density} = \frac{16}{0.5} = 32$$

**3. Using this scale from the**

Using this scale from the histogram, the frequency densities are:

$$15, 32, 23, 30, 12$$

**4. The total frequency is**

The total frequency is

$$(1)(15) + (0.5)(32) + (1)(23) + (1)(30) + (1.5)(12)$$

$$= 15 + 16 + 23 + 30 + 18$$

$$= 102$$

**Final answer**

102 watermelons

**CLASSIFIED PROBLEM**

Paper: P2H

Session: Jan 2022

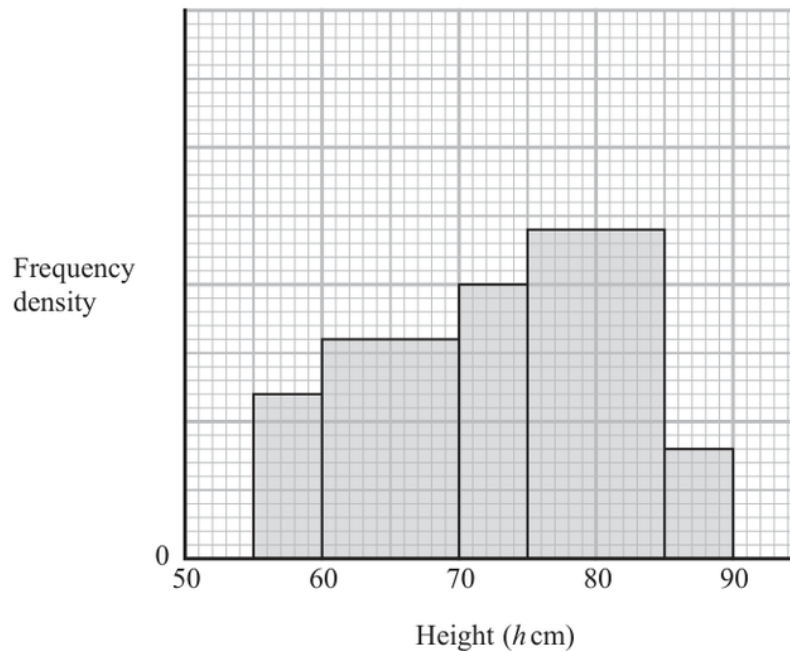
Q#: 20

Marks: 4

Topic: Histograms

Source: Jan 2022 P2H

**20** The histogram gives information about the heights,  $h$  cm, of some tomato plants.



There are 12 tomato plants for which  $75 < h \leq 85$

One of the tomato plants is selected at random.

Find an estimate for the probability that this tomato plant has a height greater than 82.5 cm

(Total for Question 20 is 4 marks)



## WORKED SOLUTION

Dr. Eslam Ahmed

Jan 2022 P2H | Question 20  
Histograms

**1. For a histogram,**

For a histogram,

$$\text{frequency} = \text{class width} \times \text{frequency density}$$

**2. For , the frequency is**

For  $75 < h \leq 85$ , the frequency is 12 and the class width is 10, so

$$\text{frequency density} = \frac{12}{10} = 1.2$$

**3. Using this scale from the**

Using this scale from the histogram, the frequency densities are:

$$0.6, 0.8, 1.0, 1.2, 0.4$$

**4. The total number of tomato**

The total number of tomato plants is

$$\begin{aligned} & (5)(0.6) + (10)(0.8) + (5)(1.0) + (10)(1.2) + (5)(0.4) \\ &= 3 + 8 + 5 + 12 + 2 = 30 \end{aligned}$$

**5. For**

For  $h > 82.5$ :

$$(2.5)(1.2) + (5)(0.4) = 3 + 2 = 5$$

**6. So the probability is**

So the probability is

$$\frac{5}{30} = \frac{1}{6}$$

**Final answer**

$$\frac{1}{6}$$

**CLASSIFIED PROBLEM**

Paper: P1H

Session: Jun 2022

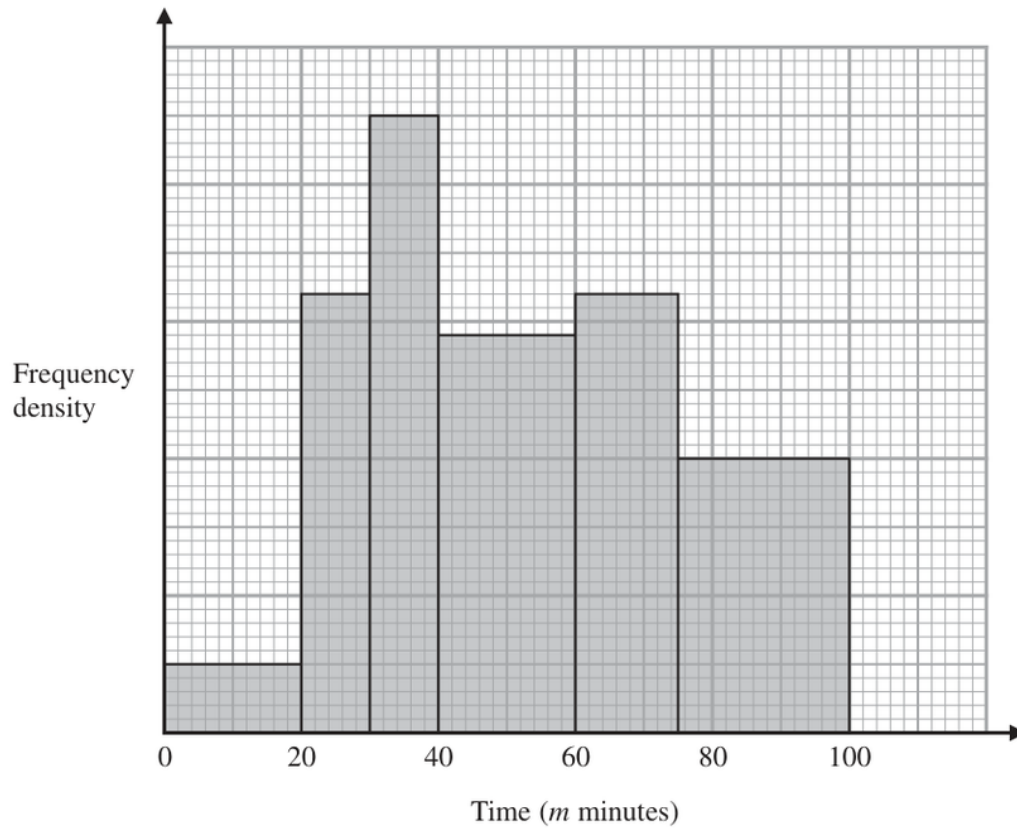
Q#: 21

Marks: 3

Topic: Histograms

Source: Jun 2022 P1H

- 21** The histogram shows information about the total time,  $m$  minutes, taken by each child in a school to walk to school every day for one week.



There are no children for whom  $m > 100$

There are 10 children for whom  $m \leq 20$

Work out an estimate for the number of children for whom  $50 < m \leq 80$

(Total for Question 21 is 3 marks)

## WORKED SOLUTION

Dr. Eslam Ahmed

Jun 2022 P1H | Question 21  
Histograms

**1. For a histogram,**

For a histogram,

$$\text{frequency} = \text{class width} \times \text{frequency density}$$

**2. For , the frequency is**

For  $m \leq 20$ , the frequency is 10 and the class width is 20, so

$$\text{frequency density} = \frac{10}{20} = 0.5$$

**3. Using this scale from the**

Using this scale from the histogram:

$$40 < m \leq 60 \quad \text{has frequency density } 2.9$$

$$60 < m \leq 75 \quad \text{has frequency density } 3.2$$

$$75 < m \leq 100 \quad \text{has frequency density } 2.0$$

**4. For , use part of**

For  $50 < m \leq 80$ , use part of each relevant bar:

$$(10)(2.9) + (15)(3.2) + (5)(2.0)$$

$$= 29 + 48 + 10$$

$$= 87$$

**Final answer**

87 children

**CLASSIFIED PROBLEM**

Paper: P2HR

Session: Jun 2025

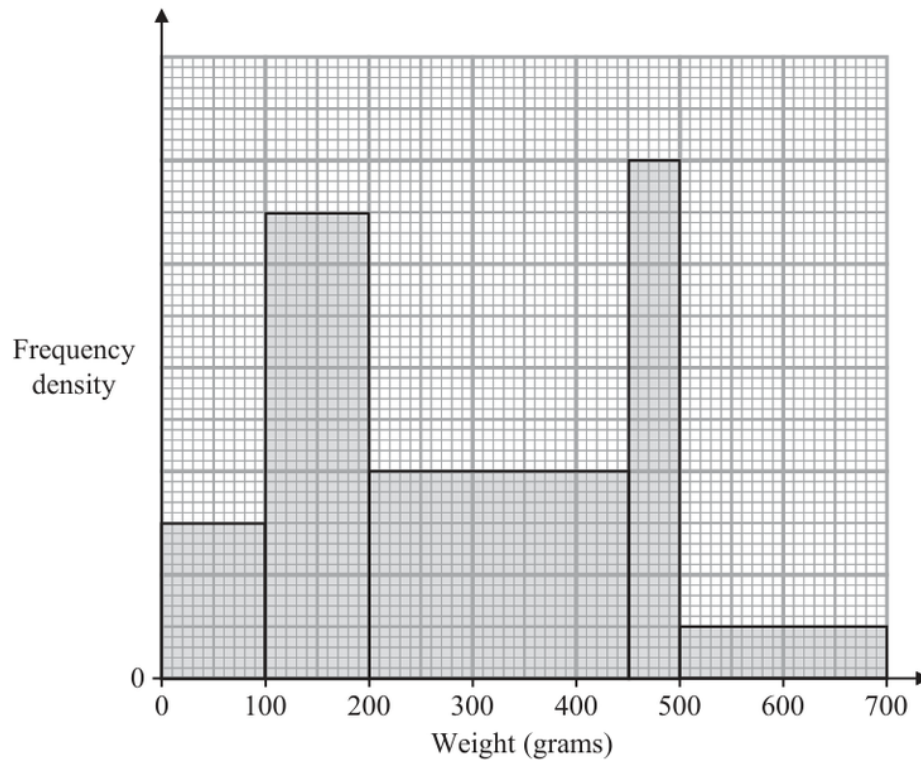
Q#: 20

Marks: 4

Topic: Histograms

Source: Jun 2025 P2HR

**20** The histogram gives some information about the weights, in grams, of some books.



75 books weigh less than 100 grams.

A book is chosen at random.

Find an estimate for the probability that this book weighs between 400 grams and 600 grams.

(Total for Question 20 is 4 marks)

## WORKED SOLUTION

Dr. Eslam Ahmed

## Jun 2025 P2HR | Question 20

Histograms

**1. For a histogram,**

For a histogram,

$$\text{frequency} = \text{class width} \times \text{frequency density}$$

**2. For books weighing less than**

For books weighing less than 100 g, the frequency is 75 and the class width is 100, so

$$\text{frequency density} = \frac{75}{100} = 0.75$$

**3. Using this scale from the**

Using this scale from the histogram, the frequency densities are:

$$0.75, 2.25, 1.0, 2.5, 0.25$$

**4. The total number of books**

The total number of books is

$$\begin{aligned} & (100)(0.75) + (100)(2.25) + (250)(1.0) + (50)(2.5) + (200)(0.25) \\ &= 75 + 225 + 250 + 125 + 50 = 725 \end{aligned}$$

**5. For weights between 400 g**

For weights between 400 g and 600 g:

$$\begin{aligned} & (50)(1.0) + (50)(2.5) + (100)(0.25) \\ &= 50 + 125 + 25 = 200 \end{aligned}$$

**6. So the probability is**

So the probability is

$$\frac{200}{725} = \frac{8}{29}$$

**Final answer** $\frac{8}{29}$  or 0.276 to 3 significant figures

**CLASSIFIED PROBLEM**

Paper: P1HR

Session: May 2023

Q#: 22

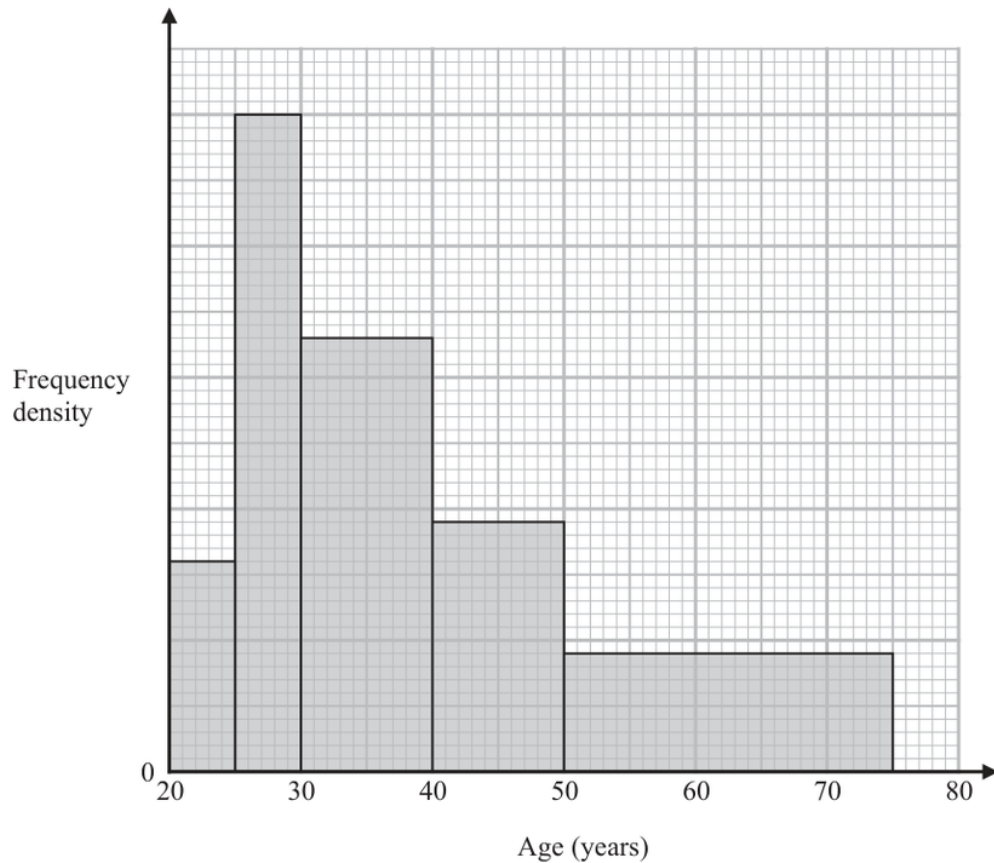
Marks: 4

Topic: Histograms

Source: May 2023 P1HR

**22** Some people attend a concert.

The histogram shows information about the ages of these people.



Work out an estimate for the percentage of these people who are aged more than 55 years.

Give your answer correct to one decimal place.

---



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.....%

(Total for Question 22 is 4 marks)

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## WORKED SOLUTION

Dr. Eslam Ahmed

May 2023 P1HR | Question 22  
Histograms

**1. The scale of the frequency**

The scale of the frequency density axis is not needed because the percentage is found from areas.

**2. Using the bar heights from**

Using the bar heights from the histogram, the relative areas are:

$$\begin{aligned} & (5)(277) + (5)(866) + (10)(571) + (10)(329) + (25)(155) \\ &= 1385 + 4330 + 5710 + 3290 + 3875 \\ &= 18590 \end{aligned}$$

**3. For ages more than 55,**

For ages more than 55, use the part of the last bar from 55 to 75:

$$(20)(155) = 3100$$

**4. So the estimated percentage is**

So the estimated percentage is

$$\frac{3100}{18590} \times 100 = 16.675 \dots$$

**Final answer**

16.7%

**CLASSIFIED PROBLEM**

Paper: P1H

Session: May 2024

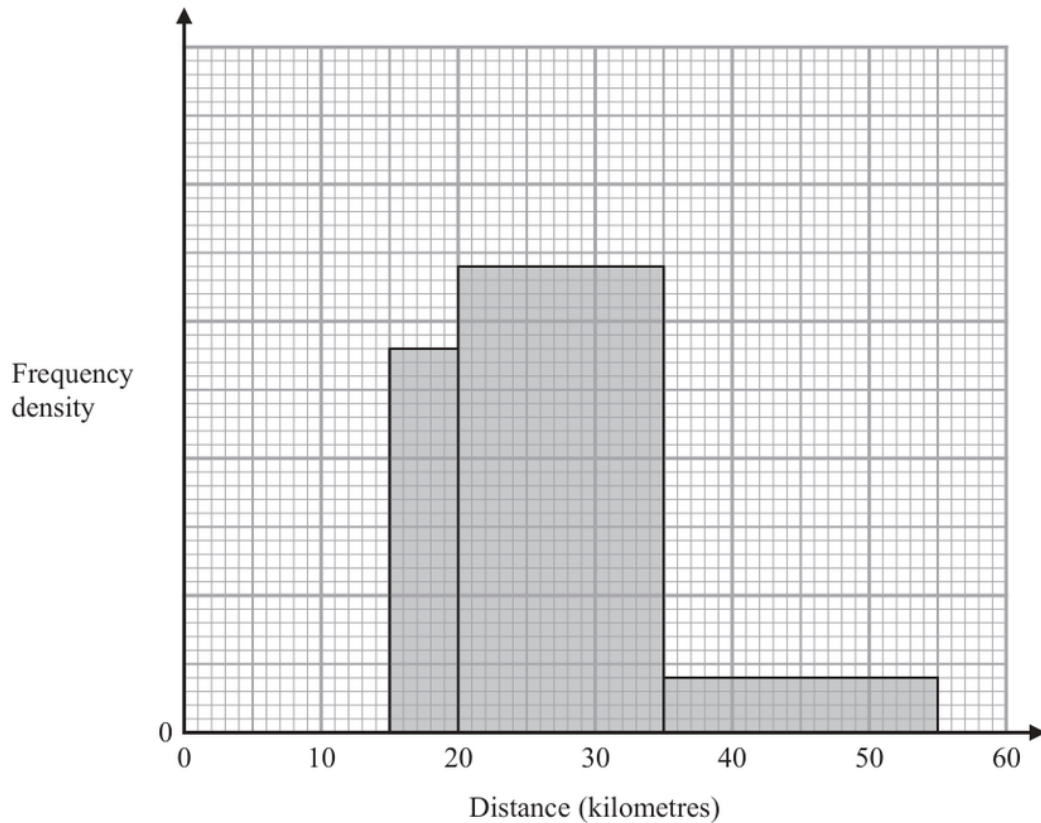
Q#: 22

Marks: 3

Topic: Histograms

Source: May 2024 P1H

- 22** The incomplete histogram shows some information about the distances, in kilometres, that 100 adults ran last week.



All of the adults ran at least 5 kilometres.  
 None of the adults ran more than 55 kilometres.  
 14 adults ran between 15 kilometres and 20 kilometres.

Complete the histogram.

(Total for Question 22 is 3 marks)



## WORKED SOLUTION

Dr. Eslam Ahmed

## May 2024 P1H | Question 22

Histograms

**1. For a histogram,**

For a histogram,

$$\text{frequency} = \text{class width} \times \text{frequency density}$$

**2. For , the frequency is**For  $15 \leq d < 20$ , the frequency is 14 and the class width is 5, so

$$\text{frequency density} = \frac{14}{5} = 2.8$$

**3. Using this scale from the**

Using this scale from the histogram:

$$20 \leq d < 35 \quad \text{has frequency density } 3.4$$

$$35 \leq d \leq 55 \quad \text{has frequency density } 0.4$$

**4. So the known frequencies are**

So the known frequencies are

$$14 + (15)(3.4) + (20)(0.4)$$

$$= 14 + 51 + 8 = 73$$

**5. There are 100 adults in**

There are 100 adults in total, so the missing frequency from 5 km to 15 km is

$$100 - 73 = 27$$

**6. The missing class width is**

The missing class width is 10, so its frequency density is

$$\frac{27}{10} = 2.7$$

**Final answer**

draw the missing bar from 5 to 15 with frequency density 2.7

**CLASSIFIED PROBLEM**

Paper: P2H

Session: Nov 2025

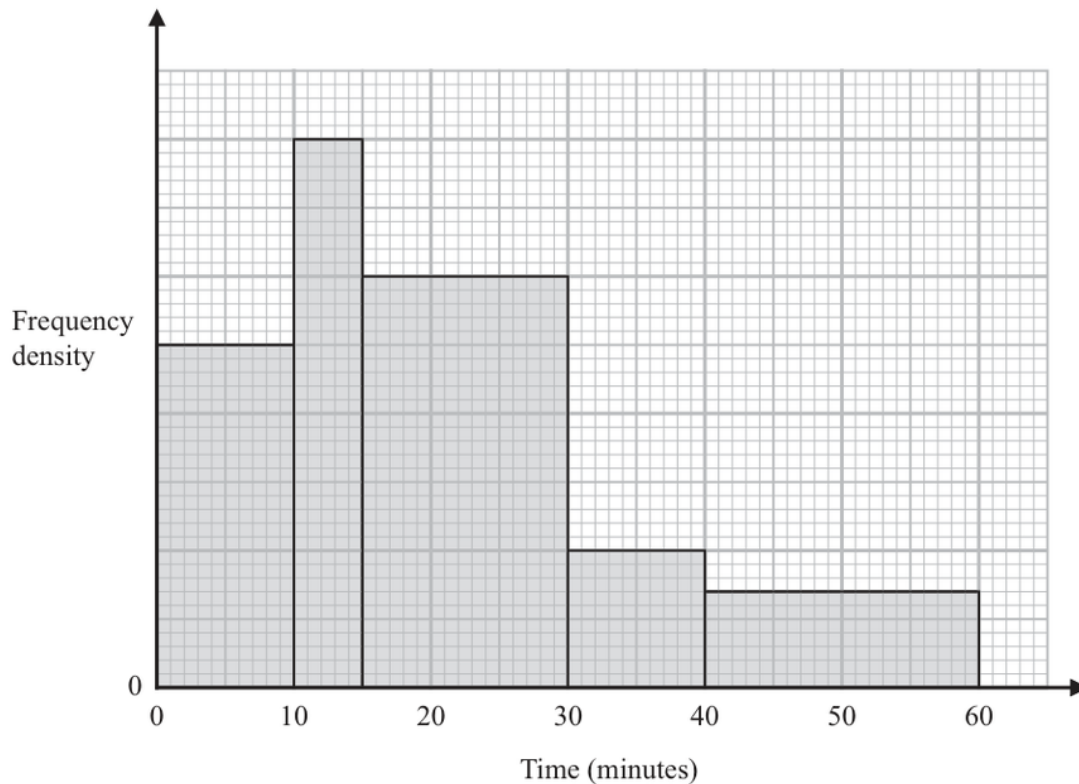
Q#: 21

Marks: 3

TOPIC: Histograms

Source: Nov 2025 P2H

- 21** The histogram shows information about the times, in minutes, that some trains arrived late at a station one day.



20 of these trains arrived between 10 minutes late and 15 minutes late.

No trains arrived more than 60 minutes late.

Work out an estimate for the number of these trains that arrived at least 25 minutes late.

**(Total for Question 21 is 3 marks)**

## WORKED SOLUTION

Dr. Eslam Ahmed

Nov 2025 P2H | Question 21  
Histograms

**1. For a histogram,**

For a histogram,

$$\text{frequency} = \text{class width} \times \text{frequency density}$$

**2. For , the frequency is**

For  $10 < t \leq 15$ , the frequency is 20 and the class width is 5, so

$$\text{frequency density} = \frac{20}{5} = 4$$

**3. Using this scale from the**

Using this scale from the histogram:

$$15 < t \leq 30 \quad \text{has frequency density } 3$$

$$30 < t \leq 40 \quad \text{has frequency density } 1$$

$$40 < t \leq 60 \quad \text{has frequency density } 0.7$$

**4. For trains at least 25**

For trains at least 25 minutes late:

$$(5)(3) + (10)(1) + (20)(0.7)$$

$$= 15 + 10 + 14$$

$$= 39$$

**Final answer**

39 trains

# Combined & Conditional Probability

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**CLASSIFIED PROBLEM**

Paper: 4WM1H

Session: Nov2025

Q#: 23

Marks: 3

TOPIC: **Combined & Conditional Probability**

Source: Nov 2025 4WM1H

**23** There are 15 buttons in a box.

3 of the buttons are red

2 of the buttons are pink

10 of the buttons are blue

Pete takes at random three buttons from the box.

Work out the probability that there is still at least one pink button in the box.

**(Total for Question 23 is 3 marks)**

## WORKED SOLUTION

Dr. Eslam Ahmed

**Nov 2025 4WM1H | Question 23**

Combined &amp; Conditional Probability

**1. There are 2 pink buttons**

There are 2 pink buttons.

**2. At least one pink button**

At least one pink button is still in the box unless both pink buttons are taken.

**3. So use the complement**

So use the complement.

**4. The probability that both pink**

The probability that both pink buttons are taken is

$$\begin{aligned} & \frac{\binom{2}{2} \binom{13}{1}}{\binom{15}{3}} \\ &= \frac{13}{455} \\ &= \frac{1}{35} \end{aligned}$$

**5. Therefore,**

Therefore,

$$\begin{aligned} P(\text{at least one pink is still in the box}) &= 1 - \frac{1}{35} \\ &= \frac{34}{35} \end{aligned}$$

**Final answer**

$$\frac{34}{35}$$

# Probability Toolkit

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**CLASSIFIED PROBLEM**

Paper: P2HR

Session: Jan 2021

Q#: 21

Marks: 6

Topic: **Probability Toolkit**

Source: Jan 2021 P2HR

- 21** A bag contains  $n$  beads.  
6 of the beads are red and the rest are blue.

Ravi is going to take at random 2 beads from the bag.

The probability that the 2 beads will be of the same colour is  $\frac{9}{17}$

Using algebra, and showing each stage of your working, calculate the value of  $n$ .

$n = \dots\dots\dots$

(Total for Question 21 is 6 marks)



## WORKED SOLUTION

Dr. Eslam Ahmed

Jan 2021 P2HR | Question 21  
Probability Toolkit

**1. There are beads altogether**

There are  $n$  beads altogether.

**2. Red beads**

Red beads:

$$6$$

**3. Blue beads**

Blue beads:

$$n - 6$$

**4. The probability of taking two**

The probability of taking two beads of the same colour is

$$P(RR) + P(BB) = \frac{9}{17}$$

**5. So**

So

$$\frac{6}{n} \cdot \frac{5}{n-1} + \frac{n-6}{n} \cdot \frac{n-7}{n-1} = \frac{9}{17}$$

$$\frac{30 + (n-6)(n-7)}{n(n-1)} = \frac{9}{17}$$

**6. Expand**

Expand:

$$(n-6)(n-7) = n^2 - 13n + 42$$

**7. So**

So

$$\frac{n^2 - 13n + 72}{n(n-1)} = \frac{9}{17}$$

**8. Cross multiply**

Cross multiply:

$$17(n^2 - 13n + 72) = 9n(n-1)$$

$$17n^2 - 221n + 1224 = 9n^2 - 9n$$

$$8n^2 - 212n + 1224 = 0$$

### 9. Divide by 4

Divide by 4:

$$2n^2 - 53n + 306 = 0$$

### 10. Factor

Factor:

$$(2n - 17)(n - 18) = 0$$

### 11. So

So

$$n = \frac{17}{2} \quad \text{or} \quad n = 18$$

### 12. The number of beads must

The number of beads must be a whole number, so

$$n = 18$$

**Final answer**

18

**CLASSIFIED PROBLEM**

Paper: P1HR

Session: Jun 2022

Q#: 24

Marks: 5

TOPIC: **Probability Toolkit**

Source: Jun 2022 P1HR

**24** Elliot has  $x$  counters.

Each counter has one red face and one green face.

Elliot spreads all the counters out on a table and sees that the number of counters showing a red face is 5

Elliot then picks at random one of the counters and turns the counter over.  
He then picks at random a second counter and turns the counter over.The probability that there are still 5 counters showing a red face is  $\frac{19}{32}$ Work out the value of  $x$   
Show clear algebraic working. $x = \dots\dots\dots$ **(Total for Question 24 is 5 marks)**

## WORKED SOLUTION

Dr. Eslam Ahmed

**Jun 2022 P1HR** | **Question 24**  
Probability Toolkit

**1. There are counters. Initially 5**

There are  $x$  counters. Initially 5 show red, so  $x - 5$  show green.

**2. After two turns, there are**

After two turns, there are still 5 red faces if:

**3. the same counter is picked**

- the same counter is picked twice, or
- one red counter and one green counter are picked.

**4. The probability of picking the**

The probability of picking the same counter twice is

$$\frac{1}{x}$$

**5. The probability of picking one**

The probability of picking one red and one green in either order is

$$\begin{aligned} \frac{5}{x} \cdot \frac{x-5}{x} + \frac{x-5}{x} \cdot \frac{5}{x} \\ = \frac{10(x-5)}{x^2} \end{aligned}$$

**6. So**

So

$$\frac{1}{x} + \frac{10(x-5)}{x^2} = \frac{19}{32}$$

$$\frac{11x - 50}{x^2} = \frac{19}{32}$$

$$32(11x - 50) = 19x^2$$

$$19x^2 - 352x + 1600 = 0$$

$$(x - 8)(19x - 200) = 0$$

**7. Since must be a whole**

Since  $x$  must be a whole number,

$$x = 8$$

**Final answer**

$$x = 8$$

**CLASSIFIED PROBLEM**

Paper: P1H

Session: Jun 2025

Q#: 21

Marks: 3

TOPIC: **Probability Toolkit**

Source: Jun 2025 P1H

**21** A box contains 20 counters.

9 of the counters are red

7 of the counters are yellow

4 of the counters are green

Alex takes at random three counters from the box.

Work out the probability that exactly two of the three counters are the same colour.

(Total for Question 21 is 3 marks)

## WORKED SOLUTION

Dr. Eslam Ahmed

**Jun 2025 P1H** | **Question 21**  
Probability Toolkit

**1. There are**

There are

$$\binom{20}{3} = 1140$$

**2. ways to choose 3 counters**

ways to choose 3 counters.

**3. Exactly two counters are the**

Exactly two counters are the same colour:

$$\begin{aligned} \binom{9}{2}(11) + \binom{7}{2}(13) + \binom{4}{2}(16) \\ = 36(11) + 21(13) + 6(16) \\ = 396 + 273 + 96 = 765 \end{aligned}$$

**4. So the probability is**

So the probability is

$$\frac{765}{1140} = \frac{51}{76}$$

**Final answer**

$$\frac{51}{76}$$

**CLASSIFIED PROBLEM**

Paper: P2H

Session: May 2021

Q#: 23

Marks: 5

TOPIC: **Probability Toolkit**

Source: May 2021 P2H

**23** Pippa has a box containing  $N$  pens.

There are only black pens and red pens in the box.

The number of black pens in the box is 3 more than the number of red pens.

Pippa is going to take at random 2 pens from the box.

The probability that she will take a black pen **followed** by a red pen is  $\frac{9}{35}$

Find the possible values of  $N$ .

Show clear algebraic working.

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**(Total for Question 23 is 5 marks)**



## WORKED SOLUTION

Dr. Eslam Ahmed

## May 2021 P2H | Question 23

Probability Toolkit

**1. Let the number of red**Let the number of red pens be  $r$ .**2. Then the number of black**

Then the number of black pens is

$$r + 3$$

**3. and**

and

$$N = 2r + 3$$

**4. So**

So

$$r = \frac{N - 3}{2}$$

**5. and**

and

$$r + 3 = \frac{N + 3}{2}$$

**6. The probability of black followed**

The probability of black followed by red is

$$\frac{(N + 3)/2}{N} \cdot \frac{(N - 3)/2}{N - 1} = \frac{9}{35}$$

$$\frac{N^2 - 9}{4N(N - 1)} = \frac{9}{35}$$

$$35(N^2 - 9) = 36N(N - 1)$$

$$35N^2 - 315 = 36N^2 - 36N$$

$$N^2 - 36N + 315 = 0$$

$$(N - 15)(N - 21) = 0$$

**Final answer**

$$N = 15 \text{ or } N = 21$$

**CLASSIFIED PROBLEM**

Paper: P1HR

Session: May 2024

Q#: 21

Marks: 5

TOPIC: **Probability Toolkit**

Source: May 2024 P1HR

**21** There are 25 counters in a bag such that

6 counters are blue

$x$  counters are orange, where  $x > 9$

the rest of the counters are pink

Maalam takes at random two of the counters from the bag.

The probability that Maalam takes one orange counter and one pink counter is  $\frac{22}{75}$

Calculate the probability that Maalam takes 2 pink counters from the bag.

Show clear algebraic working.

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(Total for Question 21 is 5 marks)

## WORKED SOLUTION

Dr. Eslam Ahmed

## May 2024 P1HR | Question 21

Probability Toolkit

**1. There are 25 counters**

There are 25 counters.

**2. Blue counters**

Blue counters:

$$6$$

**3. Orange counters**

Orange counters:

$$x$$

**4. Pink counters**

Pink counters:

$$25 - 6 - x = 19 - x$$

**5. The probability of taking one**

The probability of taking one orange and one pink in either order is

$$\begin{aligned} \frac{x}{25} \cdot \frac{19-x}{24} + \frac{19-x}{25} \cdot \frac{x}{24} \\ = \frac{2x(19-x)}{25 \cdot 24} \end{aligned}$$

**6. This is equal to ,**This is equal to  $\frac{22}{75}$ , so

$$\frac{2x(19-x)}{600} = \frac{22}{75}$$

$$\frac{x(19-x)}{300} = \frac{22}{75}$$

$$x(19-x) = 88$$

$$19x - x^2 = 88$$

$$x^2 - 19x + 88 = 0$$

$$(x-8)(x-11) = 0$$

**7. Since ,**

Since  $x > 9$ ,

$$x = 11$$

### 8. So the number of pink

So the number of pink counters is

$$19 - 11 = 8$$

### 9. The probability of taking 2

The probability of taking 2 pink counters is

$$\begin{aligned} \frac{8}{25} \cdot \frac{7}{24} &= \frac{56}{600} \\ &= \frac{7}{75} \end{aligned}$$

**Final answer**

$$\frac{7}{75}$$

**CLASSIFIED PROBLEM**

Paper: P1HR

Session: MayNov 2020

Q#: 23

Marks: 4

TOPIC: **Probability Toolkit**

Source: May-Nov 2020 P1HR

**23** In a bag, there are only

3 blue beads  
4 white beads  
and  $x$  orange beads.

Jean is going to take at random two beads from the bag.

The probability that Jean will take two beads of the same colour is  $\frac{3}{8}$ 

Find the total number of beads in the bag.  
Show clear algebraic working.

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(Total for Question 23 is 4 marks)

## WORKED SOLUTION

Dr. Eslam Ahmed

May-Nov 2020 P1HR | Question 23  
Probability Toolkit

### 1. The total number of beads

The total number of beads is

$$x + 7$$

### 2. The probability of choosing two

The probability of choosing two beads of the same colour is

$$\frac{\binom{3}{2} + \binom{4}{2} + \binom{x}{2}}{\binom{x+7}{2}}$$

### 3. This is equal to ,

This is equal to  $\frac{3}{8}$ , so

$$\frac{3 + 6 + \frac{x(x-1)}{2}}{\frac{(x+7)(x+6)}{2}} = \frac{3}{8}$$

$$\frac{x^2 - x + 18}{(x+7)(x+6)} = \frac{3}{8}$$

$$8(x^2 - x + 18) = 3(x+7)(x+6)$$

$$8x^2 - 8x + 144 = 3x^2 + 39x + 126$$

$$5x^2 - 47x + 18 = 0$$

$$(x-9)(5x-2) = 0$$

### 4. Since must be a whole

Since  $x$  must be a whole number,

$$x = 9$$

### 5. So the total number of

So the total number of beads is

$$9 + 7 = 16$$

**Final answer**

16 beads

**CLASSIFIED PROBLEM**

Paper: P1H

Session: Nov 2023

Q#: 20

Marks: 4

TOPIC: **Probability Toolkit**

Source: Nov 2023 P1H

**20** A bag contains only 10 cent coins and 20 cent coins.

Josip takes at random a coin from the bag, records its value and replaces it in the bag. He then takes at random a second coin from the bag, records its value and replaces it in the bag.

Josip finds the mean value of the two coins.

The probability that the two coins have a mean value of 10 cents is  $\frac{49}{121}$

Work out the probability that the two coins have a mean value of 15 cents.

**(Total for Question 20 is 4 marks)**

## WORKED SOLUTION

Dr. Eslam Ahmed

**Nov 2023 P1H** | **Question 20**  
Probability Toolkit

**1. Let the probability of taking**

Let the probability of taking a 10 cent coin be  $p$ .

**2. The mean is 10 cents**

The mean is 10 cents only when both coins are 10 cent coins.

$$p^2 = \frac{49}{121}$$

$$p = \frac{7}{11}$$

**3. So the probability of taking**

So the probability of taking a 20 cent coin is

$$1 - \frac{7}{11} = \frac{4}{11}$$

**4. The mean is 15 cents**

The mean is 15 cents when one coin is 10 cents and the other is 20 cents.

**5. There are two possible orders**

There are two possible orders:

$$P(10, 20) + P(20, 10)$$

$$= \frac{7}{11} \cdot \frac{4}{11} + \frac{4}{11} \cdot \frac{7}{11}$$

$$= \frac{56}{121}$$

**Final answer**

$$\frac{56}{121}$$



# Probability Diagrams - Venn & Tree Diagrams

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**CLASSIFIED PROBLEM**

Paper: P2H

Session: Jun 2025

Q#: 20

Marks: 6

TOPIC: Probability Diagramms - Venn &amp; Tree Diagramms

Source: Jun 2025 P2H

20 120 gardeners were asked if they grow carrots ( $C$ ) or potatoes ( $P$ ) or tomatoes ( $T$ )

Of these gardeners

43 grow carrots

12 grow carrots and potatoes and tomatoes

18 grow carrots and potatoes

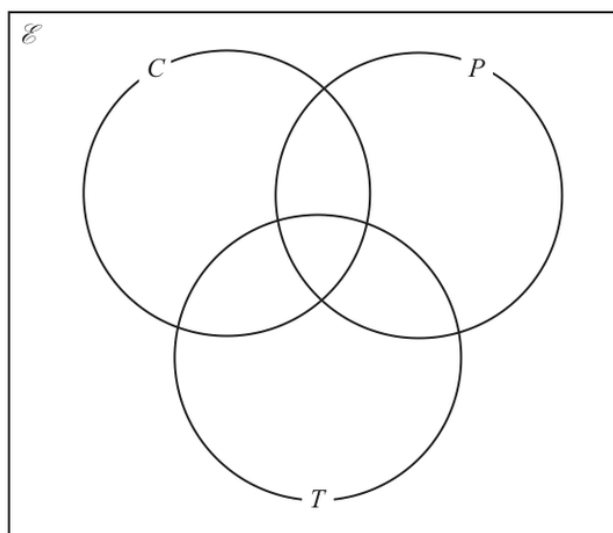
27 grow carrots and tomatoes

32 grow potatoes and tomatoes

29 do not grow carrots or potatoes or tomatoes

The number of these gardeners who grow only potatoes is equal to the number of these gardeners who grow only tomatoes.

(a) Complete the Venn diagram to show this information.



(3)

(b) Find  $n(T' \cap C)$

(1)

One of the gardeners who grows carrots is chosen at random.

(c) Calculate the probability that this gardener also grows potatoes.

(2)

(Total for Question 20 is 6 marks)

## WORKED SOLUTION

Dr. Eslam Ahmed

**Jun 2025 P2H | Question 20**

Probability Diagramms - Venn &amp; Tree Diagrams

**1. There are 120 gardeners in**

There are 120 gardeners in total and 29 are outside all three sets, so

$$n(C \cup P \cup T) = 120 - 29 = 91$$

**2. The triple intersection is**

The triple intersection is

$$n(C \cap P \cap T) = 12$$

**3. So the pair only regions**

So the pair-only regions are:

$$C \cap P \text{ only} = 18 - 12 = 6$$

$$C \cap T \text{ only} = 27 - 12 = 15$$

$$P \cap T \text{ only} = 32 - 12 = 20$$

**4. Since 43 grow carrots,**

Since 43 grow carrots,

$$C \text{ only} = 43 - 6 - 15 - 12 = 10$$

**5. Let the number who grow**Let the number who grow only potatoes be  $y$ . The number who grow only tomatoes is also  $y$ .**6. Now add the regions in**

Now add the regions in the union:

$$10 + y + y + 6 + 15 + 20 + 12 = 91$$

$$63 + 2y = 91$$

$$2y = 28$$

$$y = 14$$

**7. So the Venn regions are**

So the Venn regions are:

$$C \text{ only} = 10, \quad P \text{ only} = 14, \quad T \text{ only} = 14$$

$$C \cap P \text{ only} = 6, \quad C \cap T \text{ only} = 15, \quad P \cap T \text{ only} = 20$$

$$C \cap P \cap T = 12, \quad \text{outside} = 29$$

**8. For part (b),**

For part (b),

$$n(T' \cap C) = C \text{ only} + C \cap P \text{ only}$$

$$= 10 + 6 = 16$$

**9. For part (c), one gardener**

For part (c), one gardener who grows carrots is chosen.

$$P(\text{also grows potatoes}) = \frac{n(C \cap P)}{n(C)}$$

$$= \frac{18}{43}$$

**Final answer**

(b) 16, (c)  $\frac{18}{43}$

# Combined & Conditional Probability

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**CLASSIFIED PROBLEM**

Paper: P2H

Session: Jun 2023

Q#: 20

Marks: 3

TOPIC: **Combined & Conditional Probability**

Source: Jun 2023 P2H

**20** There are 12 counters in a bag.

3 of the counters are red

9 of the counters are green

Ameya, Jack and Ella each take at random one counter from the bag.

Work out the probability that at least one red counter is still in the bag.

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(Total for Question 20 is 3 marks)

## WORKED SOLUTION

Dr. Eslam Ahmed

**Jun 2023 P2H | Question 20**

Combined &amp; Conditional Probability

**1. There are 3 red counters**

There are 3 red counters.

**2. At least one red counter**

At least one red counter is still in the bag unless all 3 red counters are taken.

**3. So use the complement**

So use the complement:

$$\begin{aligned}
 P(\text{all 3 red are taken}) &= \frac{3}{12} \cdot \frac{2}{11} \cdot \frac{1}{10} \\
 &= \frac{6}{1320} \\
 &= \frac{1}{220}
 \end{aligned}$$

**4. Therefore,**

Therefore,

$$\begin{aligned}
 P(\text{at least one red is still in the bag}) &= 1 - \frac{1}{220} \\
 &= \frac{219}{220}
 \end{aligned}$$

**Final answer**

$$\frac{219}{220}$$

**CLASSIFIED PROBLEM**

Paper: P1H

Session: Nov 2025

Q#: 22

Marks: 3

Topic: **Combined & Conditional Probability**

Source: Nov 2025 P1H

**22** There are 15 buttons in a box.

3 of the buttons are red  
2 of the buttons are pink  
10 of the buttons are blue

Pete takes at random three buttons from the box.

Work out the probability that there is still at least one pink button in the box.

**(Total for Question 22 is 3 marks)**

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## WORKED SOLUTION

Dr. Eslam Ahmed

## Nov 2025 P1H | Question 22

Combined &amp; Conditional Probability

**1. There are 2 pink buttons**

There are 2 pink buttons.

**2. At least one pink button**

At least one pink button is still in the box unless both pink buttons are taken.

**3. So use the complement**

So use the complement.

**4. The probability that both pink**

The probability that both pink buttons are taken is

$$\begin{aligned} & \frac{\binom{2}{2} \binom{13}{1}}{\binom{15}{3}} \\ &= \frac{13}{455} \\ &= \frac{1}{35} \end{aligned}$$

**5. Therefore,**

Therefore,

$$\begin{aligned} P(\text{at least one pink is still in the box}) &= 1 - \frac{1}{35} \\ &= \frac{34}{35} \end{aligned}$$

**Final answer**

$$\frac{34}{35}$$